

RESTORING TE PĀTAKA O RĀKAIHAUTŪ BANKS PENINSULA

Raewyn Peart and
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¹ Langman and McInerney, 2020

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LIST OF ACRONYMS

Banks Peninsula	Te Pātaka o Rākaihautū Banks Peninsula
BPCT	Banks Peninsula Conservation Trust
Canterbury RPS	Canterbury Regional Policy Statement
Coastal Plan	Canterbury Regional Coastal Environment Plan
DOC	Department of Conservation
EDS	Environmental Defence Society
ETS	Emissions trading scheme
MPI	Ministry for Primary Industries
NESPF	Resource Management (National Environment Standards for Plantation Forestry) Regulations 2017
NZCPS	New Zealand Coastal Policy Statement
NZU	New Zealand Unit
ONF	Outstanding natural feature
ONL	Outstanding natural landscape
QE II	Queen Elizabeth the Second National Trust
RMA	Resource Management Act 1991
SES	Significant ecological sites

1. INTRODUCTION

Aotearoa New Zealand's distinctive natural landscapes are an integral component of our individual and national well-being. For tangata whenua, they provide a deep cultural connection with the past and future through whakapapa. Landscapes provide physical and spiritual respite, historical links and havens for biodiversity. They are a source of creative endeavour, a key component of the New Zealand brand, and a major drawcard for tourists.

Despite the importance of landscapes to the country, and this being recognised by various statutes for over 25 years, we are still seeing poor landscape outcomes. This indicates that there is institutional and regulatory failure in protecting these high value places in the public interest.

In order to address this matter, the Environmental Defence Society (EDS) has initiated a project to investigate how existing legislative and policy tools could be more effectively deployed to protect important natural landscapes, as well as how a new 'protected landscapes' model could be adopted to achieve better landscape protection in New Zealand.

The project involves case studies of landscape protection in Te Manahuna Mackenzie Basin, Te Pātaka o Rākaihautū Banks Peninsula (Banks Peninsula), the Waitākere Ranges and the Hauraki Gulf Islands. It also includes an investigation into potential linkages between tourism and landscape protection. The case study reports are available on the EDS website (www.eds.org.nz). The case studies are being integrated into an overall report, which will also contain a broader exploration of the concept of landscape within the New Zealand context.

In 2004, EDS published 'A Place to Stand', which explored the extent of loss of New Zealand's high value landscapes to inappropriate subdivision and development. The report included a number of specific case studies, including one focused on Banks Peninsula.¹

That case study described how the Banks Peninsula District Council was struggling to address landscape protection issues in its proposed Banks Peninsula District Plan which was notified in 1997. There was strong opposition from farmers to the landscape and biodiversity provisions in the proposed plan and the Council established a taskforce to help navigate a way through the competing views. The recommendations of the taskforce, which were largely adopted by the Council, mainly focused on biodiversity conservation rather than landscape protection. In the absence of protective district plan provisions, the district's landscapes were left vulnerable to increasing development pressures. The case study noted that, if amalgamation with Christchurch City proceeded (which it did in 2006), the protection of landscapes on the Peninsula could be addressed within a wider perspective.

The 2004 case study was written at a time when there was increasing nationwide demand for coastal lifestyle

living; a relevant issue to the Banks Peninsula given its location close to New Zealand's second largest city. However, the Global Financial Crisis in 2007-2008 and subsequent Canterbury earthquake sequence in 2010-2011 both had a significant dampening effect on coastal development in the Canterbury Region.

There have been considerable changes since 2004, and this Banks Peninsula case study takes a fresh look at pressures on landscape values and how they are being managed. It is based on a review of available literature, relevant statutory provisions and planning documents. We also undertook two field trips to Banks Peninsula during July and August 2020 and undertook in-depth interviews with 29 people consisting of 10 landowners, 9 agency people and experts and 10 other stakeholders. These interviews were undertaken on a confidential basis to encourage frankness. We have included some quotes from these interviews to provide 'colour' to the analysis below. However, we have not identified the source of these in order to maintain the confidentiality of the respondents.

This report is structured into six main parts. Chapter 2 provides an historical context for the case study, including describing the removal of almost all of the indigenous forest on Banks Peninsula. This is followed by a description of initiatives that started healing the land in Chapter 3. Chapter 4 describes more recent initiatives that are attempting restoration at a landscape scale. Chapter 5 reviews current and future pressures on the landscape. Chapter 6 then examines how the regulatory system is responding to these pressures. The report ends with conclusions and a set of recommendations in Chapter 7. A timeline of key events on Banks Peninsula, of particular relevance to the case study analysis, is shown in Figure 1.1.



Year	Event
1984	Removal of farm subsidies begins with election of Labour Government
1985	First dolphin eco-tours undertaken
1987	Hinewai Reserve established First Queen Elizabeth the Second covenant established on Banks Peninsula
1989	Banks Peninsula District Council formed Banks Peninsula Track established
1993	Whole of Banks Peninsula identified as an outstanding natural landscape in the Canterbury Regional Landscape Study
1997	Proposed Banks Peninsula District Plan notified
1998	Ngāi Tahu Claims Settlement Act passed Rāpaki Mātaitai Reserve established
1999	Akaroa registered as an historic area Pōhatu Marine Reserve established
2000	Koukourārata Mātaitai established
2001	Banks Peninsula Conservation Trust formed
2003	'Wildside' initiative established on south-east of Peninsula
2005	Joseph Langer Charitable Trust established
2006	Akaroa Harbour Taiāpure established Banks Peninsula District Council amalgamated with Christchurch City Council
2007	Environment Court consent order requiring Council to identify Significant Ecological Sites
2008	Process of identifying Significant Ecological Sites commences
2010	Rod Donald Banks Peninsula Trust formed with Council endowment of \$3.5 million Te Kaio and Wairewa Lake Forsyth Mātaitai established
2011	Cruise ships divert from Lyttelton to Akaroa Harbour
2014	Akaroa Harbour Marine Reserve established
2015	Proposed Christchurch District Plan notified under the Canterbury Earthquake (Christchurch Replacement District Plan) Order 2014
2017	Christchurch District Plan operative Christchurch City Council establishes biodiversity funding to support Significant Ecological Areas (\$200,000 a year) Banks Peninsula Conservation Trust launches a 2050 ecological vision for Banks Peninsula Te Pātaka o Rākaihautū Lyttelton Harbour Whakaraupō Mātaitai established
2018	14 agencies and organisations sign a memorandum of understanding for a Pest Free Banks Peninsula initiative
2019	Environment Canterbury introduces a new Pest Free Banks Peninsula rate generating \$600,000 per annum
2020	\$5.11 million funding announced for Pest Free Banks Peninsula initiative from Covid-19 Jobs for Nature fund, creating 15 full time jobs

Figure 1.1 *Timeline of key events for Banks Peninsula case study*

ENDNOTES

1 Peart, 2004

2. HISTORICAL CONTEXT

When Te-Waka-a-Aoraki failed to return, Tu-te-raki-whanoa, a mokopuna of Aoraki, outfitted a canoe and set out in search of his tipuna. He tracked the journey of his tipuna and eventually arrived at the wreck to find that Aoraki and the rest of the crew had turned into stone. Tu-te-raki-whanoa mourned for his tipuna and after much thought decided to turn Te-Waka-a-Aoraki into a place for people to live on. He noticed how the cold wind swept up the eastern side of the canoe. He made a great rake and raked the rubble into a pile to form the bays and harbours of the Banks Peninsula. Then he rested while a fellow crewman, Kahukura, took over the task of covering the peninsula with trees and shrubs and liberating birds. When this was completed another crewman, Marokura, placed various kinds of fish and shellfish in the bays, harbours, streams and swamps. He stocked Whakaraupō (Lyttelton Harbour) last. After he had deposited a wide variety of seafood around its bay, he placed a further selection around Otamahua (Quail Island). After they had completed their tasks the three men rested.¹

2.1 Geological history

"The Akaroa and Lyttelton volcanoes, producing two of the best examples of erosional craters to be found anywhere in the world, attract international attention from geologists."²

Banks Peninsula, the largest true peninsula in New Zealand, is located on the east coast of the South Island within the Canterbury region (see Figure 2.1).³ The Peninsula is around 50km long and 30km wide, covering an area of approximately 1,120km². It is an iconic landscape featuring two ancient volcanic cones and associated lava flows. The Peninsula has a rugged topography with more than 100 short and often very steep catchments. Formerly perhaps up to 3,000m high, the landform now ranges from sea level to around 920m in height. Many of the summits on the Peninsula exceed 500m.⁴

The Lyttelton and Akaroa volcanoes erupted over a period between 121 million and 5.8 million years ago creating an island off the coast of the South Island. The volcanoes remained an island for around 20 million years, with a connection to the mainland only being established permanently some 20,000 years ago, thereby forming a peninsula. This land connection was created by the enormous quantities of gravels being brought down by rivers from the eroding Southern Alps. The gravels gradually built up the extensive Canterbury Plains and, as they moved eastwards, eventually reached the volcanic island. The island origins of Banks Peninsula can still be seen in the existence of old sea cliffs in Halswell and along the Port Hills.⁵ Te Roto o Waiwera Lake Forsyth was an open harbour for most of its life, but became a lake when the shingle moving along the Kaitorete spit closed its mouth during the 1830s.⁶

Over the millions of years since the volcanoes were formed, erosion has reduced the height of the cones



Figure 2.1 Map of Banks Peninsula (Source: Google Earth)

and numerous streams have incised the flanks of the old volcanoes into steep valleys. Rising sea levels after the last ice age flooded the old craters to form the Lyttelton and Akaroa harbours. Numerous other bays formed around the perimeter of the Peninsula as the steep valleys were flooded. The sea subsequently eroded the old lava flows to form high coastal cliffs. Fine silt and glacial dust blown from the Canterbury Plains blanketed the lower slopes of the Peninsula to form the fine yellow loess soils. In some places the loess layer is up to 20 metres thick.⁷

2.2 Landcover and indigenous species

“Te Pātaka o Rākaihautū/Banks Peninsula’s volcanic and island origins make it a unique and special landform for the diversity of life found upon it, and a hotspot for endemism. The many bays, formed from the volcanics, create sheltered areas with microclimates. Loess, combined with the volcanic soils and a legacy of millions of years of burrowing sea birds, produced rich fertile soils. This combination of natural fertility, high rainfall, sheltered valleys, and added marine nutrients, created perfect conditions on Banks Peninsula for abundant growth and diversity.”⁸

Because of its long history as an island and steep, rugged topography creating numerous microclimates (with some areas being virtually frost free), Banks Peninsula “supports a unique indigenous biodiversity characterised by high levels of endemism and a high proportion of species at their distributional limits”.⁹ There are at least six species of plant endemic to Banks Peninsula including the Akaroa daisy (*Celmisia mackayi*), Banks Peninsula sun hebe (*Heliohebe lavaudiana*) and Banks Peninsula blue tussock (*Festuca actae*).¹⁰

Prior to human arrival, the Peninsula was almost entirely covered by forest. In the valley lowlands, the forest was

dominated by tōtara, matai and kahikatea. On the higher peaks there were thin-barked tōtara and native cedars. A small area of beech forest established on the south-east corner of the Peninsula. Nīkau palms grew in warm coastal valleys and snow tussock on the bare rocky high ground.¹¹ The forest environments supported a vast array of fauna. In particular, Banks Peninsula is notable for its endemic invertebrates. These include “a cockroach, at least two wētā, several beetles and weevils, spiders, flies (including a curious net-winged midge), some moths, a mayfly, a stonefly, several millipedes and no fewer than five caddisflies.”¹²

It is estimated that there were around 100 species of bird on the Peninsula at the time of first human arrival.¹³ These included numerous forest and seabirds. Petrels created nesting burrows all around the Peninsula from the mountain summits right down to the sea edge. Nesting penguins would have been present in their tens of thousands including the endemic little blue/white flippered (kororā) penguin, which was thought to have reached around 20,000 pairs.¹⁴ There would have been fur seals aplenty around the Peninsula’s shores, as well as the New Zealand sea lion. The sea around the Peninsula was also the stronghold for the little endemic Hector’s dolphin.

2.3 Early settlement

Māori tradition records three waves of historic settlement of the Banks Peninsula starting with the Waitaha people, led by Rākaihaitū, who arrived from the north on the *Uruao* canoe. Their settlement was focused around the Akaroa harbour. Today, Rākaihaitū is remembered within the harbour by the name of the peak behind Wainui (Mount Bossu) which is called Tuhiraki, referring to his kō or digging stick which was said to have been used to dig out the inland South Island lakes. He is also remembered by the name of the Peninsula itself, Te Pātaka o Rākaihaitū or food storehouse of Rākaihaitū, referencing the abundant food sources in the area including prolific forest birds.¹⁵

There was a second and longer wave of settlement on the Peninsula by the Ngāti Māmoe people, descendants of Hotu Māmoe who moved south from the Poverty Bay area. They were more aggressive than the Waitaha people and are thought to have built the first pā on the Peninsula including at Mānuka (near Tai Tapu) and Paekaroro (near Dyke Head on the southern headland of Flea Bay Pōhatu).¹⁶

The most recent wave of settlement, from around 1700, was by Ngāi Tahu people who gradually moved south from the east coast of the North Island. After several notable and successful battles against Ngāti Māmoe, the Peninsula became a strategic stronghold for Ngāi Tahu in the south, enabling the tribe to eventually exert control over most of the South Island. Ngāi Tahu built some fourteen defensive pā on the Peninsula. Settlements were established in most bays, from which was harvested the Peninsula’s rich forest, marine and freshwater resources. Kūmara cultivation was also undertaken, facilitated by the Peninsula’s rich volcanic soils and relatively mild climate. Settled areas and resource harvesting sites



Ōnuku Marae

were connected by a network of trails, tracks and waka passages. These included the Waipuna saddle, which linked settlements at Koukourāta (Port Levy) and Wairewa (Lake Forsyth), and Te Tara o Te Tangihikaia (Gebbies Pass) which linked Whakaraupō (Lyttelton Harbour) and Te Waihora (Lake Ellesmere).¹⁷

This 800 years of Māori occupation had significant environmental impacts, although they pale in comparison to the subsequent European environmental onslaught described below. Pre-colonisation, around a third of the forest was cleared and a range of bird species (mainly those that were large and edible) became extinct, including several species of moa and duck as well as an adzebill, flightless goose, black swan, giant eagle, raven and owl-nightjar.¹⁸

In 1770, Captain James Cook sailed past the Peninsula in the *Endeavour* but did not stop. He thought the land jutting out from the coast was an island and named it Banks Island after Sir Joseph Banks, the British naturalist who was on board. Ngāi Tahu's first recorded contact with Europeans on the Peninsula was in 1815 or 1816 when Captain John Grono and his crew from the sealing ship *Governor Bligh* landed in Akaroa. They were followed by other sealers and flax traders who established trading relationships with tangata whenua. These interactions were of considerable historical import. An incident in 1830 at Takapūneke, on Akaroa harbour, was a precursor to Britain deciding to assert sovereignty over Aotearoa (see spotlight).

A spotlight on Takapūneke

Takapūneke was the kāinga of the Ngāi Tahu chief Te Maiharanui. During the 1820s, Te Maiharanui established a trading base in the village to supply flax and vegetables to visiting whalers and flax traders. In 1829, he became involved in conflict with Te Rauparaha, a northern Ngāti Toa chief. Seeking to get the better of his foe, Te Rauparaha persuaded British sea captain John Stewart to take his party of warriors to what became Akaroa in Stewart's brig *Elizabeth*, as part of a plot to capture Te Maiharanui. In return, Stewart was to receive a cargo of flax.¹⁹

The *Elizabeth* anchored off Takapūneke in November 1830. Te Rauparaha's warriors remained hidden below. Stewart's interpreter John Cowell invited Te Maiharanui on board where he was held captive below decks. Subsequently his wife, daughter and other tribal members boarded the boat, being unaware of any evil intent and thinking that it was on a peaceful trading mission. The young daughter was killed by her parents to save her from outrage and her body was thrown overboard. Te Rauparaha's warriors went ashore and sacked the kāinga. Overall, at least 100 and possibly up to 200 people were killed. After a cannibal feast onshore, Te Rauparaha's men brought baskets of human flesh back onto the ship.²⁰ The *Elizabeth* sailed to Kapiti where Te Maiharanui and his wife were tortured to death by Te Rauparaha.

The survivors of the massacre moved to the next bay south, known as Ōnuku ('last resting place'). As a result of the massacre, Takapūneke is wāhi tapu, and for many years Ōnuku whānau would not venture onto the land where the blood of so many of their ancestors was shed.²¹

The *Elizabeth* affair shocked the British authorities in Sydney and London. The incident was part of the rationale for Britain deciding to assume sovereignty over Aotearoa and eventually led to the signing of Te Tiriti o Waitangi. The site is therefore of great cultural significance to mana whenua and of great historical significance to the country and internationally.

During the 1960s, a small site at Takapūneke was purchased by the Akaroa County Council for the development of Akaroa's first sewage treatment system. This started discharging into the harbour in 1964. The council subsequently purchased further land in the area and established a rubbish dump there in 1979. These developments were later described as "the ultimate in modern cultural oppression" with it being unthinkable to construct a Māori sewage works (or rubbish dump) on top of a European cemetery.²²

During the 1990s, the Council subdivided much of its landholding through a non-notified resource consent. The Ōnuku Rūnanga expressed its deep concern and outrage in a letter to the Council which resulted in a portion of the land being designated a reserve in 1998. The Akaroa Civic Trust opposed new housing directly adjacent to the Britomart Memorial. Understanding the cultural significance of the site, the Civic Trust joined forces with Ōnuku Rūnanga to stop further development in the vicinity and together they sought protection of the land. After nearly 20 years, the campaign was eventually successful, and in October 2008 the area was declared the Takapūneke Historic Reserve.²³

Although the Council agreed to close the rubbish dump in 1998, there was still the issue of the sewage treatment works. Its continued operation discharging human waste to the harbour waters was culturally offensive to Ngāi Tahu. The Council established a working party in 2008 to investigate the treatment of Akaroa's wastewater. In October 2020, the Akaroa Treated Wastewater Options Hearings Panel finally concluded its deliberations on four options that were put forward by the Council. The Panel unanimously supported the 'Inner Bays Irrigation Scheme' which will consist of a new wastewater treatment plant within Akaroa township, irrigation of treated water to native trees primarily within Robinsons Bay, and re-use through purple pipes and subsurface wetlands.²⁴

A second notable raid on Akaroa by Te Rauparaha occurred in 1832 when he set his sights on the well-fortified pā on Ōnawe Peninsula which had been built by Takatahara and two other chiefs. Te Rauparaha again used

trickery over might. He sent fifty waka to Akaroa harbour carrying his warriors as well as Ngāi Tahu prisoners from an earlier sacking of the Kaiapoi pā. "Knowing that he could not take the well fortified [Ōnawe] pā, he sent the Kaiapoi prisoners to tell Takatahara that they came in peace. When the gates opened, Te Rauparaha attacked, killing nearly 300".²⁵

From around 1835, whaling ships from America, France, England and Australia started frequenting the Peninsula. By 1844, four shore whaling stations had been established on the south coast (Island Bay, Peraki, Ikoraki and Oashore) and an additional one at the head of Lyttelton Harbour (Little Port Cooper), all under the authority of local Māori chiefs. Ngāi Tahu provided water, firewood, pork and potatoes to the whalers and in return received blankets, biscuits, firearms and alcohol. Many European whalers married Ngāi Tahu women and lived amongst the tribe.²⁶

In 1838, Jean Langlois, the commander of the French whaling ship *Cachalot*, purported to purchase the entire Banks Peninsula from its Māori owners. The Nanto-Borelaise Company was then formed in France for the purpose of establishing a settlement at Akaroa. The French government provided sufficient backing to the company to enable it to send the corvette *Aube* captained by Charles Lavaud to Akaroa, accompanied by the storeship *Comte de Paris*, in order to complete the purchase of land and establish a settlement. The ships left France in January 1840. Knowledge of these French intentions prompted the English to advance their own plans to assert sovereignty over Aotearoa. By the time the French ships were only halfway through their passage south, Te Tiriti o Waitangi was signed at Ōnuku on 30 May 1840 by two Ngāi Tahu chiefs.

When Lavaud arrived in the Bay of Islands he learnt that the British had already declared sovereignty over the country. Lieutenant Governor William Hobson swiftly despatched the HMS *Britomart* to Akaroa, and five days before Lavaud arrived with 57 French and German settlers, the Union Jack was flying over the settlement. Despite this, the French settled in what became known as French Bay (now part of Akaroa township) and the Germans went to the adjacent Takamātua Bay.²⁷ The French settlers also established flourishing gardens across the harbour at a place now known as French Farm. Akaroa became the first planned township in the South Island, and was the location of the first post office, police force, magistrate and customs house. The French Navy remained until 1846 and the French influence can still be seen in Akaroa architecture today.²⁸

By the mid-1800s, there were three purported 'purchases' of Ngāi Tahu land on Banks Peninsula. These included two French deeds of sale from 1838 and 1840 and Kemp's Deed which was signed by a group of Ngāi Tahu chiefs on board the HMS *Fly* in Akaroa harbour in 1848 and extended over a significant area of Canterbury. Ngāi Tahu denied that these gave the British or the French title over Banks Peninsula and so Walter Mantell was appointed by the British Crown to settle the matter. Eventually, he

persuaded Ngāi Tahu to sign the Port Levy and Port Cooper Deeds in 1849 and the Akaroa Deed in 1856. After losing title to much of their land most Ngāi Tahu residents left the area.²⁹ A 1848-49 census estimated the population of Māori on the Peninsula to be only 300.³⁰

It was 150 years before these injustices were addressed. In 1997, the Ngāi Tahu Deed of Settlement was signed at the Takahanga Marae in Kaikoura. The following year, the Ngāi Tahu Claims Settlement Act was passed and a formal apology was given by the Crown at Ōnuku. In that Act, the Crown acknowledged that it acted unconscionably and in repeated breach of the principles of Te Tiriti o Waitangi in its dealings with Ngāi Tahu over purchases of land. It vested various parcels of land in Te Rūnanga o Ngāi Tahu including land on the Banks Peninsula at Okains Bay, Te Waiomākua, Ōnawe Pā and Wairewa.³¹ The preamble to the Act explains:

The Tribunal found that the Crown granted the Nanto-Bordelaise Company an interest in 30,000 acres of land on Banks Peninsula, that Ngāi Tahu had not agreed to relinquish most of this land and was not compensated for its loss, and that the Crown used high-handed and unfair methods in its dealings with Ngāi Tahu over the Port Cooper and Port Levy Blocks. Significant to the Tribunal's findings on the Port Levy Purchase was the Crown's refusal to make reserves, as requested by Ngāi Tahu, at Okains Bay, Kaituna Valley and Pigeon Bay. The Tribunal further found that the Crown had dealt with land on Banks Peninsula before it had been lawfully acquired from Ngāi Tahu and that the Crown failed to meet its obligation to provide ample reserves for the existing and future needs of Ngāi Tahu.³²

Banks Peninsula contains a rich tapestry of Māori history and stories which can be found in place names, oral history and a vast array of archaeological remains in settlement and pā sites. Four Papatipu Rūnanga have marae on the Peninsula: Te Hapū o Ngāti Wheke (Rāpaki), Te Rūnanga o Koukourārata, Ōnuku Rūnanga and Wairewa Rūnanga.



2.4 Early land clearance and development

As whaling dwindled on the Peninsula during the mid-nineteenth century, due to the whale populations being decimated, attention turned to the area's rich timber stands. The Peninsula was the largest forested area close to Christchurch by far, so it soon became the source of firewood and timber for the new settlement. Many former whalers, now out of work, started bush felling and pit sawing to meet the demand. The industry soon became mechanised, with a water-driven mill established at the head of Akaroa Harbour in 1854 and a stream-powered mill in Le Bons Bay in 1857. During the 1860s, there was strong demand for timber from a Canterbury economy buoyed up by the high price of wool. There was also an expansion of railways, following completion of the Lyttelton Tunnel in 1867, providing better access to markets. Twenty-six saw mills were established on the Peninsula before 1880 and they were a feature of most of the bays. As well as logging, fires also devastated much forest. In 1869, it was estimated that fires had swept through 12,000ha of bush during the previous 10 years. By the 1870s, the timber industry was on the decline.³³

A railway from Christchurch to Little River was opened in 1886 and this supported the development of agriculture on the Peninsula. Many mill workers had made sufficient money to set themselves up on farms. The new farmers set about burning the cut-over areas and remaining bush in order to sow grass seed and run livestock. Dairy farms were established on the flats. One European grass variety did particularly well on the rich volcanic soils of the Peninsula and out-competed other grasses. It became known as 'Akaroa cocksfoot'. Rather than growing the grass for stock feed, farmers started to treat the grass as a crop itself, harvesting the seed and selling it to fast-developing North Island farms as well as overseas.³⁴

Cocksfoot seed production proved very lucrative. The considerable wealth generated is evidenced by the large homesteads and gardens that were established on the Peninsula. The booming industry also meant that a living could be made from a combination of dairying and cocksfooting on lots as small as 12-16ha. This prompted further subdivision and close settlement of the rural area. At its peak in 1904-5, it is estimated that 10,125ha of land was being harvested producing 50,000 to 60,000 sacks of Akaroa cocksfoot seed a year, with a thousand casual labourers employed.³⁵

In order to harvest the cocksfoot seed, stock had to be excluded from the paddocks, so farmers favoured the more rugged areas of their farms for its production which they cleared of native bush. This led to further deforestation, particularly on the north side of the Peninsula and around Little River where there were warmer and more sheltered conditions favouring grass growth. By 1906, the remaining forested area of the Peninsula was estimated at around just 320ha, or 0.7 per cent of the area remaining in 1850³⁶ and perhaps 0.3 per cent of the forest existing prior to human settlement.

A spotlight on the impact of indigenous forest loss on Banks Peninsula

The almost complete loss of forest was devastating for the indigenous species that lived on the Peninsula. Hugh Wilson estimates that 20 plant species became extinct, with several more species reduced to a few individuals or small vulnerable populations. Many bird species also succumbed to the loss of forest habitat as well as the introduction of predators such as rats, cats, stoats and ferrets. "Among those that vanished from the peninsula – mostly between 1850 and 1900 – are kōkako, kākā, red-crowned kākārīki, yellow-crowned kākārīki, saddleback, robin, bush wren, piopio, fernbird, New Zealand quail, yellowhead or mohua, mottled petrel, laughing owl, buff weka, New Zealand dabchick and brown teal." Tuī have more recently declined to very low numbers.³⁷

The cocksfoot industry provided fodder that supported a surge in dairying. This was the main industry on the Peninsula from around 1910 up until the Great Depression in 1930, with the number of cows peaking at 10,000. Most bays had a dairy factory and the coastal flats were intensively farmed. The depression hit the dairy industry hard and sheep started to take over as the favoured stock. The valley sides and rolling hill country were developed for extensive beef and sheep grazing operations, with homesteads established in the valleys and at the mouth of bays. By the 1960s, the number of cows had halved and sheep had increased to around 270,000, at which time Banks Peninsula had the highest carrying capacity in the South Island.³⁸

Wool became a major income source for the Peninsula, with store and breeding stock being sold to the more intensive finishing properties on the Canterbury Plains. Where the land was closely grazed by livestock, weeds were kept under control, and the native bush was not able to regenerate. But where grazing was less intense "large areas of gorse and regenerating bush rapidly took over from the grassland and dominated the landscape".³⁹ The cocksfoot industry ceased in the 1960s, and by the 1980s there was only one dairy factory left on the Peninsula. Government farm subsidies during the 1970s encouraged farmers to bring marginal land back into development, resulting in much clearance of regenerating native bush and increased stocking rates.⁴⁰

Despite the forest devastation, many scenic reserves were preserved by early conservation action and the Department of Conservation (DOC) now manages some 65 scenic, historic, recreation and other reserves on the Peninsula. They are typically small, isolated and fragmented and are mostly located on upper shady slopes along the spine of the Peninsula which were the areas least attractive for farming.⁴¹

Key messages

- Banks Peninsula is an iconic landscape formed from two ancient volcanoes and associated lava flows. It was once a largely naturally forested landscape.
- The long history of the Peninsula as an island, and its numerous different micro-climes, has meant that it supports unique indigenous biodiversity with a high level of endemism.
- There have been three waves of Māori settlement, the most recent by Ngāi Tahu. The Peninsula was rich in food resources so was heavily settled. It is a landscape of significant cultural importance to Ngāi Tahu.
- Akaroa also has considerable historical significance. Events that occurred there in 1830 helped prompt Britain to assert sovereignty over Aotearoa and were a contributing factor to the

signing of Te Tiriti o Waitangi. It is also the location of the only French settlement in New Zealand.

- Whaling was the first main industry on the Peninsula followed by logging, dairy farming and then sheep and beef farming. The production of cocksfoot seed became a major industry in the late nineteenth century.
- Logging and burning had destroyed more than 99 per cent of the forest cover by the early 1900s. This in turn adversely affected species dependent on forest habitat including around 20 plant species and many birds.
- Although there was a period of forest regeneration when farmers switched to producing wool, and areas of their farms were grazed less intensively, much of this regenerating land was cleared during the 1970s as a result of Government's farm subsidies designed to bring marginal land back into production.



Historic wharf, Akaroa Harbour

ENDNOTES

- 1 Jackson, 1990, 9-10
- 2 Olgivie, 2010, 1-2
- 3 Lake Ellersmere Te Waihora is not included within the ambit of this report.
- 4 Wilson, 2013, 11; Harding, 2003, 334
- 5 Weaver et al, 1985, 24
- 6 Olgivie, 2010, 4
- 7 Olgivie, 2010, 1
- 8 Te Pātaka o Rākaihautū/Banks Peninsula Geopark Trust, 2020, 5
- 9 Appelton, 2015, para 7.9
- 10 <https://www.doc.govt.nz/our-work/motukarara-conservation-nursery/canterbury-native-plants-by-area/native-plants-natural-to-banks-peninsula/>
- 11 Wilson, 2013, 35-38
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- 14 Banks Peninsula Conservation Trust, 2019
- 15 Christchurch District Plan, 2015, Te Pātaka o Rākaihautū/Banks Peninsula Landscape Study - Ngāi Tahu Values Addendum
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- 20 Ngāi Tūāhuriri Rūnanga et al, 2013, 40; Christchurch City Council, 2012, 23
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- 26 Olgivie, 2010, 5
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- 33 Wood and Pawson, 2008, 453-454
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3. HEALING THE LAND

3.1 Farm restructuring during the 1980s

The trend of land clearance and increased stocking during the 1970s was reversed during the economic reforms of the 1980s which, combined with a succession of drought years, resulted in considerable restructuring of farm systems. The cost of maintaining new marginal pastures proved uneconomic, land was left to revert to scrub, and stocking rates were reduced. This helped take pressure off the land. Many farms struggled for economic survival and farming families looked to diversify their income in order to survive. Things would have been worse if wool prices hadn't climbed rapidly, reaching an all-time high in 1989.¹ Overall, there was a 25 per cent reduction in livestock farmed on Banks Peninsula between 1980 and 1990. The area recorded as being in grassland, tussock or crop changed from 85,727ha in 1980 to 70,900ha in 2002, a reduction of 17 per cent. At the same time, the coverage of indigenous forest was fast-increasing and it reached 2,600ha in 2002.² Nature was making a comeback.

A spotlight on restoration of Te Pātaka o Rākaihautū by Papatipu Rūnanga

Papatipu Rūnanga have been actively involved in numerous initiatives to restore Te Pātaka o Rākaihautū. For example, Koukourārata Rūnanga have led the extensive replanting of land around the Kahukunu and Koukourāta streams with thousands of native plants, and have established a Nga Whenua Rāhui Kawenata on part of the land.³ Wairewa Rūnanga and Ngāi Tahu initiated work to bring Te Roto o Wairewa Lake Forsyth, and its fisheries, back to health through self-funding the construction of a canal to permanently link the lake with the sea.⁴ Another notable initiative was the Kaupapa Kererū project, which aimed to increase the number and range of kererū on Banks Peninsula, and was undertaken as a collaboration between Te Rūnanga o Ngāi Tahu, Lincoln University, Maanaki Whenua (Landcare Research) and DOC. It included an oral history project which recorded the stories, observations and memories of current and former Banks Peninsula residents associated with the kererū.⁵

Rūnanga have also been very active in seeking to restore marine biodiversity around Te Pātaka o Rākaihautū with the establishment of the Rāpaki Mātaitai Reserve in 1998, the first in the country, and the Akaroa Taiapūre in 2006. Rūnanga have also established mātaitai reserves in Koukourārata Port Levy (2000), Wairewa Lake Forsyth (2010), Te Kaio – which is adjacent to Wairewa (2010) and Lyttelton Harbour Whakaraupō (2017).

As well as undertaking localised projects, rūnanga have been involved in many landscape-scale initiatives on the Peninsula including the Wildside Project, the Pest Free Banks Peninsula Project, the recent Te Kāhahu Kahukura project on the Southern Port Hills (all described below) and the Whaka-Ora Heathy Harbour project focused on improving the health of Lyttelton Harbour Whakaraupō.

3.2 Banks Peninsula Track

"[The track] brought about a shift from traditional farming to regenerative farming where beautiful scenery, biodiversity, and healthy forest were valued for economic reasons."⁶

The economic pressures of the 1980s were felt most markedly by the more economically marginal farms, those in the steeper, colder and more exposed parts of the Peninsula such as on the south-east coast. There, farmers were desperate to find new income sources so they could stay on the land. In the words of one of those farmers, "a group of peninsula landowning families were talking through their problems when somebody threw out the idea of creating a walk across their lands and charging walkers for access and accommodation". The idea was sparked by the interest locals had shown over the years in walking, camping and horse trekking in the area's small coves and bays with landowner permission.⁷

Accommodation on Banks Peninsula Track



The farmers eventually developed a four-day walking track which started and ended in Akaroa. The track looped through their properties, along with areas of public land, rising from sea level to cross the Peninsula's 700m high spine. Accommodation for the walkers was provided in renovated farm buildings. The 'Banks Peninsula Track' opened in 1989 and it was the first privately-owned track in New Zealand. It is still going strong 30 years later, although its route changed when one landowner pulled out in 2017.⁸ As well as providing accommodation, one landowner offers wildlife tours and sea kayaking and there is a small shop for the purchase of supplies on another property. About 2,000 people walk the track each year, mainly New Zealanders, and it provides a significant portion of on-farm income for participating landowners.

"We underestimated what we had here. It took a few hundred trampers to tell us how amazing it all was. After that we started to look at our home with different eyes."⁹

"People wanted to see nature on the track so that was an incentive for the farmers to look after it. The track had quite an impact on this corner of the Peninsula."

As well as providing much needed additional income, the track started to change farmers' perceptions of how their land should be managed. It demonstrated, in a very tangible way, that there was money in conservation. All of the landowners participating in the track are now involved in on-farm conservation initiatives. As one farmer explains, "It gave us the excuse we wanted to properly look after this precious place. Before the track, we had to scrape every dollar out of farming. While we still run a successful working farm, forty per cent of it is now in native bush for us and future generations to enjoy."¹⁰

"... these landowners are instrumental in healing the landscape that'd been ravaged for its natural resources."¹¹

3.3 Hinewai Reserve

Not long before the Banks Peninsula Track was established, an unusual neighbour moved into the area. Hugh Wilson was a botanist who had undertaken a detailed botanical survey of Banks Peninsula during the 1980s. He had crossed the Peninsula by bicycle and foot and was amazed at what indigenous plant life still remained. "Despite the colossal destruction and fragmentation of the native vegetative cover, despite the profligate introduction of alien organisms, surprisingly few plant species have vanished altogether, even if worrying numbers of them totter on the brink of local extinction."¹²

Hugh developed a theory that, if left to its own devices, nature would reassert itself. "It dawned on me that native plants and animals were proving far more resilient and resourceful than many people gave them credit for. Kānuka and many other native forest species had been quietly regenerating faster than they were being cleared, against all odds."¹³ The removal of farm subsidies and economic decline in the rural sector was assisting with this resurgence. Many farmers could no longer afford the regular spraying and slashing of plants unpalatable to stock, such as gorse and kānuka, that was required to keep marginal land in production.

Hugh was keen to buy some land and put his theories about native regeneration to the test. However, he lacked the requisite financial resources. So he sought assistance from Maurice White who had established the Maurice White Native Forest Trust. In 1987, the Trust bought a 109ha property on the south-east arm of the Peninsula which was rundown and had proved uneconomic as a farming unit. Around 55 per cent of the land was covered in gorse, but it notably also featured a stand of old growth beech forest. The Trust named the new private reserve Hinewai meaning 'water maiden'.

Hugh planned to remove goats and possums from the property and leave the gorse to run rampant. He had initially planned to plant native seedlings but found this was ineffective and unnecessary. The surrounding farmers were very sceptical. "No-one believed us. None of them thought we would be here for more than a year or two. They thought we were city greenies coming to try out a ridiculous notion of leaving gorse to regenerate into native forest. They thought it would not happen or it would take centuries."¹⁴



A spotlight on gorse on Banks Peninsula

Gorse was brought to New Zealand by early European settlers as a hedging plant. By the 1860s, gorse was widely used by farmers as a cheap form of 'live fence' to contain their stock. It was also used to form shelter belts and to mark the boundaries of properties. Gorse grew well on poor soils, and its fast growth and dense form made it ideal for stock fences, particularly where other materials were not readily available. Initially seen as a useful plant, laws were created to protect and promote the use of gorse. In the early 1850s, a government regulation required smallholders leasing land from the Crown to fence their properties with either gorse or hawthorn. In 1872, the Otago Provincial Council formally protected gorse hedges.¹⁵

However, it soon became apparent that gorse also had a propensity to spread as a weed. In particular, it could survive fire, which gave it a competitive advantage over other plants when large scale burning was undertaken to clear land for farming.¹⁶ This led to a change in attitude towards the plant. Leaving gorse to grow outside well-contained hedges became contrary to both the prevailing farming culture and council regulations. "From the 1970s onwards gorse had come to be widely regarded as New Zealand's worst weed, or at least the most expensive to control."¹⁷

More recently, attitudes towards gorse have started changing again, as the positive attributes of the plant have been more widely recognised. Gorse fixes nitrogen into the soil (thereby improving soil fertility), helps control soil erosion on bare slopes, provides a canopy for native seedlings (thereby facilitating native forest regeneration), and provides a pollen source for bees. As gorse does not succeed in the shade, when the native plants get taller, the gorse is shaded out.

Hugh Wilson describes the battle that some farmers had with gorse on Banks Peninsula. "Gorse-infested Ōtānerito Station (earlier known as Long Farm) had thwarted the plans of family after family ever since Charles Narbey more or less abandoned the property in the late 1920s, at the start of the Depression. The land reverted to the 'Narbey estate', and also to gorse, bracken, broom and kānuka. Subsequently a succession of briefly hopeful owners and managers struggled to make a living from the scrub-covered farm. 'They kept burning and spraying and doing everything else they could think of to subdue the gorse,' another neighbour told me, 'but it just kept growing right back up around their ears.'" The land was eventually incorporated into the Hinewai Reserve and the gorse was left alone.¹⁸

"It was thought to be a dereliction of masculine duty not to fight gorse. There was almost a religious duty to fight gorse."

Spraying is often used by farmers to get on top of gorse, but if used indiscriminately, it can make the situation worse. "There is a huge amount of spray being used on gorse by farmers on their land and by the council on the roadside. If you use a helicopter to spray it kills everything and the gorse regenerates quickly on the bare ground."¹⁹

For over a century, gorse has been controlled as a noxious weed under various pieces of legislation, the most recent being the Biosecurity Act 1993. The Canterbury Pest Management Plan 2018-2028, prepared under that Act, still treats gorse as a pest plant. Occupiers of hill country on the Banks Peninsula are required to eliminate gorse infestations on their land within 10m of an adjoining property boundary. This is designed to manage its spread to adjacent occupiers. Occupiers are also required to eliminate gorse infestations that cover up to 50m² on their land in order to "maintain the past investment by occupiers in establishing areas clear of gorse within their properties".²⁰

In 1991, the Nature Heritage Fund together with the Maurice White Native Forest Trust purchased part of the adjacent Ōtānerito Station which substantially enlarged the Hinewai Reserve. With subsequent small additions of land, Hinewai now covers most of three catchments – Ōtānerito, Stony Bay and Sleepy Cove – and three prominent summits overlooking Akaroa. The land runs from a subalpine 806m to near sea level. Although the reserve itself has yet to reach the coast, an easement has been granted over the adjacent farm in Stony Bay to provide coastal access. Hinewai contains around 50ha of old growth forest which is mostly red beech. There are also some surviving ancient tōtara, mataī and kahikatea. More than 60 species of fern are featured amongst a total of 327 species of native plants. Hinewai is now the largest private reserve in New Zealand, extending over 1,250ha of land.²¹



Reverting land on Banks Peninsula

A spotlight on the Nature Heritage Fund

The Nature Heritage Fund is a contestable Ministerial fund controlled by the Minister of Conservation. It was originally established in 1990 to achieve the objectives of the Indigenous Forest Policy. This vision was expanded in 1998 and now enables the protection of all indigenous terrestrial ecosystems that are locally or nationally important, and that represent the full range of natural diversity originally present in the landscape.

The Fund receives an annual allocation of funds from the Government (and the Vote Conservation budget) which it uses to help meet the cost of protecting areas of high ecological value. In the early 2000s, up to \$10 million was allocated annually. However, this subsequently declined by about 80 per cent, with a low of \$2 million being allocated in 2016. This funding enables the protection of important landscapes either through direct acquisition (in full or as a contribution towards purchase by other agencies) or to assist with the costs of covenanting.

In its first 25 years, the National Heritage Fund protected 341,881ha of conservation land across 750 sites. Of this, 88 per cent was through direct purchase.²² Eleven of the total of 750 sites nationally

are on Banks Peninsula and include contributions to Hinewai Reserve, Ellangowan Scenic Reserve, Palm Gully Scenic Reserve and Carews Peak Scenic Reserve among others.²³

There is no replanting at Hinewai. "We have a minimum interference policy. We basically let nature do the hard work and put the bush back. It's doing it incredibly efficiently and better than we would. Every plant we put in would probably be in the wrong place and nature is planting a million. The idea of planting 1,200ha of marginal hill country is absurd."²⁴ During 2009 and 2010, 72 tūi were transferred to Hinewai Reserve from Maud Island in the Marlborough Sounds and it appears the population on Banks Peninsula is gradually increasing.²⁵

There are numerous exotic plants on the property, but the policy has been to largely accept them. Many are benign and others are eventually shaded out by the natives. However, some exotic plants are problematic including the sycamore maple, old man's beard and *Pinus radiata*. There were some pine plantations on Hinewai when the Trust took over the land, but these have since been removed, including a 11ha block which was felled in 1997-98. But wildling pines are still a problem. There is a pine plantation on a neighbouring property. The trees run along Hinewai's boundary and seedlings spread into the reserve. "Farmers were going on about gorse on our

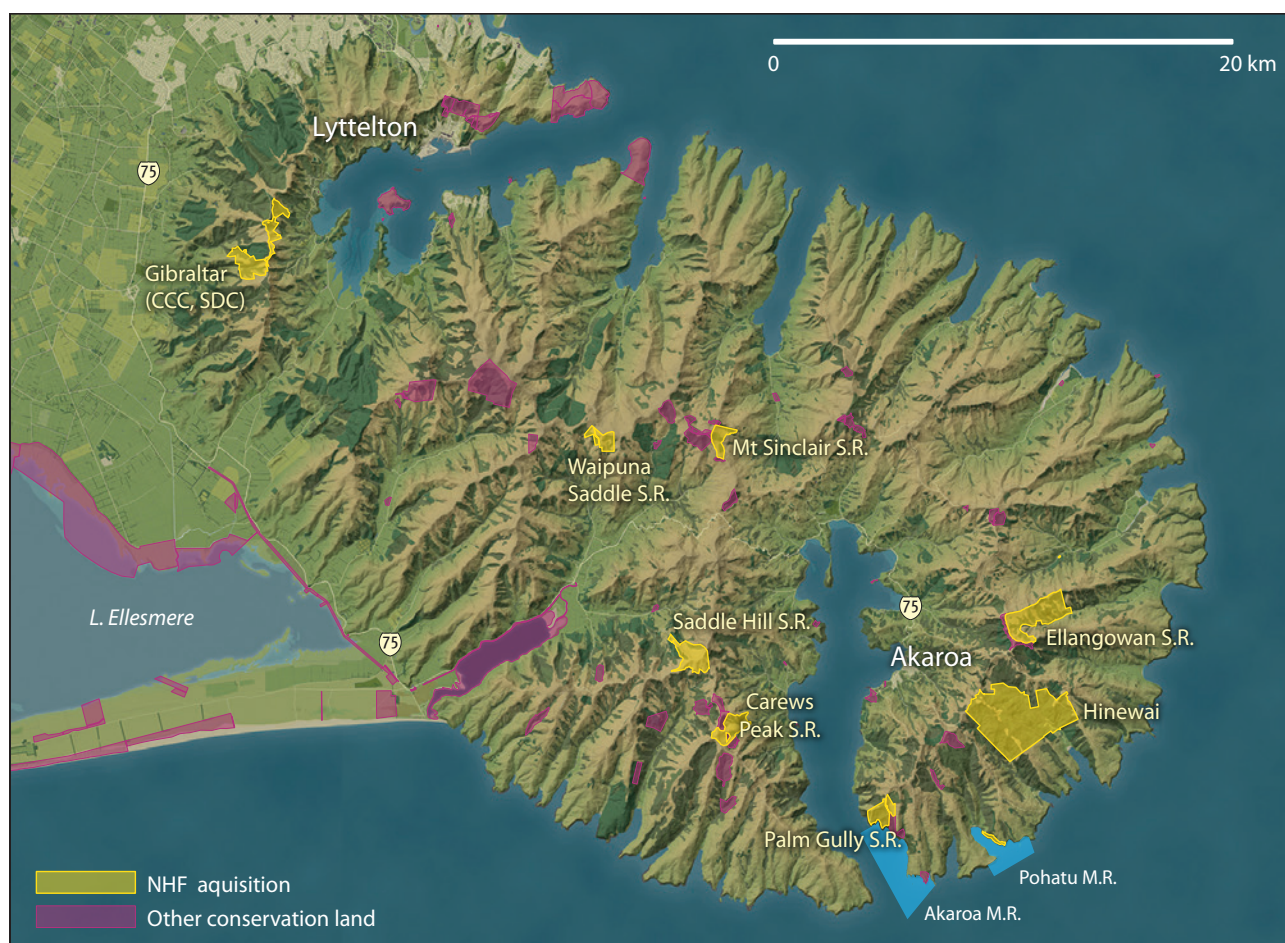


Figure 3.1 Map showing Nature Heritage Fund and other conservation land on Banks Peninsula (Source: Geographx)

boundary, but we have *Pinus radiata* which is our worst weed, up on our fence.”²⁶

Hinewai is now part of the Banks Peninsula Track which provides a small income. The reserve’s extensive track network is also freely open to the public. About half the property is registered under the emissions trading scheme (ETS) and the carbon credits received by the Trust are sold to businesses that want to reduce their carbon footprint. Because the credits are sourced from native forest regeneration, Hinewai is able to sell these credits at above market price. The revenue generated provides an important contribution to the management of the reserve.

The evident success of the Hinewai Reserve has shifted the thinking of other farmers about gorse and native bush regeneration. It has proved to be an inspiring practical demonstration of what can be achieved on the land. It has also shown that private conservation land can generate income through a combination of tourism and carbon credits (although the latter has proved challenging as described below).

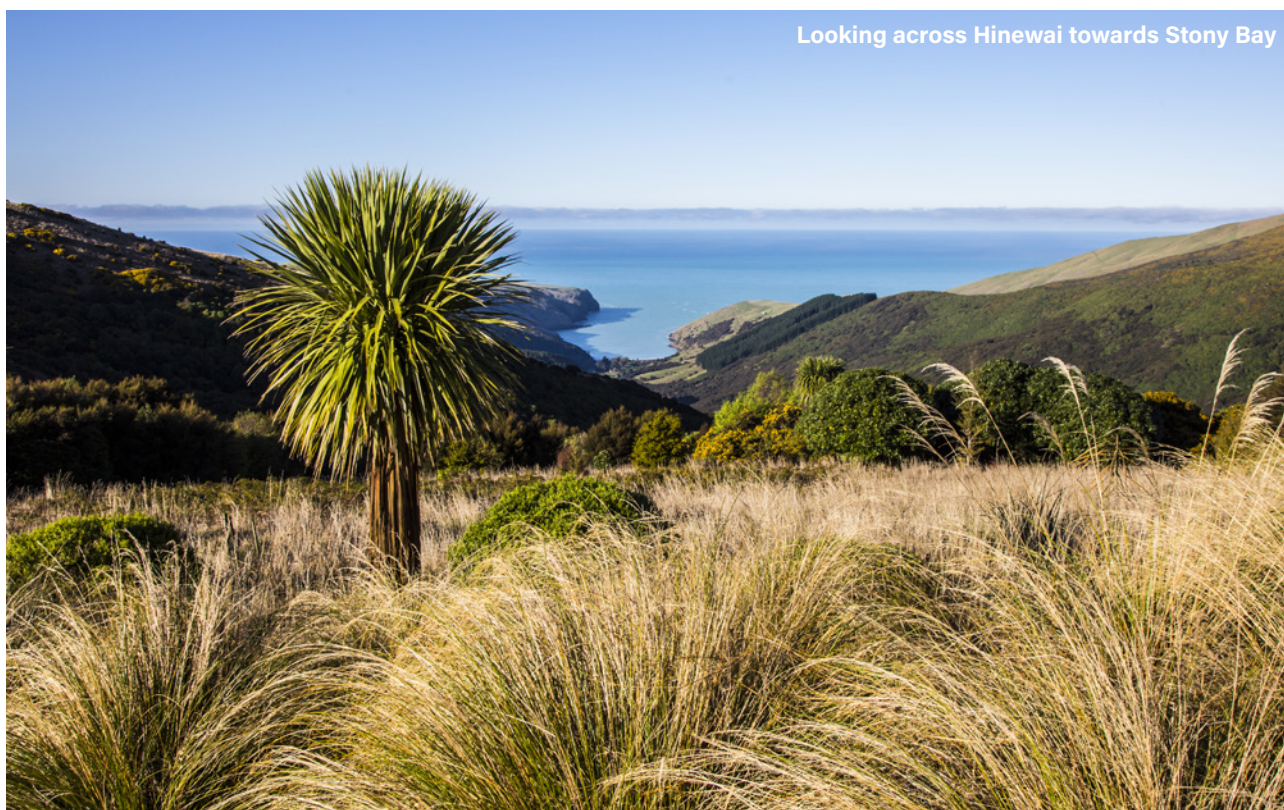
“Hinewai changed the attitudes of farmers. When Hugh arrived, people thought he was a nutter. He wanted to farm gorse. They thought he wouldn’t last more than a couple of years. But he stuck it out. A few years after he started, people could see what was happening. Now no one on Wildside is not on board with Hinewai. They appreciate what he is doing there and he is a very loved part of the community.”

3.4 Banks Peninsula Conservation Trust

“Normally if you asked me whether voluntary measures work over time I would be a bit of a sceptic. But the work of the Banks Peninsula Conservation Trust with 60 covenants and growing all the time is outstanding.”

The Banks Peninsula Conservation Trust (BPCT) has provided a real boost to farmer efforts to restore the land. The Trust was set up by the former Banks Peninsula District Council, in the wake of conflict over provisions in the proposed Banks Peninsula District Plan. The plan had included extensive coastal and landscape protection areas that potentially restricted what farmers could do on their land (see Chapter 6). In the face of a strong backlash from the farming community, the Council largely backed down from imposing statutory controls, instead establishing the BPCT to support landowners in adopting voluntary measures. The Trust started its work in 2001 with a small \$5,000 seeding grant from the Council and a further \$5,000 grant from DOC.²⁷ In 2003, it was granted covenanting powers under the Reserves Act 1977, and it is still the only non-governmental organisation in the country to have such powers. The Trust has flourished over the past two decades. It currently has a budget of around \$450,000 with Christchurch City Council and Environment Canterbury being key contributors.

“A lot of the landowners on Banks Peninsula are multi-generational landowners, so there is real passion for looking after the land that you don’t get everywhere.”



Looking across Hinewai towards Stony Bay

A spotlight on the Queen Elizabeth the Second National Trust

The Queen Elizabeth the Second National Trust (QEII) is an independent charitable trust established by statute to promote the protection, preservation and enhancement of open space. QEII works with individual landowners to protect sites of natural and cultural significance on their property by covenanting them in perpetuity. Covenanted land is regularly monitored to ensure conditions, such as ensuring the area is properly fenced and prohibiting the planting of exotic species, are being complied with. The protections offered by QEII covenants are legally robust and have been successfully defended in the courts against development or material alteration of the protected area.²⁸

To date, QEII has covenanted over 180,258ha of land across New Zealand. On the Banks Peninsula, QEII has worked with 46 landowners to covenant 1,511ha across 87 covenants (with some landowners having more than one QEII

covenant).²⁹ Of these, 30 per cent have been on farms, 30 per cent on lifestyle blocks and 24 per cent with conservation projects or trusts.

Landowners on Banks Peninsula were early adopters of QEII covenants, with the first covenant registered in 1987. Conservation-minded farmers have used QEII covenants to protect remnant forest patches on their farmland. This is reflected in some of the historic QEII covenants on properties such as Flea Bay Pōhatu and Stony Bay.

The BPCT covenanting programme has gone from strength to strength and often works in partnership with QEII. The two trusts are closely aligned, with both now covenanting in perpetuity. Many landowners have both QEII and BPCT covenants, and choose which entity to covenant with, on the basis of what suits them best at that time.

The BPCT is a local entity, which we heard was preferable to some landowners, and is also more adaptable to changes in conditions when requested by landowners. In addition, there is a difference in the funding arrangements



Figure 3.2 Map showing QEII covenants on Banks Peninsula (Source: Canterbury Maps)³⁹

between the two trusts. The BPCT seeks to cover 100 per cent of the surveying and fencing costs of covenanting. However, landowners often contribute when they are able. In comparison, QEII typically covers only half of the fencing costs – by matching either monetary or in-kind landowner contributions. The BPCT's more generous financial support has proved attractive to landowners, and its ability to covenant which is running at around seven to eight covenants a year, is over-subscribed. This has resulted in the process being slower than creating a covenant with QEII, as it takes time to source the necessary funding and support. The BPCT is supported by considerable pro bono assistance from commercial land surveyors, ecological consultants and surveyors. Each covenanted area has a management plan, and the BPCT has recently developed a covenant engagement and support programme to provide ongoing support to landowners, including the goal of visiting each covenanted area around every three years.³⁰

"Funding for QEII covenants on Banks Peninsula is stuck in the dark ages. These people are setting aside land for conservation and should be 100 per cent funded."

As at September 2020, the BPCT had entered into 76 covenants over 1,502ha of land (see Figure 3.3 for a recent map showing the location of covenants).³¹ Many of the covenants are small and dispersed, which is evidenced by the total area protected being similar to that within the

privately-owned Hinewai Reserve. But covenanting can be a first positive step taken by landowners which starts them down a path of engaging in conservation initiatives. To date, the covenants have mainly focused on protecting small remnants of old growth forest, but the Trust is now looking to expand its ambit to protect broader areas of land around such remnants and also to create corridors between them. Some of the larger areas of old growth forest protected by BPCT covenants include a 334ha covenant in the Kaituna Basin, 10ha of remnant podocarp forest in Koukourārata Port Levy and a 42.4ha block in Sally's Bay.³²

When we asked farmers why they had covenanted their land with BPCT we received a range of responses including "because I think it is the right thing to do" and "to protect against the next owner". We were also told that covenanting enabled farmers to obtain financial support for the cost of fencing off blocks that were problematic on the farm anyway, such as bluffs and dense gully areas.

Although covenanting has been the focus of the BPCT for many years, the Trust has more recently played an important facilitation role in a range of landscape-scale conservation initiatives on the Peninsula. This has enabled it to leverage off the positive relationships it has built up with landowners. BPCT has been a major player in the Wildside project (see Section 4.1) and the Pest Free Banks Peninsula initiative (see Section 4.3). It is also leading the Te Kāhahu Kahukura Port Hills Biodiversity Hub initiative (see spotlight).³³

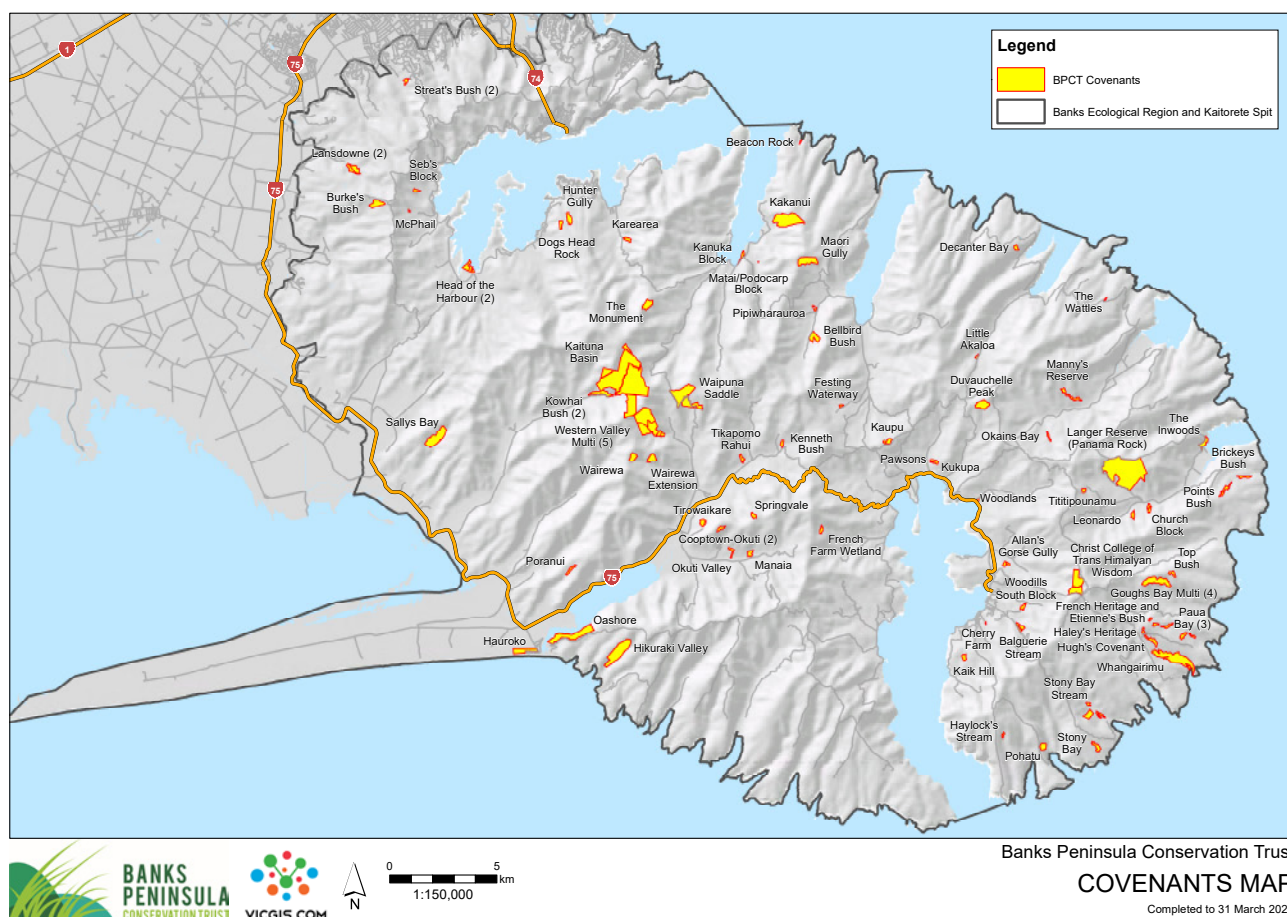


Figure 3.3 Map of Banks Peninsula Conservation Trust covenants (Source: BPCT)

"A lot of established farmers have been here a long time, have done a lot of things and know how the community works. You need them on board for anything to happen."

"A lot of landowners don't like the Council, Environment Canterbury or DOC, for whatever reason, but they will generally get on with the Trust."

In 2017, BPCT launched a '2050 Ecological Vision' for Banks Peninsula which is "to create an environment in which the community value, protect, and care for the biodiversity, landscapes and special character of Banks Peninsula/Te Pātaka o Rākaihautū". The vision includes eight conservation goals for 2050:³⁴

1. All old growth forest remnants (more than 1ha in area) of Banks Peninsula Te Pātaka o Rākaihautū forest cover are protected and appropriately managed.
2. Rare ecosystems are protected and appropriately managed.
3. The connections between land, freshwater and marine habitats are managed to support viable populations of species that depend on them.
4. Four core indigenous forest areas of more than 1000ha each have been protected.
5. Land and freshwater primarily used for production and for settlement also supports thriving indigenous biodiversity.
6. Rare and common indigenous flora and fauna of the Peninsula are increasingly abundant.
7. At least two locally extinct species have been reintroduced.
8. Banks Peninsula Te Pātaka o Rākaihautū is effectively free of pest animals.



Fencing for BPCT covenant

A spotlight on Te Kāhahu Kahukura

Te Kāhahu Kahukura is a large-scale collaboration of landowners, residents, organisations and agencies seeking to co-ordinate native forest regeneration on the southern Port Hills. The project's core area extends south from Victoria Park towards Gebbies Pass and then Lyttelton Harbour and incorporates Ōtamahua Quail Island. Around this core area, landowners are invited to participate by planting natives and undertaking pest and weed control. This is to support the growth of indigenous forest and enable corridors for native biodiversity to move through the landscape.

The southern Port Hills and Ōtamahua Quail Island area was chosen as the core focus for the project due to the presence of significant areas of existing and regenerating native forest and the ability to provide a large forested area near to the city of Christchurch. The Port Hills area was also badly damaged by the 2017 Port Hill fires. Te Kāhahu Kahukura assists in the regeneration of the area and builds on the collaborative action already underway.

The project is coordinated by the BPCT and is part of the Trust's 2050 Ecological Vision. While all eight goals of the Vision are relevant and interrelated, Te Kāhahu Kahukura particularly supports Goal 4 which is to protect four core areas of indigenous forest of more than 1000ha each. The project also contributes more generally to the wellbeing of the community of greater Christchurch.

A spotlight on private conservation on Banks Peninsula

The success of Hinewai Reserve has inspired others to buy up marginal land on the Banks Peninsula to protect it for conservation purposes. As well as providing a boost for conservation, this also provides a market for farmers to sell marginal land, and potentially use the proceeds to further invest in their productive farming areas and/or to fund their retirement. Because many farms on the Peninsula are already in multiple titles (due to its history of close settlement during the late Nineteenth Century) such land disposition does not always require subdivision.

Bob Webster and Carol Jensen are part of this new wave of property owners on the Peninsula. They bought a block of land located between Port Levy and Western Valley in 2012. As Bob explains, "Carol always thought it would be nice to own some land to protect on the Peninsula with some bush and a stream. We found 75ha above Little River. I saw the land while on a run. It was for sale and I had always loved that part of the world. It's on the top ridge of

Banks Peninsula and has 18 different titles. A third was original totara, with some matai, and a third regenerating bush. It was a run off block, but it was not good for grazing as it was too cold and it would have been difficult to muster stock."

"We covenanted the whole property in perpetuity with the Banks Peninsula Conservation Trust. They were quite accommodating in terms of agreeing to what we wanted to do. Their fencing advisor came up and they sought funding to upgrade the fences to keep the stock out. We drafted up our own management plan which is an agreement with the Trust about what we will do."

Management of the land is largely passive, focused on keeping the stock out, getting rid of any weed species and then keeping pest numbers down. "The land has been ungrazed by livestock since we purchased it in 2012. When you take the stock off you remove the disturbance to the grass which then forms a thick thatch and slows down gorse seed germination. The natives find it difficult to seed in the grass, but the gorse shades out the grass and then the natives come away. There are plenty of seed sources and a huge number of seedlings popping up around the margins of existing forest areas. You often get bracken coming in first, and it competes with the grass, and then native seedlings come away under it."

The plot has been monitored each year from 2013 to 2017. This shows that native seedlings have increased generally across the covenanted area. However, native establishment is generally slow when grass competition is strong, although kākū and five finger are doing well in some grassed areas. Native species that are susceptible to stock browsing are establishing on the margins of existing bush where there is less grass competition due to shade. In addition, areas that had been pugged bare by cattle have been invaded by natives. Overall the number of native plants recorded almost tripled over a five-year period.³⁵

Bob and Carol now gain carbon credits for their regenerating native bush under the ETS but found the process challenging. "If you want to apply for carbon credits you have to prove it is new forest land and that it wasn't forest land in 1989. To qualify as forest land, you need a sufficient distribution of forestry tree species, so that when they mature you will have a 30 per cent canopy or more."

"There is a high variability in species composition, distribution and age with natural regeneration, but MPI [Ministry for Primary Industries] requires quantitative evidence of the forest status of all land areas. You can't assume anything, and you have to prove everything. You have to prove it was not forest land before 1990, but the only aerial photos available

from around that time don't show sufficient detail to clearly determine whether dark areas were trees, other shrubby vegetation, gorse, bracken, or even just shadow. Those images certainly don't show the distribution of tree seedlings."

"It is very hard to determine if the land reached 30 per cent cover before or after 1990. Some wouldn't have reached that point, and some might have qualified, but it is hard to tell. It takes a lot of field sampling work to prove anything. MPI require you to cut down the oldest tree at each sampling point to show how old it was. In the end, only 16ha out of our 75ha was approved."

"We do a return every five years and use a look-up table to determine carbon accumulation over the last five years. We then get credits on the Emissions Trading Register and we can decide when to sell."

"There is a premium for carbon that's grown on land that's protected for conservation. There is a voluntary market and we have found that prospective purchasers can be as interested in the conservation management of the land as they are in carbon accumulation. They want to pay money to help people protect land for conservation purposes and that means they pay us more per unit than for pine forest credits."



Okains Bay farm nursery

3.5 Farm restructuring during the 2000s

The success of the BPCT and other efforts to promote native bush regeneration on Banks Peninsula have been supported by the drop in coarse wool prices. Prior to Covid-19, coarse wool was selling for as little as \$2.50/kg, and due to the impact of the pandemic, this has now dropped to around \$1/kg.³⁶ The cost of shearing sheep now exceeds proceeds from the sale of the wool, but shearing is usually required for animal welfare reasons, even if the wool cannot be sold.

A spotlight on the price of wool

Wool prices have been volatile over the past century, and being prime sheep country, this has affected farming practices and associated land management on the Banks Peninsula. After a period of low export prices, wool prices spiked in 1951 due to the US military requiring large quantities of wool for its winter campaign in the Korean War. There was

another, more sustained, increase in prices between 1971 and 1989. The wool price peaked in 1989 and then declined due to the global slowdown. The dissolution of the former Soviet Union in 1991 also affected prices as the federation had been a major consumer of wool. During the 1990s, there was growing competition from synthetics. In particular, the growing popularity of synthetic carpets reduced the demand for New Zealand's coarse wools, which were mainly used for carpet making. The volume of wool exported by New Zealand halved between 1989 and 2011.³⁷

Wool prices are now largely affected by the Chinese market, and increased demand helped wool reach a high of over \$6/kg in 2011 and again in 2016. The price has been in decline since that time. During the same period, there have been sustained high export prices for red meat due to African swine fever severely affecting China's pork production and drought and bushfires affecting Australian red meat production.³⁸



Woolshed, Okains Bay

A spotlight on Banks Peninsula Wool Growers

In 2009, a group of farmers on the Banks Peninsula decided to pool their resources in an attempt to improve the value of wool and strengthen the farming community. The idea arose within a farm discussion group comprising some 19 farmers who were disillusioned with the prices they were getting for their wool. They decided to market their wool differently, so that instead of farmers competing against each other, they collectively marketed their wool as Banks Peninsula Wool Growers. By doing this, they managed to remove the wool broker from the supply chain, and the wool now goes directly to the spinner and then on to the manufacturer.

The farmers wanted to create a distinct voice and capture the story of the wool grown on the Peninsula. This involved telling the story of the people, the product and the place. There is a strong focus on traceability, with end-users able to identify the farm on which the wool was grown, and sustainability. The farms are low pesticide, low fertiliser, free range farming operations surrounded by the spectacular Banks Peninsula landscapes. Special care is taken in the preparation of the wool clip and farmers are required to adopt a strict on-farm quality control programme.

The company, now with 44 shareholders, markets its wool under the Banks Peninsula Farms brand and sees the possibility extending the brand over time to other high quality products produced on the Peninsula.

With the collapse of the coarse wool market, farmers on Banks Peninsula have placed greater emphasis on sheep and beef meat production. This has increased the focus on improving lambing percentages and growth rates through selective breeding and providing better nutrition for stock. This, in turn, has resulted in lower stocking rates and the development of more nutritious pastures, fencing of land into smaller paddocks and a higher degree of animal management and husbandry. When the emphasis was on wool production, stocking rates were much higher, as the rate of animal growth was not so important.

Farming is now generally focused on the more manageable and accessible areas of land, with less grazing (and more indigenous bush reversion) in the steeper and more difficult areas. Higher performing stock can no longer be used as weed control agents, as heavy grazing of less palatable weed species gives a lower nutritional return. Lower stocking rates have also taken pressure off the land more generally.

Key messages

- Farm restructuring during the 1980s resulted in much marginal land reverting to indigenous bush as it was uneconomic to keep it in production.
- Papatipu Rūnanga have long played a strong kaitiaki role in restoring the Peninsula, both on land and sea.
- The Banks Peninsula Track was established in 1989 by a group of farmers who needed to generate additional income to stay on the land. It has been highly successful and has demonstrated that there is money to be made from conservation.
- Hinewai Reserve was established in 1987 and tested the theory that gorse-infested land, left alone, will naturally (and relatively quickly) regenerate into indigenous forest. The experiment was highly successful, the Reserve has been significantly expanded, and it has changed the thinking of farmers on the Peninsula about the management of gorse.
- A new wave of private landowners on the Peninsula are buying rural land for the purpose of biodiversity conservation.
- QEII and the Nature Heritage Fund have made an important contribution to protecting land on the Peninsula for conservation purposes.
- The BPCT has supported landowners to covenant old growth forest and other areas of ecological significance on their land. The Trust has more recently acted as a facilitator of landscape-scale biodiversity regeneration initiatives building on its positive relationships with rural landowners.
- The recent collapse in the price of wool has prompted farmers to prioritise meat production and this has resulted in a need to provide better nutrition for fewer stock. This has meant that steeper and more difficult areas are of less value to the farming operation and have been left to revert to native forest in many cases.

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4. RESTORATION AT SCALE

During the 2000s, restoration on Banks Peninsula evolved from individual property efforts into landscape-scale initiatives where mana whenua, landowners, not-for-profit entities and government agencies pooled their efforts. This has enabled results to be scaled up significantly.

4.1 On the 'Wildside'

"On the Wildside, there is often only one or two farms in each bay, and the landowners were looking after the native biodiversity themselves before conservation was a thing."

The Wildside project was initiated by Robin Burleigh, who moved to Le Bons Bay in 1985, and bought a 7ha property there in 1988. Robin started working for DOC in 1989 and during his time with the Department he managed a comprehensive survey of penguin populations on Banks Peninsula. The survey revealed that the best examples of remaining native biodiversity on the Peninsula were on the south-east coastal flanks, running north from the eastern head of Akaroa Harbour to Le Bons Bay.¹ Here could be found the Akaroa daisy, Cooks scurvy grass and the Banks Peninsula tree weta – which is New Zealand's rarest indigenous tree weta and can only be found in the vicinity of the Wildside area (and has been unofficially nick-named the 'Wildside weta'). The south-east area contained the only naturally established beech trees on the Peninsula and nikau palm. Little blue/white flippered penguins (endemic to Banks Peninsula) and Hoiho (yellow-eyed penguins) frequented the area and there were numerous seabirds nesting on the coastal cliffs.

By the time Robin Burleigh started his survey, several landowners were already involved in protecting the penguins and seabirds that frequented their shores. This was in response to a reduction of Government funding for rabbit control during the 1980s and the disbanding of rabbit boards in 1989 which led to an explosion of rabbit numbers on the Peninsula, accompanied by an increase in mustelids which fed on them. The enlarged mustelid population also turned on the coastal birds with devastating consequences.

Mark and Sonia Armstrong are the current custodians of the farm in Stony Bay which has been in the Armstrong family for five generations. Mark began trapping mustelids in the 1980s when they started ravaging Stony Bay's resident colony of little blue/white flippered penguins. He managed to save around 40 pairs. Other landowners pitched in and set trap lines up the valleys, and DOC subsequently followed, creating an extensive 'wall' of traps.

"Some tourists wanted to see the penguins. So Mark lifted the lid on a nesting box and saw a ferret attacking the chicks. They killed the ferret and found seven penguins inside."

The mustelids were also attacking the colony of tītī (sooty shearwaters) that nested on the steep cliffs in Stony Bay. Mark decided to take action when the colony was reduced to just three pairs. As it turned out, it was the last remaining tītī nesting site on the Canterbury mainland. Together with his farm worker, one of the first predator proof fences was erected around three sides of the colony, with the 120m cliff to the sea forming the lower side. He also put in a line of predator traps.²

Shireen and Francis Helps, who farm the adjacent Flea Bay Pōhatu, also started trapping predators to protect the penguins. "We didn't have much money but whenever we had some spare cash we'd buy a couple of traps to help keep the stoats and ferrets away. We also created nesting sites, monitored the penguins and rehabilitated any that were struggling to survive."³ The Helps started offering penguin tours in 1988 with the proceeds used to support their conservation work.⁴

In 2003, while working for DOC, Burleigh contacted several of the landowners, the councils, rūnanga and science interests with a proposal to create a 'Wildside' project on the Peninsula's south east. The idea was that efforts would be combined to provide more intensive predator control and management practices that encouraged forest regeneration. The idea took hold. The initial focus of the Wildside group was on better protecting the seabirds that the landowners had helped save. Outside assistance was brought in and money was raised to contract a trapper and also to construct a predator proof fence around the tītī colony.⁵ These efforts have been highly successful with the penguin colony increasing at Flea Bay Pōhatu from 717 pairs in 2000/01 (when the first survey was undertaken) to 1,304 pairs in 2012. By 2018, the tītī colony had increased from three pairs to 50 pairs attempting to nest and 33 chicks being produced.⁶

"The focus of Wildside has moved on from the initial protection of pelagic seabirds to become a truly collaborative whole landscape restoration project within a living and working environment."⁷

Since that time, the Wildside project has gone from strength to strength, addressing terrestrial and marine issues. It currently protects over 2,500ha of regenerating forest (see Figure 4.1). It includes the Hinewai Reserve, land owned by public agencies (including the 489ha Misty Peaks Reserve bought by the Christchurch City Council in

2007) and private land covenanted under the BPCT and QEII. It is working towards restoration of landscape and biodiversity at scale (Goal 3 in BPCT's 2050 Ecological Vision) by connecting up areas that are already managed for conservation purposes. Over a quarter of the land is under some form of legal protection, a large proportion in a working farm landscape. The area also contains the only two marine reserves in the seas around Banks Peninsula and is the location of the Banks Peninsula Track.

A spotlight on marine reserves around Banks Peninsula

There are currently two small marine reserves along the coast of the Banks Peninsula, the Pōhātu and Akaroa marine reserves. The Pōhātu Marine Reserve, which is just 2.2km² in size, was established in 1999 after a proposal by recreational fishers. The Akaroa Marine Reserve was established in 2014 and is more than twice the size at 5.1km². Both reserves abut the Akaroa Taiapūre and joint monitoring is undertaken between the three areas.

The Akaroa Marine Reserve was only established after years of controversy and legal battles. The reserve was formally proposed by the Akaroa Harbour Marine Protection Society in January 1996. The proposal was opposed by recreational fishers and the local rūnanga was concerned about the loss of customary fishing and mana. The processing of the application was effectively halted by the proposal by recreational fishers for an alternative (and much smaller) marine reserve in Flea Bay Pōhātu. In addition Te Rūnanga o Koukourārata, Ōnuku Rūnanga and Wairewa Rūnanga proposed a taiāpure in Akaroa harbour. These were both subsequently created, but excluded the initial area proposed for the Akaroa marine reserve thereby leaving open the possibility that it could proceed.⁸

Once the Akaroa Taiāpure was created in 2006, DOC continued the processing of the initial marine reserve application and undertook a second round of public consultation. This generated 77 submissions with 25 opposed. The Conservation Minister declined the application in August 2010 on the basis of effects on recreational fishing. The Akaroa Harbour Marine Protection Society judicially reviewed the Minister's decision in the High Court, and was successful on 8 May 2012, when the High Court quashed the Minister's decision. On reconsideration by a new Minister of Conservation, a slightly smaller reserve was approved in 2013, and it was created in 2014.⁹

Overseeing activities at Wildside is a committee which brings together local landowners, rūnanga representatives, people from the councils and DOC, and other scientific specialists. It is chaired by a local landowner. The BPCT coordinates the committee and the DOC Community Fund

supports a three-day a week facilitator, who coordinates and facilitates conservation, protection and enhancement in the area.

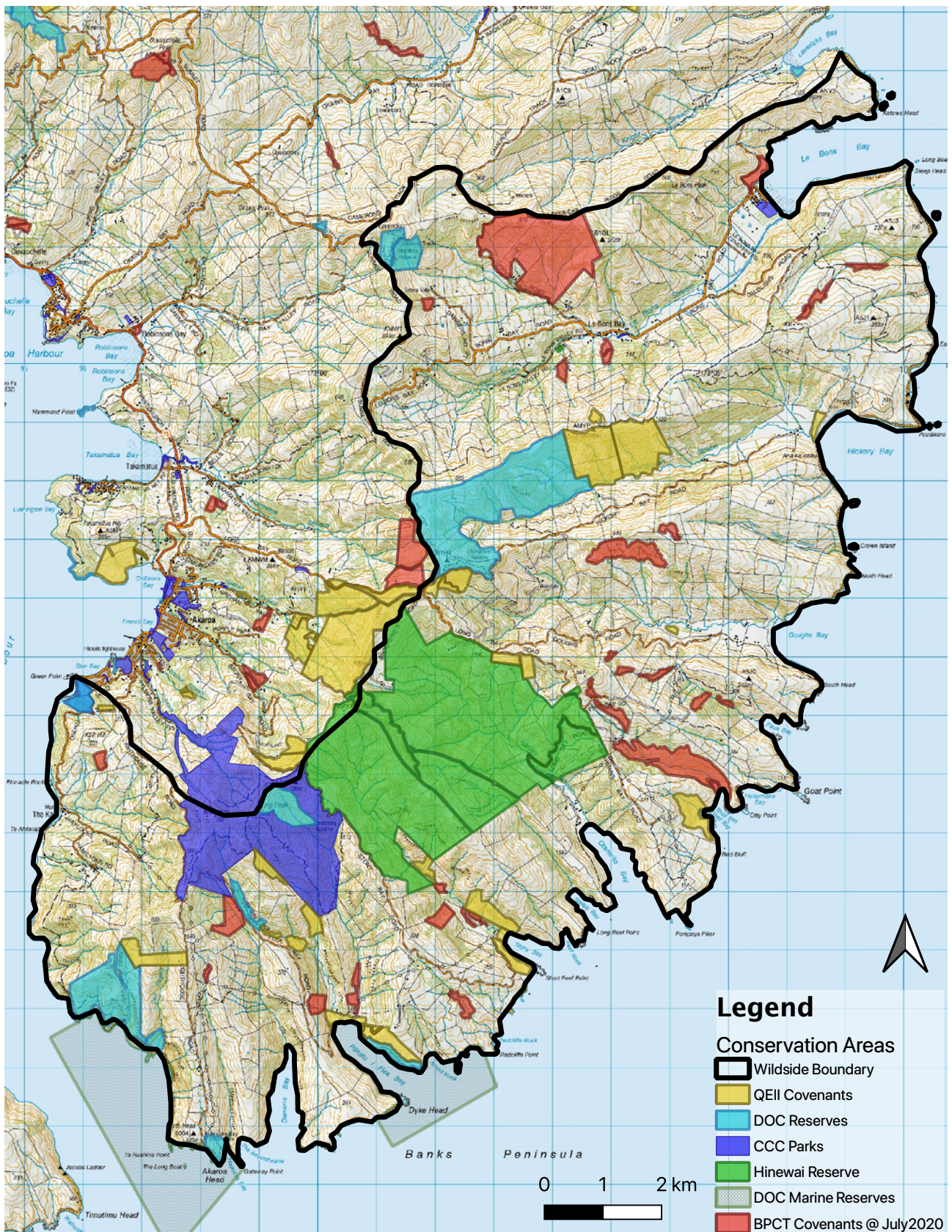
"Everyone on the Wildside committee is an enabler. We discuss and argue about what is most important and how to do things. The whole point of the committee is to do that. But we get outcomes because everyone is genuinely supportive of trying to get to a positive outcome to protect and enhance the native biodiversity."

Additional funding for Wildside has been provided by the Josef Langer Charitable Trust. The Trust was formed in 2005 as a result of a bequest from the estate of Josef Langer, a European immigrant who had worked as an electronic engineer for Air New Zealand and who had a passion for nature. After meeting the other requirements of the estate, the remaining funds were made available for projects on the 'Wildside'. The Trust has funded a wide range of initiatives including bush restoration, covenanting, predator and possum control (including through the engagement of a contract trapper), nature publications, tuī reintroduction, weed control, the predator fence at Stony Bay, the Wildside Coordinator employed by the BPCT, biodiversity workshops and school education programmes. Most of the money in the original bequest has now been spent and the Trust is not currently in a position to fund further projects.¹⁰

A spotlight on the Joseph Langer Reserve

In 2009/2010 the Josef Langer Trust purchased a 48ha block of land in Le Bons Bay which includes the striking volcanic dyke and dome structure known as Panama Rock (Kellers Peak). It also includes a large area of regenerating bush, with the area having been totally cleared during the 1800s apart from a few remaining tōtara trees under a rocky bluff. There is a small lodge on the land which is now available for use by members of the community. In 2012, an adjoining 60ha was added and, in 2014, another neighbouring block of 100ha was purchased. There are walking tracks and public access to all of the 200ha reserve. The area also includes the little gentian (*Gentionella serotina*) which is the only known location of the plant on Banks Peninsula. Burleigh, who left DOC in 2006, now manages the area on behalf of the Trust.¹¹

Mammalian predators are now being suppressed over 7,000ha of Wildside land through the trapping of stoats, ferrets and feral cats. There is also possum control across 5,000ha around the core habitat of Misty Peak Reserve and Hinewai Reserve. Trapping has helped bird life, and in particular nesting seabirds, but there are continual re-invasions. Without a landscape approach to predator control, it is likely to be a long, ongoing fight.



The Wildside - Protected Conservation Land

Protected Conservation Land on the South-eastern tip of Banks Peninsula - The Wildside.
As at November 2020.

Figure 4.1 Map of the Wildside (Source: BPCT)

4.2 Rod Donald Banks Peninsula Trust

The Rod Donald Banks Peninsula Trust was established in 2010, in the wake of the disestablishment of the Banks Peninsula District Council, in 2006. The Council owned several endowment farms in Ashburton which it had inherited from previous councils on the Peninsula. Not wanting these assets to be swallowed up within Christchurch City upon the merger, the Council decided to place the proceeds from the sale of the endowment properties into a trust where they would be used to create an enduring legacy for the Peninsula. The Trust was named after Rod Donald, who had served as a Green Party Co-Leader and Member of Parliament from 1996 until his untimely death in 2005 at 48 years of age. Rod had lived in Christchurch and had a close association with the Peninsula.

The Trust was set up with \$3.5 million in funds. It was given a wide mandate to promote sustainable management and nature conservation; recreation, culture and heritage; research projects; and public education in the area formerly administered by the Banks Peninsula District Council. The Trust is governed by nine trustees, seven who are appointed by Christchurch City Council and two who are co-opted. The Trust operates as a council-controlled organisation and each year it agrees a statement of intent with the Council.

The Trust works on a range of projects to foster public access, support biodiversity restoration and improve knowledge of the Banks Peninsula area. Early on the trustees decided to spend down the Trust's capital, rather than live off the interest, in order to acquire strategic areas of land to facilitate access and recreation. This was either through outright purchase or through securing easements. Its flagship project is Te Ara Pātaka – the Summit Walkway which is a 2.5-day tramping track linking the Lyttelton and Akaroa craters. The track passes over a mix of public and private land, including unformed legal roads. The Trust is now seeking to extend the walking route so it links Akaroa to Christchurch.¹²

The Trust also has a broader programme focused on encouraging people to explore the Peninsula on foot. It

convenes an annual Banks Peninsula Walking Festival, which provides a series of guided walks in different parts of the Peninsula over four weekends in November. The guides are local residents who have in-depth knowledge of the area and are passionate about their local places. Around 500 tickets are sold each year. The Trust has mapped all of the Peninsula's walks, regardless of land tenure, and has put this information together into a comprehensive directory. It works with statutory agencies to improve signage and interpretation. It also produces brochures for the walks, and operates a website, both of which provide interpretative information. This provision and promotion of walking trails across the Peninsula helps Christchurch residents, and visitors from further afield, to better engage with the landscape through a form of 'slow' tourism.

The Trust has also taken a broader role in promoting conservation outcomes on the Peninsula and has adopted the same ecological goals as the BPCT. In particular, it has led work investigating carbon offsetting and exploring how it can be used to support biodiversity regeneration (see Section 5.2). In addition, the Trust has supported other organisations including funding the BPCT in its early stages so it could pay administration staff and supporting the Pest Free Banks Peninsula initiative described below. It has also provided funding to the Quail Island Ōtamahua Restoration Trust, Orton Bradley Farm Park, Maurice White Trust, Living Springs and the Geopark proposal. In addition, it has contributed funds to the purchase of the Saddle Hill Reserve, Curry Reserve and Kaituna Reserve addition; and has signed a sale and purchase agreement to establish a 500ha Conservation Park (Te Ahu Pātiki) which encompasses Mount Herbert and Mount Bradley.

4.3 Pest Free Banks Peninsula

"Predator Free 2050 really struck a chord here in the community. It galvanised community support around a cause. It's still growing. More and more people are putting in trap lines."

Banks Peninsula has a wide variety of pest animals including possums, rats, mustelids, feral deer, goats, pigs, rabbits, hares, feral cats and hedgehogs. As already indicated, pests really took a hold on the Peninsula when the government abolished the rabbit boards in the 1980s, leading to an explosion in rabbit and mustelid numbers. Canada geese are another emerging pest.

Since 2003, there has been a collaborative effort to control feral goats, through the establishment of the Feral Goat Eradication Working Group with representatives from the BPCT, Environment Canterbury, Christchurch City Council and DOC. In 2018, feral goats were listed as a pest in the regional pest management plan. A large-scale mustering and hunting operation was also launched. The south-eastern side of the Peninsula is now feral goat free. Six sites with feral goat populations remain on the north-western side. The goal of the Group is to eradicate goats from the Peninsula over the next five years.¹³



The BPCT has been a leader in the Pest Free Banks Peninsula initiative. In October 2018, it brokered the signing of a memorandum of understanding between 14 agencies and organisations on the Peninsula who formed the 'Pest Free Banks Peninsula/Te Pātaka o Rākaihautū and Port Hills Partnership'. The goal of the Partnership is for the Peninsula to be "effectively free of pest animals" meaning they have been eradicated or are kept at such low levels that they do not threaten biodiversity values.¹⁴

A spotlight on membership of the Pest Free Partnership

This Pest Free Banks Peninsula/Te Pātaka o Rākaihautū and Port Hills Partnership includes the BPCT, Cacophony Project, Environment Canterbury, Christchurch City Council, DOC, Living Springs (a not-for-profit camp and conference centre in Governors Bay), Ōnuku Rūnanga, Rod Donald Banks Peninsula Trust, Selwyn District Council, Summit Road Society Incorporated, Te Hapū o Ngāti Wheke (Rāpaki) Rūnanga, Te Rūnanga o Koukourārata, Te Taumutu Rūnanga, Onuku Rūnanga and Wairewa Rūnanga. They have collectively established a Project Oversight Group which will monitor delivery of the project, ensure continued agency and funder support and provide a risk and audit function, with a Project Management Group to be responsible for delivery.¹⁵

A spotlight on the Pest Free Banks Peninsula targeted rate

In 2019, Environment Canterbury introduced a new Pest Free Banks Peninsula targeted rate which is providing a long-term sustainable funding source for pest control on the Peninsula. This replaced a previous possum control rate which applied to the rural part of the old Banks Peninsula District. The new rate applies to all rateable properties on the Port Hills, Banks Peninsula and Kaitorete Spit. It is being combined with revenue from general rates to provide \$600,000 a year for pest control activities in the area.¹⁶

The Partnership made a funding application to Predator Free 2050 but was unsuccessful. However, government funding associated with the Covid-19 economic recovery effort created an opportunity which they seized. In August 2020, the Minister of Conservation announced that \$5.11 million would be allocated to the Pest Free Banks Peninsula project. This will fund large-scale pest control over around 28,500ha on the expanded Wildside area which includes land to the south-east of a line from Duvachelle on Akaroa Harbour across to Okains Bay. Stage 1 will focus on the eradication of possums, suppression of mustelids and ramped up research into rat control. Rat control is particularly challenging on Banks Peninsula, with limitations on the use of aerial 1080 due to people living within the landscape. It is also very steep,

extreme country, so that bait stations are often not feasible. The project will also fund pest control on the spit and dune systems of Kaitorete.¹⁷ Although other species still pose a threat, for example feral pigs and deer, there is no funding allocated to control these species in order not to 'water down' the work that can be done on the focus species.

The funding, which is sourced from the Government's 'Jobs for Nature' package, is expected to create around 15 full-time jobs. Jobs for Nature was established in response to Covid-19 and includes \$76 million for investment in community predator free projects through Predator Free 2050 Limited.¹⁸

A spotlight on the Cacophony project

Cacophony is a not-for-profit initiative led by mechanical engineer and Akaroa resident Grant Ryan. Grant is deploying his engineering skills onto the problem of developing better methods to eradicate pests. He is approaching the problem from an engineering perspective rather than from the usual biodiversity science viewpoint.

"Why are we pouring money into automating traps or communicating trap lines when we know that most predators just walk past them? Why put more money into archaic trapping methods? We need something better."

Grant believes it is not possible to control predators at scale with the currently available tools. The use of thermal cameras shows that about 99 per cent of predators run around traps and are not being caught. This means that the use of traps might actually be making things worse, as by only catching the small proportion of predators that go into traps, trap-adverse animals are being selectively bred into the predator populations.

The Cacophony Project is investigating how thermal cameras and artificial intelligence can be used to make a step change in predator control effectiveness. This is through using such techniques as luring invasive predators with sound and light, observing predators using a thermal camera, identifying predators automatically using machine learning algorithms and eliminating those that are positively identified. The project is also looking at measuring impact through monitoring bird song over time.¹⁹ Particularly promising is the potential to use a thermal camera to positively identify a predator and then a paint ball gun to dispense poison onto its fur, although funding has yet to be secured to develop this idea further. Although these devices would be expensive to build, not many of them would be required if they were used as a 'rolling front'.

"We think this approach has the potential to dramatically increase trapping efficiency, long term."

4.4 Te Pātaka o Rākaihautū Banks Peninsula Geopark

"The rocks, soils, landscapes and waterways are all inherently derived from the geology. All biodiversity is controlled by geodiversity. If you preserve geodiversity you protect biodiversity at the same time."

The term 'Geopark' is used internationally to refer to sites and landscapes of international geological significance which are managed in order to promote protection, education and sustainable development. There is a UNESCO network of 161 'Global Geoparks' spanning 44 countries.²⁰ The idea of applying the Geopark concept to the Banks Peninsula arose shortly after the Christchurch earthquakes in 2010-11 which had generated renewed interest in the geology of the area. Researchers from the University of Canterbury's Department of Geological Science started investigating the Geopark idea and held two hui in 2018 to discuss the concept with the local community.

The Peninsula "has some of the best exposed volcanic features in the world" with erosion uncovering "the hearts of these ancient volcanoes."²¹

The initiative is still in its early stages and is now being progressed through a collaboration between Papatipu Rūnanga of Te Pātaka o Rākaihautū, Te Rūnanga o Ngāi Tahu, Frontiers Abroad Aotearoa (which supports American graduate students to experience New Zealand's geological environment) and the Department of Geological Sciences at the University of Canterbury.²² The Rod Donald Banks Peninsula Trust is also investigating effective ways in which it can assist with progressing the Geopark.

The Geopark proposal is based on the idea that the underlying geology of the Peninsula is the base layer on which all other significant features and elements sit. It aims to inform locals and visitors about the region's geology, landscape, flora, fauna, archaeology, histories and communities. This would include the establishment of 'geopoints' at specific sites, 'geosites' which are collections of geographically related sites, and 'geotrails' which are pathways connecting the geopoints and geosites by walking, kayaking, boating or cycling. Interpretative panels would provide information about the locality, describing the physical location and what had occurred at the site.²³

"We want to make connections between the specific rock type, to the specific fern; or a promontory to a specific oral tradition or resource found there."

The Geopark idea seeks to encourage locals to engage with the Peninsula more deeply through trails, sign boards, open air learning, field trips, research, experiences and digital interfaces as well as attracting national and international visitors. It could also provide the basis for branding local products and producers and connecting them with markets.²⁴ A first step in bringing the Geopark to fruition is likely to be establishing interpretative panels on publicly-owned or publicly accessible conservation land if funding for this can be secured. The focus on geology could serve to compliment the other initiatives which are more targeted at the biodiversity that naturally drapes the geodiversity – both of which contribute to the natural landscape attributes.



Looking down the volcanic slopes into Akaroa Harbour

4.5 Collaboration

A key theme that came out of our interviews for the project was the power of collaboration. There were many instances of organisations working together on Banks Peninsula and collectively achieving more as a result. The quotes below reflect some of the sentiments expressed to us on this point.

"I thought, 'so this is what collaboration is, genuine collaboration'."

"The thing that excites me most is the willingness of the community to work on conservation initiatives and the ability of the big agencies and iwi to also work together to underpin that community aspiration."

"I think what accelerated working together is relationships – investing in relationships, getting some things done together and building trust and experience together. It takes a lot of effort, and you have to invest in it, but the pay back is enormous."

"When you work together the bang for the buck you get is enormous ... the collective pulling power is phenomenal."

Key messages

- During the 2000s, several landscape-scale biodiversity regeneration initiatives were established on Banks Peninsula.
- The Wildside project, established in 2003, combined efforts on the south-east coast of the Peninsula to control predators, particularly to protect seabirds. It has since expanded into landscape restoration and community education, and there is now a Wildside Coordinator that works with landowners and the community and operates under the auspices of the BPCT. It has been supported by significant funds from the Joseph Langer Trust, DOC Community Fund and most recently Pest Free Banks Peninsula.
- The Rod Donald Banks Peninsula Trust, established in 2010, provided significant funds for the purchase of land for the provision of public access on the Peninsula. The Trust has also promoted walkways on the Peninsula and has contributed to other biodiversity initiatives.
- A partnership to facilitate the Pest Free Banks Peninsula initiative was established in 2018 and \$5.11 million was announced in 2020 for the project from the Government's 'Jobs for Nature' package.
- Another landscape scale initiative is the proposed Te Pātaka o Rākaihautū Banks Peninsula Geopark which seeks to use the underlying geology of the Peninsula as the framework for communicating information and stories about the place.
- Effective collaboration has been a key element of the success of landscape and biodiversity initiatives on Banks Peninsula. Several well-resourced trusts (such as the Rod Donald Banks Peninsula Trust and Josef Langer Trust) have also provided important facilitation and funding assistance in support

ENDNOTES

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- 2 Cronshaw, 2009; Banks Peninsula Conservation Trust, 2018, 14-15
- 3 Country Calendar, Episode 3, Penguin paradise at <http://tvnz.co.nz/content/1672163/2527715.xhtml>
- 4 <https://www.avenues.net.nz/all-stories/2019/7/26/saving-banks-peninsula-penguins>
- 5 Burleigh, 2019
- 6 Banks Peninsula Conservation Trust, 2018, 14-15
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- 8 <https://www.doc.govt.nz/news/media-releases/2013/decision-made-on-akaroa-marine-reserve/>
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- 10 Burleigh, 2019
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- 12 Rod Donald Banks Peninsula Trust, 2020
- 13 Banks Peninsula Conservation Trust, 2018, 25; Banks Peninsula Conservation Trust, 2019, 25
- 14 Banks Peninsula Conservation Trust, 2019, 24
- 15 Pest Free Banks Peninsula & Port Hills Partnership Memorandum of Understanding
- 16 Banks Peninsula Water Zone Committee, 2019
- 17 Minister of Conservation, 2020
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- 22 <https://www.bpgeopark.com/our-journey>
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5. CURRENT AND FUTURE PRESSURES

5.1 Farming

Farming systems on Banks Peninsula are closely linked to the natural attributes of the land, soil and climate. The volcanic soils on the upper slopes have high natural fertility as they are derived from basaltic volcanoes which are rich in phosphate. The lower levels of the Peninsula (mainly below 150m) are dominated by loess soils which are finer and have less natural fertility. These soils are highly erodible, particularly when the woody vegetation cover is removed and the soil disturbed.¹ The soil also becomes unstable when wet which can result in tunnel gullying and soil slides.²

"The Peninsula has naturally highly fertile soil so a lot of farmers use few inputs. We top-dress annually but a lot of farmers don't do that. Most properties have little cultivatable land."

Banks Peninsula has a relatively wet climate with higher annual rainfall than the Canterbury Plains. Jutting out from the east coast of the South Island, the Peninsula is hit by moist weather systems coming from the south and east. However, most rain falls in winter and the soils can dry out over the summer months. There is also a considerable

rainfall gradient across the Peninsula with more rain falling on the ridge tops and higher slopes and less on the more sheltered slopes, particularly in the north-east.

"The summer is dry. But because of the topography, where the land goes from sea level to high latitudes, we are not really drought exposed like Hawkes Bay and the Canterbury Plains."

Water supplies on the Peninsula are derived from rainfall which fills small aquifers in the shattered volcanic rock. These, in turn, supply numerous springs feeding a network of small streams and rivers. Over 8,000 springs have been identified on the Peninsula. Their recharge zones (ie the land area from which water drains into them) are not aligned with catchments, which makes their management complex.³ During dry periods, when there is little rainfall, the springs can run dry. Unlike the Canterbury Plains, which sit over extensive gravel aquifers, it is not possible to irrigate farmland on the Peninsula as there are no large and reliable sources from which water can be drawn. With a five to 15 per cent decrease in summer rainfall projected for Banks Peninsula by 2090, summer droughts and restricted water supplies are likely to become an increasing problem.⁴



Farmland on the flanks of Akaroa Harbour

"We are bound by natural rain and climate and can't afford to push the property too hard as we hit the wall very quickly. It is low input natural farming."

The other main restriction on farming on the Banks Peninsula is topography. Much of the land is very steep and is not readily accessible by tractor or other large farm equipment. This means that it cannot be cultivated and used for intensive pasture production or to grow crops. On the small flatter areas at the head of valleys, or on some headlands, farmers will often grow a pasture crop such as rape, lucerne or clover for supplementary feeding.

Ploughing to plant new crops can be problematic on the loess soils due their susceptibility to erosion, and so some farmers have moved to direct drilling, where the seeds are directly inserted into the ground. However, this requires spraying to kill the previous crop. Spraying is also still used to remove regenerating kānuka, plus gorse and other weeds.

"We used to plough but it didn't go very well. Big waves of soil would blow away. On some of the steep slopes with loess the soil would just go. We haven't ploughed for 30 years."

The application of inputs such as fertiliser and seed on steep country can be expensive as it often requires the

use of a helicopter. In addition, there are numerous gullies which cannot be grazed intensively. On the positive side, because of the variety of topographies on the Peninsula, stock can be moved to areas with different characteristics depending on the conditions. Typically, stock are kept on the warmer and drier lower paddocks during winter months and are sent up to the higher and wetter paddocks over the summer.

All these factors have resulted in low intensity farming systems which are largely managed within natural constraints. It is a relatively simple farming system, requiring little investment in equipment, and one that attracts farmers who enjoy working with nature.

"They choose to farm on the Peninsula for lifestyle as much as for economic viability. It's different to other parts of Canterbury where farmers have a much more hard-nosed approach."

Banks Peninsula has traditionally been regarded as good stock breeding country. We were told that only about 10 per cent of farmers finish their stock (ie take the stock through to slaughter). To achieve this, farmers need high quality feed for the lambs once they are weaned, as well as good genetics which favour fast growth. For the bulk of farmers, lambs and calves are usually sold at weaning. A lot of stock, especially lambs, leave the Peninsula in late November and early December, when there is a reliable market for them,

Eroding loess soils on Banks Peninsula



and before the risk of pastures drying out. We were told that the more valuable irrigated flat land is used for fattening stock as this is where the highest added value occurs.

"You can get extra income off your lambs by taking them through to slaughter but then you have to compromise something else. It is easier to be a breeding operation, as it is too risky to try to take stock through the summer period when it gets dry, without extra feed."

Historically, there has been a close linkage between farms on the Peninsula and the nearby Canterbury Plains. Stock have typically been bred on the more rugged Peninsula country and then fattened on the flat irrigated farms on the Plains over the summer. In addition, many farmers on the Peninsula have had arrangements with farms on the Plains, allowing them to move their stock there in the event of drought.

But this 'balance' between the two farming areas has been gradually upset over the past 30 years with the conversion of many Canterbury farms to intensive dairying. We were told that young stock from the Peninsula are now being trucked to destinations as far away as Buller, Central Otago and Western Southland to be fattened.

There has been a recent trend towards amalgamating farms on the Peninsula to form larger economic units. In addition, some long-standing farming families now lease their land to adjoining owners so they can retain ownership of the land, continue to live on it and generate some income, without actively farming. This can be an attractive option when the next generation are not interested in becoming farmers.

The constraining natural characteristics of Banks Peninsula means that there is little likelihood of a shift to intensive farming models or dairying in the future. Although farming clearly has some landscape impacts, farming systems are generally of low intensity with few inputs and are generally sympathetic to the landscape and natural values of the area.

"The current level of economic returns from farming systems appropriate to Banks Peninsula means that it is unlikely that there will be any commercial drivers towards large scale development of reverted areas of exotic weeds or native scrub land in the medium term future."⁵

5.2 Forestry

Exotic plantation forestry can create significant adverse effects on landscapes. Native forests comprise a variety of vegetation with a range of sizes, colours and ages. On the other hand, plantation forests are typically single (exotic) species monocultures, planted in a linear form and at the same time yielding single year classes. Once harvested, the area is left scarred and the steep slopes and fine loess soils (which are common on Banks

Peninsula) are susceptible to erosion. Plantation forestry also reduces water yield, resulting in less water availability in waterways.⁶ As already indicated, water supply is currently an issue on Banks Peninsula as it is spring-fed, and climate change will likely place further pressure on the shallow aquifers. Conversion of pasture to plantation forestry would place further stress on water in the future.

There are currently only small pockets of exotic forestry on Banks Peninsula. The larger plantations are predominantly in the Lyttelton harbour area, where the trees are closer to the port, thereby reducing cartage costs. Many of these were established during the 1990s, when there was significant expansion in the area planted throughout New Zealand, as a result of the doubling of log prices overseas.⁷ This means many trees are now entering the harvesting cycle. There are also numerous small pockets of pine trees on farmland dotted around the Peninsula, as, for a period of time, many farmers were encouraged to plant pines by farm advisers and there were government incentives to do so.⁸

"Every farmer here at some point planted a little block of pines. At the time, there was a government incentive to plant two hectares of pines and they were told this would be their pension. But now we can't get our pines out. No-one will take them out as it is uneconomic."

Pine and other exotic trees grow well on the same marginal land that is currently regenerating into native bush. There is also the possibility that exotic plantations could replace land currently in pasture if the economic incentives are strong enough, and this is already starting to occur as illustrated in the spotlight on Le Bons Bay below. The future naturalness of Banks Peninsula landscapes will largely depend on whether exotic plantations win out over these other two options over time. This will, in turn, depend on the economic and regulatory incentives in place and which land use they favour. We discuss three main pieces of government policy which impact the incentives for forestry on Banks Peninsula below.

National Environmental Standards for Plantation Forestry

The Resource Management (National Environmental Standards for Plantation Forestry) Regulations (NESPf), which were promulgated in 2017 and came into effect in May 2018, are the key regulatory tool for managing plantation forestry greater than 1ha. The regulations seek to control environmental outcomes associated with plantation forestry activities nationally, while also increasing certainty and efficiency for foresters.⁹ The NESPf prevails over the rules in a district or regional plan. Planning provisions may only be more stringent than the NESPf in limited circumstances (such as to protect matters of national importance),¹⁰ and may not be more lenient.¹¹

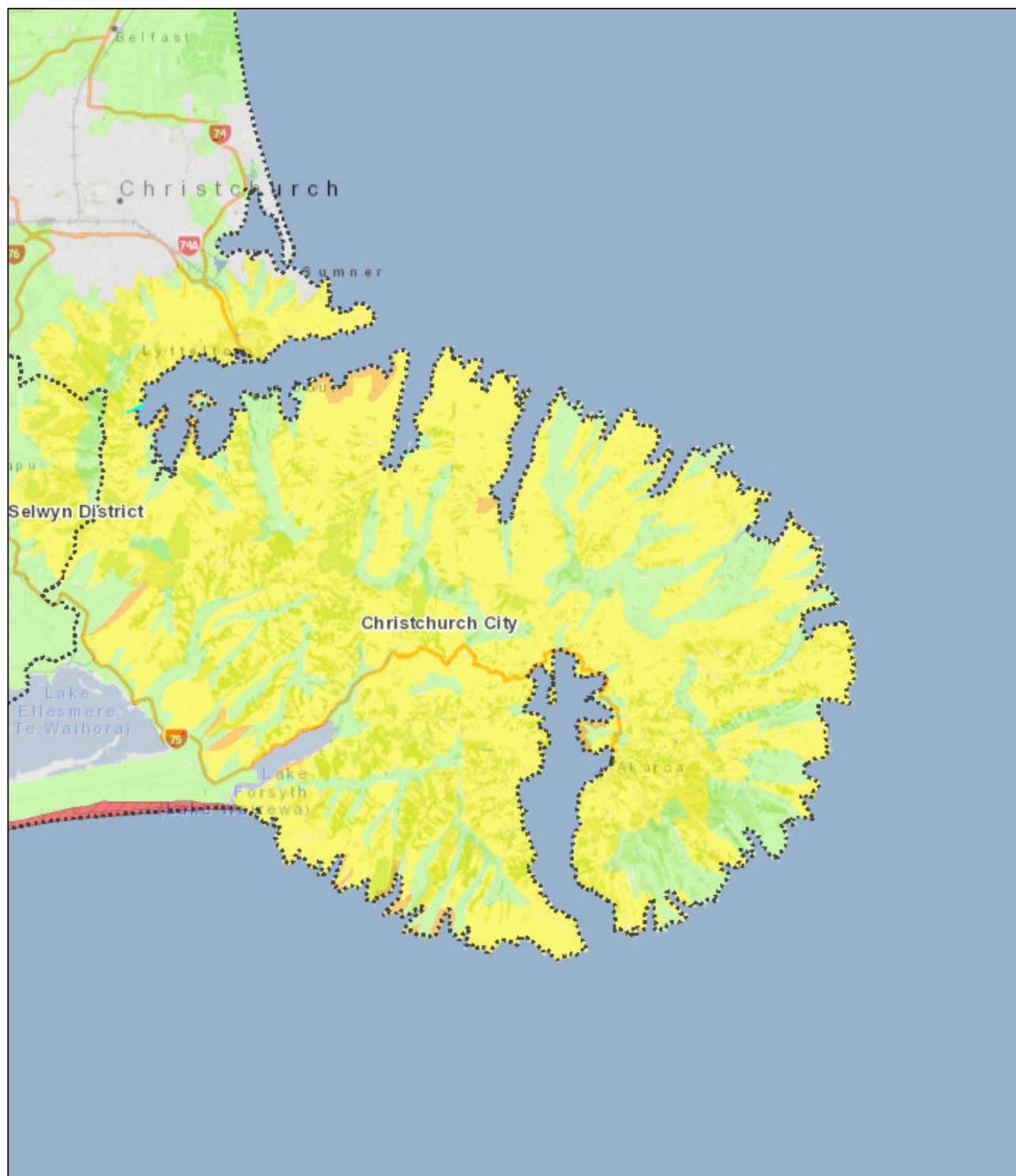
Prior to the NESPf coming into force, the Christchurch District Plan provided the rule framework for plantation

forestry on Banks Peninsula. The Plan stipulated that all plantation forestry within the Rural Banks Peninsula Zone was a restricted discretionary activity and required a resource consent.¹² Adverse effects on significant features and amenity landscapes, indigenous biodiversity and cultural landscapes were included as matters of discretion. In the Rural Amenity Zone (which covers much of the Peninsula's rural area) plantation forestry was discretionary. Within outstanding natural landscapes (ONL), significant features, areas of outstanding high and outstanding natural character, and areas of natural character in the coastal environment, plantation forestry was a non-complying activity.¹³ However, most of these rules have now been superseded by the much more lenient ones in the NESPF.

The NESPF sets a permissive baseline, with the underlying premise being that the establishment and operation of plantation forestry is a permitted activity subject to compliance with standards. The Rural Banks Peninsula Zone is largely mapped as yellow zoned land (indicating moderate erosion risk) using the Erosion Susceptibility Classification System utilised by the NESPF (see Figure 5.1). This means that afforestation and replanting are now permitted activities. The application of yellow zoning does not appear to adequately take into account the steep slopes and highly erodible loess soils on Banks Peninsula. This means that planting plantation forestry trees on land where there are no existing plantation forests, or replanting the same species, does not require resource consent. There is an exception to this rule in the NESPF, with plantation forestry within significant natural areas and ONLs requiring a restricted discretionary consent.¹⁴



NES-PF Erosion Susceptibility Classification Report



02/12/2020

Legend

Class 8e land Very High

Erosion Susceptibility

Other

Low

Moderate

High



Te Uru Rākau
Forestry New Zealand

Figure 5.1 Map showing erosion susceptibility for Banks Peninsula (Source: Te Uru Rākau)

As mentioned above, Regulation 6 of the NESPF also enables councils to apply greater stringency to their planning provisions in limited circumstances. These include to provide for the protection of ONLs or significant natural areas; or to give effect to either the National Policy Statement on Freshwater Management or the New Zealand Coastal Policy Statement 2010 (NZCPS). In the Christchurch District Plan, greater stringency has been applied, and afforestation within the Coastal Environment overlay and ONL overlay is non-complying.

However, to fall within the ONL exception of the NESPF, the landscape must be identified in a regional policy statement, regional plan or district plan as outstanding, and that planning document must include the identification of its location by way of mapping, a schedule or description. Although the entire Banks Peninsula is identified as an ONL in the Canterbury Regional Policy Statement (Canterbury RPS),¹⁵ specific identification (and mapping) is deferred to the district council. As discussed in section 6.3 below, events during the preparation of the Banks Peninsula District Plan (and subsequent adoption into the Christchurch District Plan) resulted in the ONL overlay being severely limited. It applies mostly to only prominent ridgelines and the coastal edge. The balance was mapped as rural amenity landscape, which now covers the majority of the Peninsula (see Figure 6.4). These planning decisions made over a decade ago are now playing out in an unintended way within the new regulatory framework for plantation forestry under the NESPF.

The unforeseen consequences of this are, quite frankly, alarming. Subject to a few weak conditions relating to wilding control and setbacks (for shading, but not fire resilience), the planting of exotic pines and other commercial species is now provided for over most of the Banks Peninsula as of right, with Council's discretion being limited to those matters listed in the NESPF. An example of the impact of the NESPF on Banks Peninsula is highlighted in the spotlight below.

A spotlight on forestry at Le Bons Bay

In August 2020, a new forestry development was approved at Le Bons Bay on the eastern side of Banks Peninsula. The site, shown in Figure 5.2, is currently a large pastoral landholding with pockets of indigenous vegetation. It is described as a wild peninsula landscape including an absence of development, impressive landforms, highly natural coastlines, presence of extensive native vegetation and displaying a high degree of aesthetic values.¹⁶ Once planted, the 250ha site will be largely covered in *Pinus radiata* and redwoods, thereby destroying the natural character and landscape values of the area.

The majority of the development will occur within the rural amenity landscape overlay. As such, it is a permitted activity and no resource consent was required. This formed the permitted baseline and influenced the decision to grant a non-complying consent on a non-notified basis for afforestation over the remaining 17ha, which sits within the Coastal Environment and ONL overlays.

The Resource Management Act 1991 (RMA) affords discretion to disregard the adverse effect of an activity if it is provided for as a permitted activity (the permitted baseline).¹⁷ As such, the anticipated adverse effects of 175ha of permitted plantation forestry was not taken into account when determining the resource consent for the balance. The assessment found that, cumulatively, forestry within the ONL and Coastal Environment overlay is relatively small when compared with the overall forestry within the site and would have a very low degree of adverse cumulative effects. This was on the basis that the forestry extending into the coastal environment would be visually cohesive with, and largely screened by, the permitted forestry plantation.



Figure 5.2 View from the eastern extent of the Le Bons Bay site towards the coastline (Source: Resource Consent application)⁶³

The application shows the failure of the NESPF to protect the landscapes of the Banks Peninsula, both within the rural amenity landscape and in the adjoining ONL and coastal areas.¹⁸

The adverse effects of plantation forestry relate both to the impacts of the trees on landscape and biodiversity values when they are *in situ* and their removal through clear fell harvesting (see spotlight below). The permissive nature of the NESPF seems likely to pose a significant threat to Banks Peninsula landscapes in the near future. Fortunately, the Labour Party has indicated that it will undertake a further review of the NESPF during the current term of Government, and this is an opportunity to rectify these glaring loopholes.

A spotlight on the impacts of forestry harvesting

Forestry harvesting can have disastrous consequences if things go wrong. For example, the activities of Waiake Forestry on land above the Loudon farm near Teddington resulted in tonnes of slash washing down the Waiake stream on 21 July 2017 during a period when a month's worth of rain fell in just two days. "... tonnes of slash – logging off-cuts – that washed down nearby Waiake Stream was scattered up to six metres high, flattening fences and ruining grazing paddocks. More debris blocked the stream for three kilometres. Summers' cottage was flooded and logs and sediment washed into Lyttelton Harbour." The forestry company escaped censure as it was told incorrectly by the Council that logging next to the stream was permitted and consent was not required.¹⁹

Emissions trading scheme

The economic future of farming on the Peninsula is uncertain and landowners are looking to derive some additional income off their marginal land. This has resulted in many turning to carbon credits under the ETS. While we heard that many would like to do this through native forest regeneration, the impediments within the ETS (discussed below) may see more people turning to exotic plantation forestry as a source of carbon credits.

International and domestic climate change obligations

New Zealand is a party to the United Nations Framework Convention on Climate Change and is a signatory to a number of important agreements under it. Notably these include the Kyoto Protocol and Paris Agreement. The Kyoto Protocol operationalises the Convention by committing industrialised countries to limit and reduce greenhouse gas emissions in accordance with agreed targets.

Of particular importance to this case study is Article 3(3) of the Kyoto Protocol which establishes that forestry activities, such as afforestation and reforestation,

can be measured as verifiable changes in carbon stocks in each commitment. This forms the basis on which foresters can accrue emission credits.

"...promote sustainable management, and promote and cooperate in the conservation and enhancement, as appropriate, of sinks and reservoirs... including biomass, [and] forests..."²⁰

New Zealand ratified the Kyoto Protocol in 2002 and committed to a reduction in net greenhouse gas emissions to 1990 levels between 2008 and 2012. In response to these obligations, the Climate Change Response Act 2002 was enacted, which establishes the ETS. New Zealand met its first target and reported a surplus of 123.7 million units. Forestry contributed significantly to this period, sequestering 71.6 million tonnes of carbon dioxide.²¹

For the period of 2013 to 2020, New Zealand committed to a reduction of net greenhouse gas emissions to 5 per cent below 1990 levels (the 2020 target). As at April 2020, New Zealand is projected to meet this target, with forestry contributing a reduction of 108 million tonnes of carbon dioxide over the seven year period.²²

New Zealand's post 2020 commitment to reducing greenhouse gas emissions is found in the Paris Agreement, which was adopted by parties to the Convention in December 2015. The Paris Agreement aims to limit the global temperature to 1.5 to 2°C above pre-industrial levels during this century. New Zealand's target is to reduce greenhouse gas emissions by 30 per cent below gross emissions for the period 2021 to 2030. This is equivalent to 11 per cent below 1990 levels for that same period.

In 2019, the Climate Change Response (Zero Carbon) Amendment Act 2019 was enacted to bring New Zealand in line with the global ambition in the Paris Agreement. It adopts a two-baskets approach and sets in law New Zealand's new domestic climate change targets for 2050 of:

- Net zero emissions of all greenhouse gases other than biogenic methane by 2050
- By 2050, 24 to 47 per cent below 2017 biogenic methane emissions, including 10 per cent below 2017 biogenic methane emissions by 2030

The Act also set up a Climate Change Commission to provide independent expert advice to the Government and to create a system of five-year emissions budgets to act as stepping stones to the long term targets.

In addition to reducing greenhouse gas emissions, carbon sequestration plays a critical role in meeting these international commitments. This can be achieved through exotic or indigenous, planted or regenerating, forestry. This is discussed further in the body of the text below.

The ETS is the Government's main tool for meeting domestic and international climate change targets. It puts a price on greenhouse gas emissions by setting a regulatory limit on emissions from sectors. That limit translates into a market price which is designed to incentivise behaviour to reduce emissions.

The ETS requires polluters to surrender a tradable emission unit known as a New Zealand Unit (NZU) to the government for each tonne of emissions for which they are liable. The government limits the supply of NZUs into a trading market which then sets the emission price based on unit supply and demand. Where demand for units increases and supply remains constant, the price of units will generally increase. The price will decrease where there is an abundant supply of units. This is often referred to as the 'carbon price'.

New Zealand is the only country that currently includes forestry in an ETS.²³ The ETS creates a financial incentive for landowners to earn money by planting forests that sequester carbon. As the forests grow, the Government gives eligible foresters NZUs for the carbon dioxide that is absorbed by the trees. These units can then be sold on the ETS market, thereby providing an income from

the forest. However, the rules determining eligibility are somewhat complex, and proving land is eligible for the ETS is the most difficult aspect.

To be included in the ETS, the land must be 'forest land' defined as being at least 1ha of forest species and at least 30m in width. It does not need to have continuous heavy canopy cover, but at maturity, it must have 30 per cent canopy cover and stand at least five metres tall.

The ETS classifies forests differently depending on whether they were first established after 1989 or pre-1990. Forests that were planted pre-1990 are considered baseline forests (as 1 January 1990 is the baseline for New Zealand's international climate change obligations) and as such do not count towards New Zealand's obligations. New Zealand is the first, and only country to include pre-1990 forest owners as mandatory participants in the ETS. Therefore, while forest owners cannot earn NZUs from a pre-1990 forest, they will face liabilities for deforestation or permanent removal of the trees.²⁴

Forests that were established after 1 January 1990 are considered new forests and can be counted towards New Zealand's climate change obligations. Owners can earn



NZUs for their carbon sequestration. This includes both permanent and rotational forests.

Rotational forests earn NZUs as they grow and absorb carbon. However, these are surrendered upon harvest. Under the ETS, rotational forests will only earn NZUs for sequestration during the first rotation (usually around 20 years for *Pinus radiata*). Following harvest, these do not need to be repaid, provided the land is replanted. However, if the land is not replanted, the owner will be liable to repay the credits. By contrast, permanent forests are not intended for harvest and will continue to earn carbon credits until the forest reaches maturity (upwards of 50 years for natives).

"If we could get ETS credits for natives this would change our framework of farming. Fencing is a big investment and if we could get a financial benefit from fencing and allowing regeneration we would do more of it."

There is interest from many landowners on Banks Peninsula in the potential to earn income from carbon sequestration. However, the process is not straightforward, particularly for naturally regenerating areas.²⁵

Landowners can only earn carbon credits for forests that have been established since 1990. As such, they must prove the age of the forest. Unlike single age-class pine plantations, where the trees are planted at a specific time, a high degree of temporal and spatial variability occurs as a result of natural regeneration. As discussed in the spotlight in section 3.4, this means it is difficult to prove the age of the forest. This has resulted in a number of areas, which may in fact have established since 1990, not qualifying for carbon credits under the ETS.

The carbon sequestration rate is also initially slower for native forest. Natives have a very slow start (over the first

three years) followed by rapid growth (3–27 years) before the curve slows into a steady growth pattern as trees establish and become dominant in the forest.²⁶ This is shown in Figure 5.3.

By comparison, exotic species such as pine or eucalyptus grow faster and sequester carbon at a higher rate for the life of the forest. However, these species reach their maximum carbon storage capacity in about 20 years.

When taking a short-term view, exotics look preferable. However, if a broader longer-term view is adopted, natives are a better option. They continue to sequester carbon for much longer, and sequester more carbon overall compared to exotics²⁷. They also have other environmental and ecosystem service benefits including erosion control, fire and drought resistance and water purification. Native forests also provide significant biodiversity benefits by supporting a range of indigenous species and habitats.

"The whole system is designed around pines. This creates a monoculture which is killing off the biodiversity."

Participants in the ETS must file emissions returns measuring the amount of carbon sequestered and measured over a five-year accounting period. For areas under 100ha, lookup tables taken from the Climate Change (Forestry Sector) Regulations 2008 are used to estimate the amount of carbon likely to have been sequestered. These tables give pre-calculated values for a forest's carbon stock based on its age, region and forest type. If the forest is over 100ha, a Field Measurement Approach must be used. This requires the landowner to physically measure the growth and sequestration of their forest.²⁸ Landowners must measure the trees at specific sample plots and submit this information to Ministry for Primary Industries (MPI) to create a participant-specific

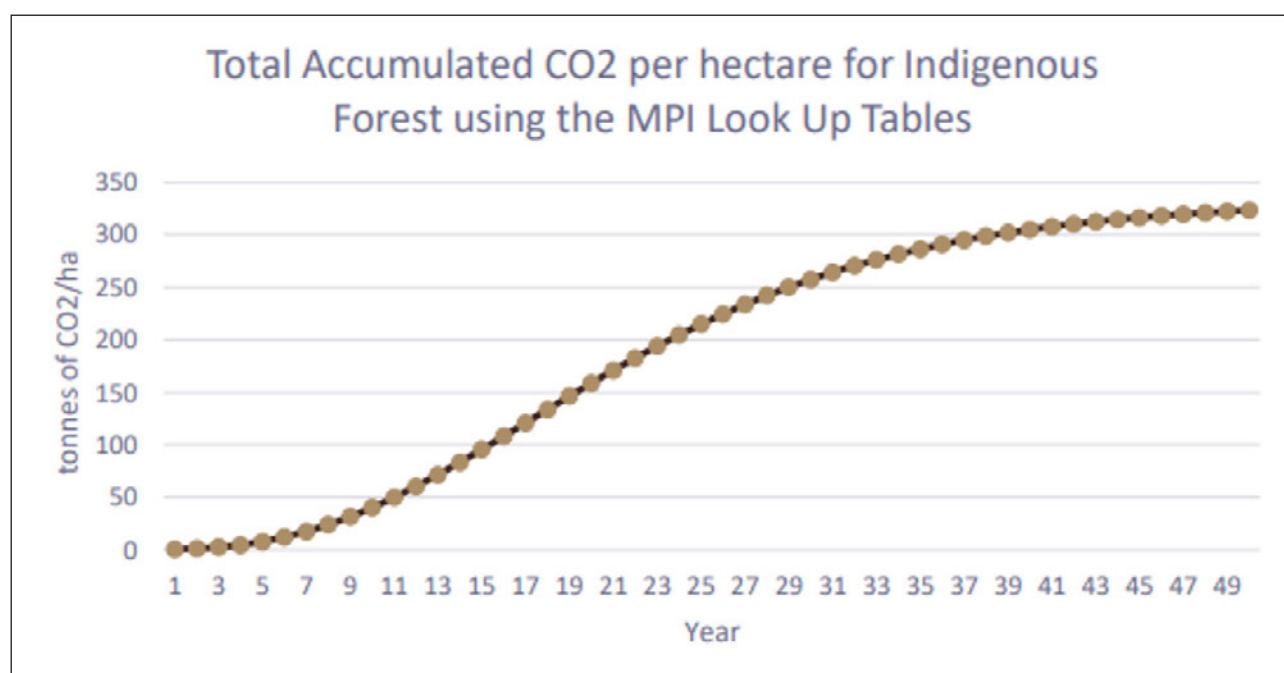


Figure 5.3 Total accumulated CO₂/ha for indigenous forest (Source: MPI)

lookup table.²⁹ This requires the establishment of upwards of 30 permanent sample plots to estimate the total biomass of the forest. For naturally regenerating forests this is an onerous task, as the composition of the forest does not allow easy access, unlike exotic or native plantation forestry.

"The monitoring is unbelievably tedious. There are randomly placed plots that need to be measured every 5 years... you need to scramble through gorse and regenerating bush to even get there."

One Billion Trees

As described above, New Zealand is committed to reducing carbon emissions to net zero by 2050. Carbon sequestration by trees will play an important role in reaching this target. Modelling undertaken by the Parliamentary Commissioner for the Environment, in 2018, suggested that land use will need to change significantly to achieve this goal. Between 1.3-2.8 million hectares of forestry will need to be planted within the next three decades.³⁰ Currently, the low carbon price does not incentivise the planting rates required to meet New Zealand's climate change obligations.

The One Billion Trees fund was launched in November 2018 to increase tree planting across New Zealand. It aims to double the current planting rate to achieve one billion trees planted by 2028.

"The idea of putting pines on the Peninsula would be a lost opportunity and would destroy the benefit of natural amenity that could be brought about by good native replanting and regeneration."

The catchphrase for the One Billion Trees programme is the 'right tree, right place, right purpose'. The programme seeks to encourage both permanent and rotational forests (the right tree) in areas well integrated with the landscape (right place) and in a manner that is well-planned and considers the long-term maintenance of the trees (right purpose). It is anticipated that commercial foresters will plant 500 million trees over the period, with the other 500 million being planted by individuals or the community. The Government has allocated \$240 million for direct grants to landowners to achieve this purpose.³¹ It was agreed by Cabinet that two-thirds of the new planting budget under the One Billion Trees programme should be native trees. To date, this target has been largely met across the country overall (see spotlight).



Indigenous regeneration on Flea Bay Farm

A spotlight on progress towards planting One Billion Trees

The One Billion Tree programme is now into the final year of its initial three years of funding. The latest statistics show that, as of 7 September 2020, 258.5 million trees have been planted with almost 37 million of these directly funded by the programme. Grants have been made to 495 landowners for 24,681ha of planting and 142 applications are in the pipeline for consideration. They are spread across all New Zealand regions including the Chatham Islands with 61 per cent of funding going to natives and 39 per cent to exotics. Funding has also been agreed for 84 Partnerships (\$67.14m) and 23 applications are under consideration (\$14.4m).³²

There are four categories of direct grants available to landowners. These are: native planting (\$4,000/ha), native reversion (\$2,000/ha), mānuka/kānuka planting (\$1,500/ha) and exotic planting (\$1,500/ha). There is also an additional \$500/ha available for plantings on erosion prone land.³³

In order to be eligible for a grant, a number of criteria need to be met, relating to the land and project proposal. The land must not be 'forest land' at the time of the application and must not have been 'forest land' in the five years prior to applying. As such, the funding is effectively only available for bare land. Also, the land must not be in another government-related scheme, including the ETS. Areas already receiving carbon credits are not eligible for One Billion Trees funding, nor are those which already have a 30 per cent canopy. However, applications



Riparian planting, Okains Bay

can be made for areas that qualify but are not yet receiving credits.

The project also needs to meet specific criteria including minimum size requirements (1ha for native planting, 5ha for all other categories); planting of species that are capable of reaching 5m at maturity (unless as part of a native planting project where smaller trees and shrubs are accepted as part of the planting mix); and appropriate planting density. For exotic and native plantation forests this is likely to be 1,000-2,500 stems/ha. However, for native planting projects this is instead 2,500-10,000 stems/ha. In addition, all activities must be capable of meeting the establishment standard of a minimum 750 stems/ha successfully growing after planting.³⁴ We were told by farmers on the Peninsula that planting at these densities is quite difficult and expensive to establish. Native planting also requires significant pest control measures for hares and deer which eat the seedlings.

A spotlight on grants approved on Banks Peninsula under the One Billion Trees programme

On Banks Peninsula, 20 Direct Grants have been approved totalling \$1,748,375. This funding will be used to plant 218,346 (33%) native trees and 453,052 (67%) exotic trees, noting that native trees are usually planted at a higher stem count per hectare. The resulting breakdown of planting type per hectare is shown in Figure 5.4.

For comparison, a breakdown of the Direct Grants for planting types for each region is shown in Figure 5.5. This shows the average for natural reversion nationally is much lower (10.97%) and the average for native planting higher (36.20%). Exotic planting remains similar (52.83%).

Planting Type	Approved Planting (Hectares)	% hectares
Mānuka/Kānuka	0	5% of hectares funded is for native planting
Mixed indigenous species	64.3	
Other exotic	272.4	49% is for exotic planting
Pinus radiata	316.6	
Reversion (Indigenous)	539.8	45% is for reversion
Grand Total	1,193	100%

Figure 5.4 Direct Grants approved by Te Uru Rākau on Banks Peninsula (Source: MPI)

Region	Reversion Native (Hectares)	Indigenous (Other) (Hectares)	Mānuka/ Kānuka (Hectares)	Other Exotic (Hectares)	Radiata Pine (Hectares)
Auckland		266.63	29.1	7.51	
Bay of Plenty	10.42	207.49	68.33	1.36	65.63
Canterbury	1,370.41	882.53		2,297.16	4,212.34
Chatham Islands		142.08		10.37	
Gisborne	29.03	192.87	94.01	11.06	246.76
Hawkes Bay	476.49	649.18	1,446.20	372.14	865.03
Manawatu/Whanganui	207.11	180.50	1,622.64	113.74	72.54
Marlborough	89.64	175.26	1.79	24.63	876.71
Nelson		77.83			
Northland	5.68	553.87	102.58	98.00	244.96
Otago	181.81	870.31	25.90	480.90	1,221.08
Southland	66.56	269.14		177.11	451.22
Taranaki	255.76	8.03	558.70	168.88	
Tasman	47.19	134.66	25.35	124.33	616.28
Waikato	13.34	211.66	28.75	16.24	121.16
Wellington	49.04	305.54	82.10	171.62	407.69
West Coast		30.39		4.57	8.42
Total	2,802.48	5,157.97	4,085.45	4,079.62	9,409.82

Figure 5.5 Direct Grants approved by Te Uru Rākau nationally (Source: MPI)



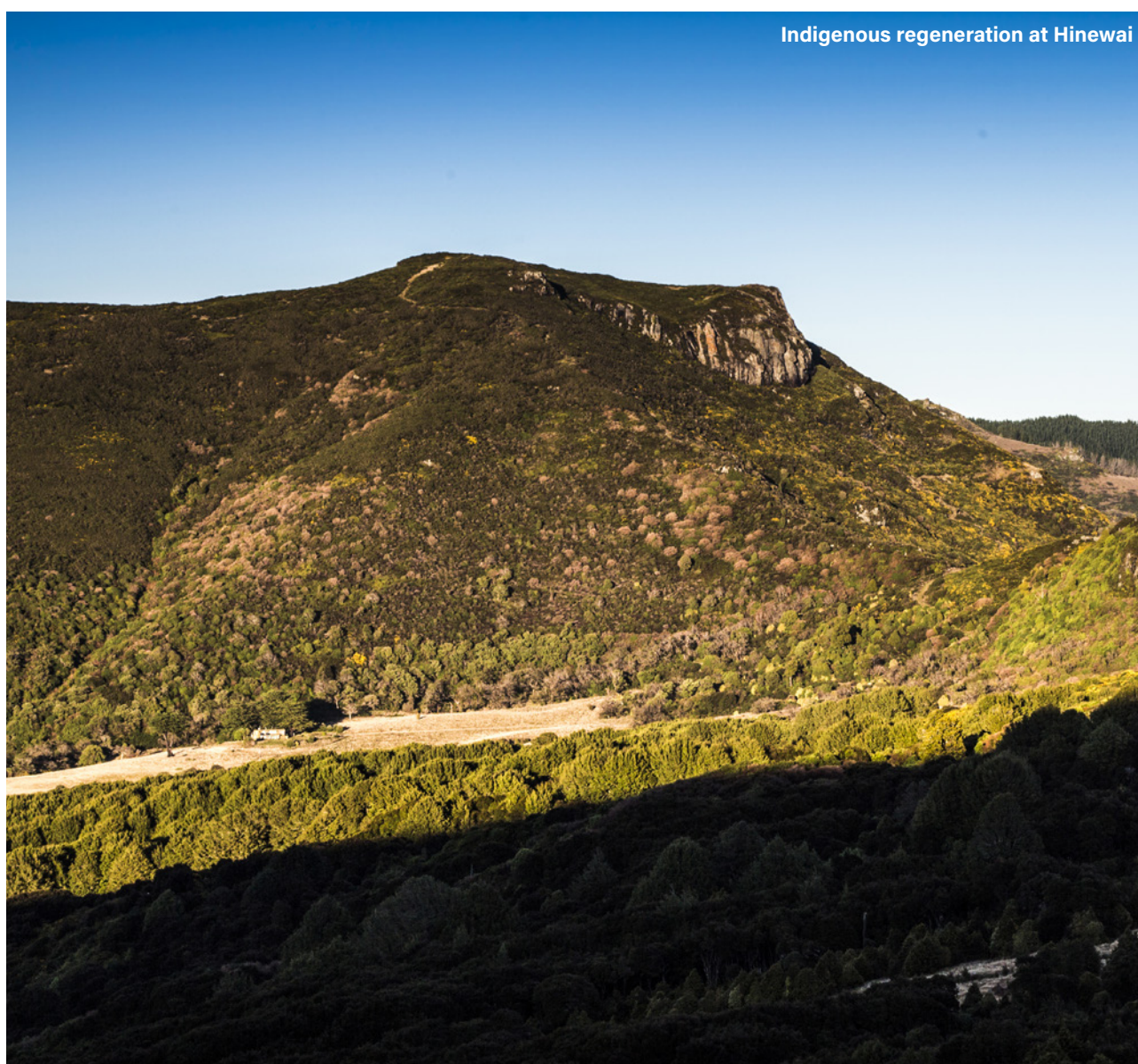
In a place like the Banks Peninsula, where mother nature is already onsite and rates of regeneration are high, natural reversion/regeneration seems an obvious choice. All that must be demonstrated is that the land is being protected (eg pest and weed control carried out) and that there is an adequate seed source nearby. Based on the availability of the seed source, supplementary planting may be recommended to meet the establishment standard of 750 stems. However, the timeframes and establishment requirements of the One Billion Trees programme are not well equipped to deal with the characteristics of regeneration.

Payments for all categories of planting are made on the achievement of three milestones. The first milestone (30 per cent) is paid once the project is underway, the second payment (50 per cent) is paid on the establishment of a successful forest (eg 750 stems/ha successfully growing free from pest and weed competition). It is anticipated that this will occur within four years from start of planting. However, we were told that this was unrealistic for native seedlings, and particularly for natural reversion and regeneration. The final payment (20 per cent) is for

the successful maintenance of the project and must be claimed within eight years of the start date.

The One Billion Trees programme creates a number of hurdles for natural regeneration that may see landowners switching to exotic planting. This is because planting native forest is not necessarily a good alternative. First, it is expensive. Seedlings are costly to source and are more susceptible to animal browsing, grass and weed competition and drought than their exotic counterparts. Monitoring is more difficult due to the composition of the forest and carbon sequestration rates are slower, albeit more sustained. This means that exotic planting is likely to be seen as the most attractive option from a financial perspective. In addition, we were told that it was more economic for farmers to periodically spray off large areas of regenerating bush (mostly kānuka) to enable them to farm the land, rather than letting the bush grow. If native regeneration was adequately incentivised this would not be the case.

Overall, as described in the preceding sections, the current system either intentionally or unintentionally



skews the balance towards exotic plantation forestry. The NESPF largely removes the ability of council to control new plantation forests on the Peninsula. The One Billion Tree fund and ETS make it significantly more difficult to establish and monitor a naturally reverting forest compared to a plantation forest.

On Banks Peninsula, where marginal grassland is already reverting or has reverted to woody scrub on its own, native forest will regenerate if landowners are provided with the right incentives. In areas such as this, natural regeneration provides the opportunity to convert large areas of barely productive farmland into carbon-sequestering perpetual indigenous forests with associated environmental benefits.³⁵ On the contrary, if these incentives are not addressed and corrected, we may see large areas of Banks Peninsula converted to plantation forestry which will significantly degrade its landscape and biodiversity values.

A spotlight on reducing greenhouse gas emissions on farmland

Integrated farm management plans have the potential to support carbon zero farming through avoiding and mitigating emissions. Such plans map the current farm regime and then identify alternative land-use and management options (such as native regeneration on marginal land) to reduce or mitigate climate change effects. Greenhouse gas emissions that cannot be avoided are addressed through inseting (on-farm works) rather than offsetting.³⁶

Through the use of inseting and offsetting, farmers on Banks Peninsula have the potential to help meet greenhouse gas emissions targets through native forest regeneration.

5.3 Urban development

The two largest settlements on Banks Peninsula are Lyttelton with a population of 2,982 and Akaroa with a population of 756 (as at 2018). The two towns are very different in function and character. Lyttelton is a port town and effectively operates as a suburb of Christchurch. For many years Akaroa was a rural service town. It now also serves as a holiday destination and tourism hub. At the time of the 2018 census, the township had 327 occupied and 612 unoccupied dwellings highlighting the large number of holiday homes in the town, comprising almost two thirds of the total. Including those houses under construction at the time of the census, there was an increase of 36 houses since 2013 (or an average of around 7 new houses a year).

Both towns are constrained by topography, being located between steep hills and a harbour, and there is consequently little room for additional growth. There are similar constraints on the smaller settlements around Lyttelton harbour, with the exception of Diamond Harbour where there is an extensive area of Council owned farmland that is zoned residential and may be developed in the future. The small amount of urban growth that has occurred in recent times on Banks Peninsula has largely been at Little River, which is still within commuting distance of Christchurch, while providing a high quality rural living environment.

Urban encroachment

Residential development can adversely affect the rural landscape, for example, by reducing its naturalness and openness. To varying extents, depending largely on the scale of development, the introduction of buildings and structures can modify the landscape to the extent that the human-made elements of the landscape become dominant. The addition of gardens, driveways and



Urban development in Akaroa

pathways – the curtilage effects – can also degrade the naturalness of the landscape.

The Christchurch District Plan review process indicated that the Port Hills remain under pressure from urban expansion, with a number of requests for development beyond the existing urban area. The current Canterbury RPS precludes unplanned expansion within the Greater Christchurch area, and the development proposals were consequently declined.³⁷

A spotlight on landscape protection on the Port Hills

The Christchurch Port Hills, which form the western flank of Banks Peninsula, provide an important backdrop to both Christchurch and Lyttelton harbour. The Summit Road, created as a result of Harry Ell's vision, provides access around the Port Hills. In 1948, the Summit Road Protection Society was formed to further this vision and to enhance, preserve and protect the nature, beauty and open character of the Port Hills and provide access to the public through a network of walking tracks.

The Society owns three reserves on the Port Hills – Ohinetahi Bush Reserve, Omaha Bush and Gibraltar Rock, and Linda Woods Reserve. Much of the regenerating bush and tussockland of the reserves is protected by QEII covenants and is open for the public to enjoy. Funding for the acquisition of these reserves comes from the Harry Ell Summit Road Memorial Trust, formed in 2002, with the purpose of enabling the funding of further purchases of land for reserves and the maintenance of the tracks, vegetation and other facilities.³⁸

In 2001 the Summit Road (Canterbury) Protection Act was enacted to further this purpose by providing for the preservation and protection of the scenic amenity associated with the Summit Road and associated protected land. It also provides for the improvement of facilities for public enjoyment of the scenic and natural amenities.³⁹ The Act provides for the establishment and continuation of the Summit Road Protection Authority, a joint committee of Christchurch City Council, Selwyn District Council and the then Banks Peninsula District Council, which is deemed to be a local authority under the RMA. The Act prescribes that any person seeking to develop or subdivide protected land in the Port Hills area be required to seek written consent of the Authority.⁴⁰ The Society, together with the Authority, therefore provide a watchdog role – seeking to minimise urban encroachment into the Port Hills area.

Outside of the district plan review process, there have been a number of appeals in relation to proposed subdivision and construction of residential buildings on Banks Peninsula. These cases have been important in moderating developer expectations for the area and serve

to illustrate the Court's reluctance to allow residential development where it will unacceptably degrade landscape values. For example, in 2010, the Akaroa Civic Trust successfully appealed a two-lot subdivision at Red Point, Takamatu Hill, on the western side of Akaroa Harbour. Part of the land was located in the coastal protection area and the balance was in the Rural Amenity Landscape Zone. Despite the proposal to include a 36ha conservation covenant, the visual dominance of the site and the adverse effects on natural character values meant that consent was overturned.⁴¹

In 2004, the Environment Court refused a proposal for a seven-lot subdivision and the erection of seven dwellings on the basis of adverse landscape effects. The proposal related to a 140ha site on the slopes and gullies of Mount Herbert, above Diamond Harbour. The site was predominantly used for pastoral farming and identified as an ONL. It was accepted that the proposed subdivision would unacceptably detract from the landscape values.⁴²

In another appeal, a proposal to construct a large holiday house on the western side of Akaroa Harbour underwent formal mediation. The proposed house was to be situated on a visually prominent edge at the top of the hill and was obtrusive when viewed from the water. It would have been the first invading house on a natural stretch of coastline directly opposite the location of the earlier appeal at Red Point. Mediation resulted in consent being granted for a building that was relocated to a less intrusive position on the headland.⁴³

The last substantial subdivision on the Peninsula was in 2009. It created 14 lots at the northern end of Akaroa. In 2008, there was also a large eight-lot subdivision in the Lyttelton Basin in the Te Pohue Camp Bay Area, where land was subdivided into a series of lots of between 6-14ha running along the foreshore. Open space covenants were created on the balance areas. The Banks Peninsula District Council also approved a non-notified 11.2ha subdivision at Black Point in Lyttelton Harbour in 2006. In addition, there has been a steady stream of farmers carving off small lifestyle blocks and placing an open space covenant over the balance lot. This is particularly so in the Akaroa Harbour area and is described further in relation to District Plan provisions below.



Historic heritage

Both Lyttelton and Akaroa have managed to retain much of their historic fabric. However, sadly, many heritage buildings were lost in Lyttelton as a result of the Christchurch earthquakes. There has been a long history of community activism to protect the historic heritage of Akaroa, including the early French influence on the settlement of the town.

A spotlight on the Akaroa Civic Trust

The Akaroa Civic Trust was established in 1969 due to concerns that development was harming Akaroa's charm and character. The Civic Trust prevented the town's streets being widened and then successfully persuaded the Akaroa County Council to establish a Design and Appearance Committee in order to provide advice on plans for new buildings in Akaroa. Design guidelines were drawn up during the 1970s and revised in 2001-02. In 1999, The Civic Trust successfully secured the registration of the heart of Akaroa as a historic area under the Historic Places Act 1993. It then fought to obtain adequate recognition of the historic values of the town in the district plan. One notable success was achieving lower height limits on new buildings in the town centre to prevent the construction of more large apartment blocks. The Civic Trust also worked with the Ōnuku Rūnanga to have the land at Takapūneke set aside as a reserve (see earlier spotlight).⁴⁴

After the Banks Peninsula District Council merged with Christchurch City Council in 2006, there was concern that the focus on protecting Akaroa's heritage might be diluted. We were told that, since amalgamation, less attention has been paid to heritage issues. For example, standard solutions for street furniture and traffic management, that had been applied in other parts of Christchurch, were being proposed for Akaroa.

"It's one size fits all. What they do at Hagley Park they want to do at Akaroa. It makes their job easier. But it's changing the character of Akaroa, making it more urban and formal, and turning it into a generic part of Christchurch."

In addition, the Akaroa Design and Appearance Advisory Committee had become inactive and is in the process of being wound up. The Committee will be replaced by the Akaroa Design Panel, which unlike the Committee, will not be a sub-committee of the Community Board. The Panel will also not be bound by the requirements of the Local Government Official Information and Meetings Act 1987, which provides more flexibility and means that the Panel can be convened for meetings as and when required. Meetings will be confidential between the applicant and members of the Panel.

The use of the existing Committee to provide input into resource consent applications once lodged, was frustrated

due to difficulty in reconciling resource consent timeframes with those in the local government legislation. The new Design Panel, which has been structured to overcome that problem, will hopefully result in more development proposals in Akaroa benefiting from the expert assessment of the Panel. As with the former Committee, the Panel will comprise a mixture of design experts and members of the community with expert local knowledge.

5.4 Tourism

Tourism has become increasingly important to the economy of Banks Peninsula, particularly in Akaroa and surrounding rural communities. Over the past few decades, the town has attracted day-trippers from Christchurch as well as increasing numbers of visitors from other parts of New Zealand and overseas. There are various nature-based cruises around Akaroa, tramping experiences, bed and breakfast accommodation, sea kayaking tours and wildlife cruises/tours (to see Hector's dolphins, penguins, seals and occasionally orcas) operating in the area. Many farming properties are involved in tourism initiatives providing Air BnB's, farmstays, glamping pods and the like.

Marine mammal tourism has become the mainstay of the tourism economy in Akaroa. It was first established in 1985, when Black Cat Cruises started taking people out onto Akaroa harbour to see the small endemic Hector's dolphins. In 1988, the Banks Peninsula marine mammal sanctuary was created to provide protection for the dolphins from set netting, and this helped to support population numbers in the area. Dolphin-based tourism has grown significantly over the years, with other operators entering the market, and swimming with dolphins introduced. Prior to Covid-19, Black Cat Cruises was the largest employer in town. A 2018 economic analysis found that Hector's dolphin tourism directly generated \$19.5 million for the Canterbury economy and supported 416 jobs.⁴⁵

The activity is closely managed by DOC under the Marine Mammals Protection Regulations 1992. There is currently a moratorium on the issue of further commercial tourism permits in order to restrict the impacts of vessel and swimming activity on the dolphin population. In addition, permit holders pay a levy which goes into a trust, and this funds research on the impact of tourism on the dolphins. The levy was disestablished after the Christchurch earthquakes, to reduce the financial burden on the tourism operators, but will shortly be reinstated. The moratorium will be reviewed in 2026, when results from current research are available, as will the potential renewal of existing permits.

Tourism has been generally welcomed in the Akaroa area as a valuable addition to the rural economy. But over the last 10 years, the advent of cruise ships and freedom campers has threatened to overwhelm the small town and its struggling water and wastewater infrastructure.

Residents in Akaroa became concerned about the impact of freedom campers on the town when visitors started

defecating in public places and dumping rubbish. In response, Christchurch City Council introduced a freedom camping by-law in 2014, and this has since been reviewed twice. The most recent by-law, which came into force on December 2018, bans freedom camping in Akaroa apart from in one designated area which provides for 18 self-contained campers.⁴⁶

"There was a proliferation in numbers, the camper vans got bigger, and the way freedom campers behaved got worse. This made it a big issue in Akaroa."

The impact of cruise ships on Akaroa was of much greater magnitude. As a result of the Christchurch earthquakes in 2010 and 2011, the Lyttelton Port was badly damaged and was no longer able to receive cruise ships which diverted to Akaroa Harbour. Akaroa then became a very popular destination for ship visits with around half of cruise ship passengers to New Zealand visiting the scenic natural harbour. During the 2013-14 cruise season, there were 73 ship visits by 19 vessels carrying a total of 127,080 passengers and 59,265 crew. Five years later this had increased by more than a quarter to a total of 93 trips by 21 vessels carrying 145,914 passengers and 64,564 crew.

Usually the cruise ships visited one at a time but there have been multiple occasions when two, three or even four ships have been anchored in the harbour together. Once anchored, the vessels keep their engines running to maintain power. The engines can be powered by heavy fuel oil which on combustion releases sulphur, particulate matter and trace metals into the air.⁴⁷

The ships are typically in the harbour for the daylight hours of a single day, arriving early in the morning and leaving later in the evening.⁴⁸ This means that the up to 2,000 – 4,000 passengers and crew on board have only one day to see the sites of Akaroa, the broader Banks Peninsula area and also Christchurch should they wish to visit the city. This is the antithesis of slow tourism.

"Locals would have a schedule of cruise ship visits on their fridge door and if a ship was coming in they would not go into town."

"On a cruise ship day you can't get into the door of the library, as all the passengers go there for the free wifi, as it's too expensive on the ship."



Due to Akaroa's small resident population of less than 800, during a cruise ship visit day, there could be multiple tourists in town for every one resident. Statistics New Zealand has reported that, during the 2017/18 season, the 210,478 cruise passengers and crew visiting Akaroa spent a total of close to \$35 million while onshore (which may have been in Akaroa or any other destination that they visit during their stopover such as Christchurch) which is an average of \$270 per passenger. This provides considerable additional income to the region. However, it has also changed the nature of retailing, with several quality gift shops and galleries in Akaroa being replaced by souvenir shops.

"People say we have lost the personal New Zealand made quirky and arty gifts and that we have gone to little plastic dolls and tea towels. But that's what sells."

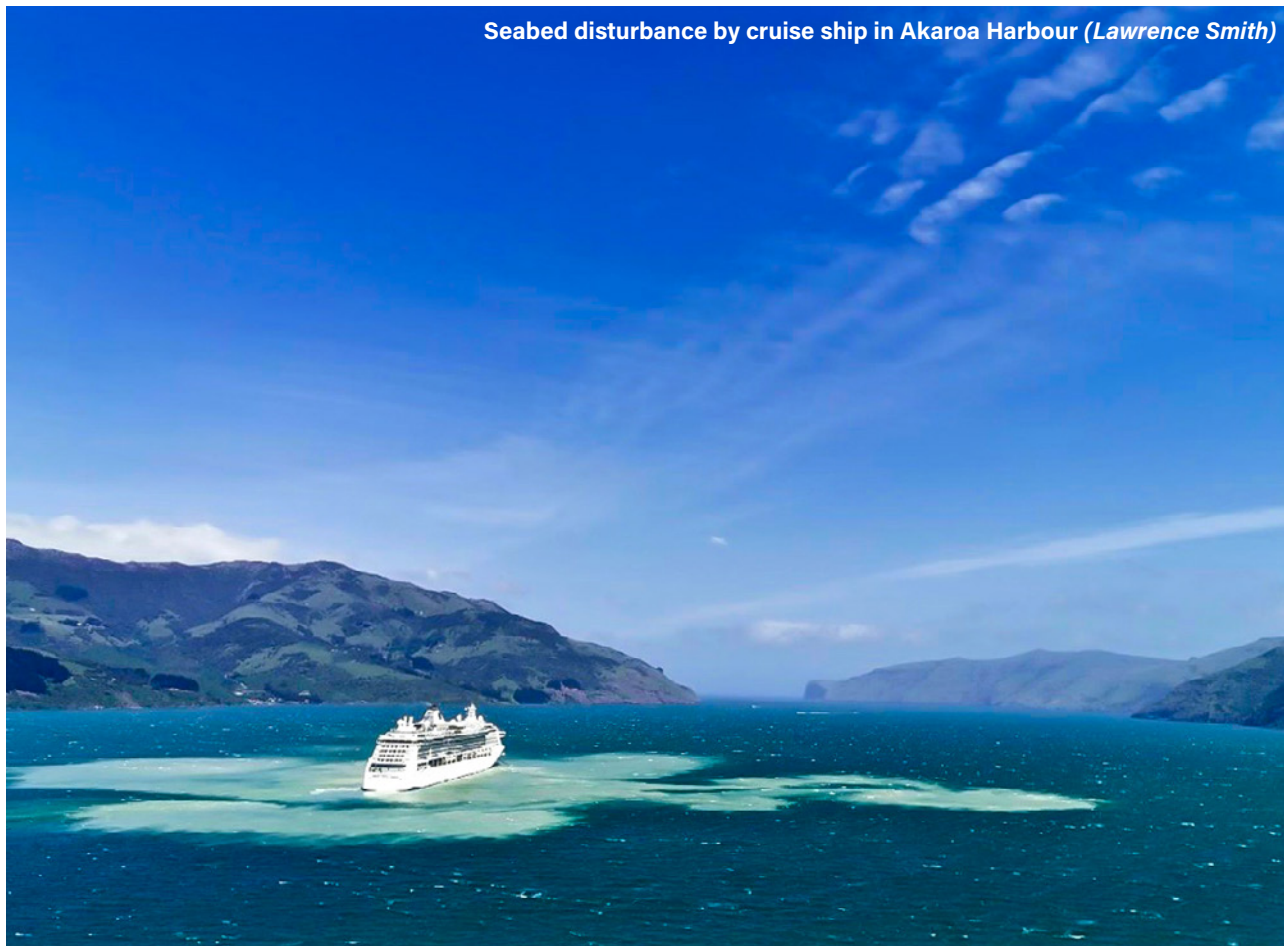
Many cruise passengers would take a bus to Christchurch for the day. As the buses are housed in Christchurch, each bus would drive from Christchurch to Akaroa, wait to pick up passengers, drive them to Christchurch, then later in the day drive the tourists back to Akaroa and then drive back to Christchurch for the night. This means there were four trips along the narrow winding roads between Akaroa and Christchurch for each passenger load. There are reports of up to 40 to 50 tour buses servicing cruise ships in Akaroa during a single day which would represent something like 160 to 200 bus trips along the Akaroa-Christchurch road.⁴⁹

A spotlight on cruise ship tourist buses in Akaroa

One Akaroa resident described the experience of having tour buses in the town. "The buses would come out from Christchurch and wait in town. They would marshal outside the school with their engines running. When the children came to school there would be five or six buses outside the school. Then one or two would go to pick up passengers and then go to Christchurch. They would drive back to Akaroa at the end of the day and then drive back to Christchurch. If I wanted to go to Christchurch I would need to leave before 8.30 to 9 am and not go back after 5pm. There were also the camper vans on the road. It turned the lovely drive into an overtaking nightmare. There was crazy overtaking behaviour in both directions."

There have been some high-profile accidents on the Akaroa-Christchurch route. In February 2017, a bus taking a group of cruise passengers back to Akaroa crashed over the cliffs. The 32 passengers were returning to the *Celebrity Solstice* after spending the day in Christchurch. At about 5pm the bus veered into oncoming traffic and smashed into a car. It then crashed over the bank into dense bush and was stopped by a tree. Twenty passengers were injured with six being taken to hospital. The drivers of the bus and the car were also hospitalised.⁵⁰

Seabed disturbance by cruise ship in Akaroa Harbour (Lawrence Smith)



Cruise ships are charged for the use of the Akaroa wharf to land their passengers via the ship's tenders. For a ship with a capacity of 2001-2500 passengers the basic charge is \$13,160 per visit or around \$5-6 per passenger. The money was paid into Christchurch City Council's general fund rather than being used to specifically support infrastructure in Akaroa. After pressure from the Council, cruise operators agreed to also pay an amenity contribution which increased the wharf charge by just over \$1,000 per visit for a similar sized vessel. This was estimated to provide the Council with an additional \$90,000 per season to cover the costs of such items as the provision of additional portable toilets and cleaning when ships were visiting. Previously such costs had been absorbed by the Council.⁵¹

Due to concerns about the ecological effects of cruise ships on the sensitive environment of Akaroa Harbour, Environment Canterbury commissioned Cawthron Institute to undertake an ecological risk assessment, which was released in October 2019. In particular, there was concern that the use of thrusters for manoeuvring the ships was resulting in plumes of sediment being driven up from the seabed. Plumes up to 1km from the vessel had been observed.⁵² There were also concerns about the impacts of the vessels on wildlife including the threatened and nationally vulnerable Hector's dolphin and multiple

threatened seabirds. The ecological effects assessment concluded that the anchoring and propulsion of cruise ships have the potential for significant or unacceptable adverse ecological effects on the very high ecological values of Akaroa harbour.⁵³

On 24 June 2020, Environment Canterbury issued guidance on anchoring cruise ships in Akaroa Harbour. The guidance is operative from 1 November 2021 and no cruise ship will be allowed to anchor in the harbour unless it can demonstrate compliance with its conditions.⁵⁴ The guidance states that no vessel greater than 50,000 tonnes displacement or 260 metres in length can anchor in the harbour without consent or scientific evidence of no adverse effects. In addition, no more than two vessels greater than 100m in length can anchor in the harbour at any one time. The guidance also sets a cap on the number of allowable ship visits during a 12-month period.⁵⁵ These controls will likely exclude large ships from anchoring in the harbour and reduce overall ship visit numbers if and when cruise ships are permitted to return to New Zealand waters. In addition, Lyttelton harbour's new \$56 million cruise berth will be able to accommodate the larger vessels.⁵⁶ This will also encourage larger ships to visit Lyttelton rather than Akaroa thereby helping to avoid the situation occurring prior to the pandemic.



"There is no oversight of the tourism industry."

"If we got a smaller number of higher value, more discerning tourists to the Peninsula with an interest in the environment, culture and heritage, you would probably get the same if not better economic outcomes, without the more negative effects on the environment and community."

Tourism in Akaroa and the broader Banks Peninsula has been largely unmanaged. The Covid-19 pandemic, and diversion of future large cruise ships away from Akaroa, provides an opportunity for pause and rethink about the tourism model which might best fit the Banks Peninsula landscapes and communities. There appears to be considerable opportunity to strengthen the Peninsula's image as a sustainable tourism hub, where its remarkable geology, biota, landscapes, seascapes and cultural heritage is matched by community efforts to heal the land and farm sustainably.

"If we could market all our farms and all of our tourism operations as carbon neutral and cover the money involved by doing things like planting trees, reducing sedimentation and looking after the marine environment, it would be a nice story. We could say 'come to the Banks Peninsula and see how life could really be like'. Imagine if national and international tourists came to the Banks Peninsula as it had a reputation as a place where people live and work in harmony with the environment."



5.5 Marine farming

Marine farming remains a small but viable industry on Banks Peninsula, with a focus on greenshell mussels, king salmon, pāua and marine algae.⁵⁷ Interest in new species such as seaweed, including kelp, has the potential to put demand on limited space in the harbours of the Peninsula, and increase the potential for negative impacts on landscape and seascape values, and natural character.

A spotlight on marine farming in Pigeon Bay

Pigeon Bay Aquaculture Limited v Canterbury Regional Council was the first case about marine farming in the bays of Banks Peninsula to be argued before the Environment Court. The 1999 decision provides key jurisprudence on landscape interpretation and its findings remain relevant today.

The case concerned the establishment of two marine farms in Pigeon Bay and the amenity effects of the proposal. The Court found that "Banks Peninsula as a whole may well be an outstanding natural feature and landscape".⁵⁸ In regard to Pigeon Bay, which was predominately pastoral, it was described as an area of high natural character in a working landscape sense.

The Court ultimately approved the marine farms, on the basis that they would not have significant adverse effects on the amenity values of the coastal marine area. However, in making its assessment, the Court acknowledged the need to distinguish between natural character (such as the naturalness of the landform) and the assessment of aesthetic values.

The Court, led by Judge Jackson, utilised the factors in the Canterbury Regional Landscape Study 1993⁵⁹ to provide a list of seven factors to be taken into account when assessing the significance of landscape. These are now colloquially known as the 'Pigeon Bay' criteria:

- The natural science factors – the geological, topographical, ecological and dynamic components of landscape
- Its aesthetic values including memorability and naturalness
- Its expressiveness (legibility): how obviously the landscape demonstrates the formative processes leading to it
- Transient values: occasional presence of wildlife; or its values at certain times of the day or of the year
- Whether the values are shared and recognised
- Its value to tangata whenua
- Its historical associations.

The factors illustrate the mix of physical, perceptual and associative factors that make up landscape values, including the richness of human response to landscapes. They also serve to highlight that landscapes are an important component of the environment and that they exist in the context of other activities and cannot readily be separated from them. While these factors have been amended slightly in *Wakatipu Environmental Society Inc v Queenstown Lakes District Council*,⁶⁰ they remain pivotal in landscape jurisprudence and practice today.

In 2005, the Environment Court approved a proposal to establish a marine farm in Port Levy. The court accepted that both headlands of Port Levy constitute an ONL and the coastline was high in natural character. But it concluded that the proposal could be adequately absorbed into the landscape to avoid adverse effects.⁶¹

Since 2004, a total of 25 marine farms consents have been applied for, of which three are still being processed. Two applications were withdrawn or returned, and 10 terminated or replaced. Seven of the consents were issued and are active, and three consents were issued but remain inactive.⁶² This indicates that there is continuing demand for marine farming in the marine area surrounding Banks Peninsula.

Key messages

- Farming systems on Banks Peninsula generally operate within the natural constraints of the landscape and are largely low impact. This situation is unlikely to change in the foreseeable future.
- Exotic plantation forestry, if undertaken at scale, is a significant threat to landscape and biodiversity values on the Peninsula. Lax regulation and financial incentives driven by government policy are favouring exotic plantations over indigenous regenerating forest. Water supply is also a key issue on Banks Peninsula, and conversion to plantation forestry will put further stress on the availability of water.
- There has been only modest pressure for urban development on Banks Peninsula with several potential developments declined on the basis of landscape impacts. However, rural subdivision is still occurring with landowners choosing to carve off lifestyle blocks and place an open space covenant over the balance.
- Tourism pre-Covid-19 overwhelmed the small Akaroa township and was largely unmanaged. The current pause in visitors provides the opportunity to reimagine the model underpinning tourism on the Peninsula.
- There is continuing demand for marine farming around the Peninsula.



Looking down into the upper Akaroa Harbour

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- 6 Boffa Miskell, 2014, 11
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- 16 Resource Consent Application to the Christchurch City Council: RS and LS Holloway Farming Partnership: resource consent for forestry development at 392 Dalglishs Road, Le Bons Bay, June 2020, 68
- 17 RMA, Section 104(2)
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- 40 Summit Road (Canterbury) Protection Act 2001, Sections 7-8 and 12
- 41 *Akaroa Civic Trust v Christchurch City Council* [2010] NZEnvC 110.
- 42 *Manger v Banks Peninsula District Council* [2004] NZEnvC 114.
- 43 *Pacific Investment Trust v Banks Peninsula District Council* [2000] NZEnvC 86.
- 44 Akaroa Civic Trust, 2019, 5-6
- 45 Yeoman et al, 2018,
- 46 Law, 2018
- 47 Johnston, 2019, 17
- 48 Johnston, 2019, 8
- 49 Letter from the Akaroa Civic Trust and Akaroa Ratepayers and Residents Association to Prime Minister Jacinda Adern, 9 June 2020
- 50 Mitchell and Sherwood, 2017
- 51 Christchurch City Council, 2019
- 52 Johnston, 2019, 45
- 53 Johnston, 2019, 61
- 54 Environment Canterbury, 2020a
- 55 Environment Canterbury, 2020b
- 56 Hayward, 2019
- 57 <https://www.mbie.govt.nz/immigration-and-tourism/tourism/tourism-funding/international-visitor-conservation-and-tourism-levy/how-the-ivl-works/>
- 58 *Pigeon Bay Aquaculture Ltd & Ors v Canterbury Regional Council* C32/99, Paragraph 55
- 59 Boffa Miskell and Lucas Associates, 1993
- 60 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59
- 61 *Southern Seas Marine Farms Ltd v Canterbury Regional Council* [2005] NZEnvC 119
- 62 Local Government Official Information and Meetings Act Request 1927C, 12 May 2020.
- 63 Resource Consent Application to the Christchurch City Council: RS and LS Holloway Farming Partnership: resource consent for forestry development at 392 Dalglishs Road, Le Bons Bay, June 2020, 4

6. STATUTORY RESPONSES

6.1 Legislative framework

The RMA is currently the primary legislative tool for the management of New Zealand's environment and resources (including landscapes). The purpose of the RMA is to "promote the sustainable management of natural and physical resources". A range of policy and planning documents are prepared under the RMA to implement this purpose including national policy statements (including the NZCPS), regional policy statements, regional plans and district plans.

In achieving the purpose of the RMA, all persons exercising functions and powers under it are required to recognise and provide for various matters of national importance.¹ These matters are sometimes referred to as section 6 matters. The protection of ONLs and outstanding natural features (ONF) from inappropriate subdivision, use,

and development is a key section 6 landscape matter, as are the following inter-related matters:

- Preservation of the natural character of the coastal environment, wetlands, lakes and rivers and their margins from inappropriate subdivision use and development (section 6(a))
- Provision for the relationship of Māori and their cultures and traditions ancestral with their lands, water, sites, waahi tapu, and other taonga, which is important in the context of Māori cultural landscapes (section 6(e))
- Protection of historic heritage from inappropriate subdivision use and development which is important in the context of historic cultural and heritage landscapes (section 6(f))



Akaroa Harbour

A spotlight on heritage landscapes

Section 6(f) of the RMA requires the protection of historic heritage from inappropriate development and use. To date, this provision has largely been used to protect individual historic sites or buildings, whilst ignoring the wider landscape in which they are situated.² The concept of 'heritage landscapes' has been argued in the Environment Court on a number of occasions – with responses ranging from acceptance of the concept to full rejection.³ As there is no 'outstanding' or 'significant' qualifier in section 6(f), the provision is to be read in line with the introductory words of section 6; that it is a 'matter of national importance'.⁴ This means that protection is only likely to be extended to surrounding land of historic sites or places (eg the wider heritage landscape) if there is sufficient intensity of heritage fabric woven into the landscape to be of national importance.⁵

While some cases have accepted the concept of a heritage landscape, few have actually found one to exist. A lack of probative evidence is one reason for this. But, perhaps most peculiarly, so too is a resistance of the Courts to accept the concept. This has been in spite of strong evidence demonstrating the presence of a historical landscape in some cases. As a result, heritage landscapes have largely failed to be protected under the RMA.

Underpinning this failure are concerns about double-counting; that heritage landscapes are already protected by other sections of the RMA. For example, Māori values are already recognised and protected under sections 6(e), 7(a) and 8 of the RMA. Heritage values also inevitably overlap with landscape and often heritage values will contribute to the conclusion that a landscape is outstanding under section 6(b). However, as landscapes protected under this section are required to be 'natural', this prevents protection on the sole basis of the (often human constructed) heritage values present. This led to the concept of heritage landscapes being rejected in

one Environment Court proceeding on the basis that it would introduce a new concept not recognised explicitly by the RMA. It was thought that this would add to the already complex web of the Act and make matters more confusing.⁶

This does not however shut the door on protecting heritage landscapes, and as said by retired Environment Court Judge Kenderdine in her address to a New Zealand Institute of Landscape Architects conference in 2005, the inclusive nature of section 6(f) overall "provides considerable scope for asserting an area, place, feature, as a 'landscape' contributes to historic heritage in planning and resource consent processes."⁷

The Christchurch District Plan includes a number of heritage areas, which are areas of inter-related historic places, buildings, structures and/or sites that make a significant contribution towards an understanding and appreciation of Christchurch District's history and cultures. This includes the entirety of Akaroa township, as illustrated in Figure 6.1. However, there is no reference to cultural heritage landscapes. This is despite the 2007 Boffa Miskell study, described further below, identifying cultural heritage landscapes within the rural areas of the Peninsula.

The rules relating to heritage areas are weaker than those afforded to ONLs and require management only, not protection. Built form guidelines exist for the commercial areas of Akaroa⁸ as well as for defined character areas within Akaroa's residential zone.⁹ However, these do not operate on a landscape-scale basis; rather they look at individual building design within the context of the existing built environment.

Further work needs to be undertaken by Christchurch City Council to recognise the importance of heritage landscapes in the Banks Peninsula context, given its history and importance to Canterbury's identity, and to provide for their management across integrated urban and rural areas.



Akaroa waterfront

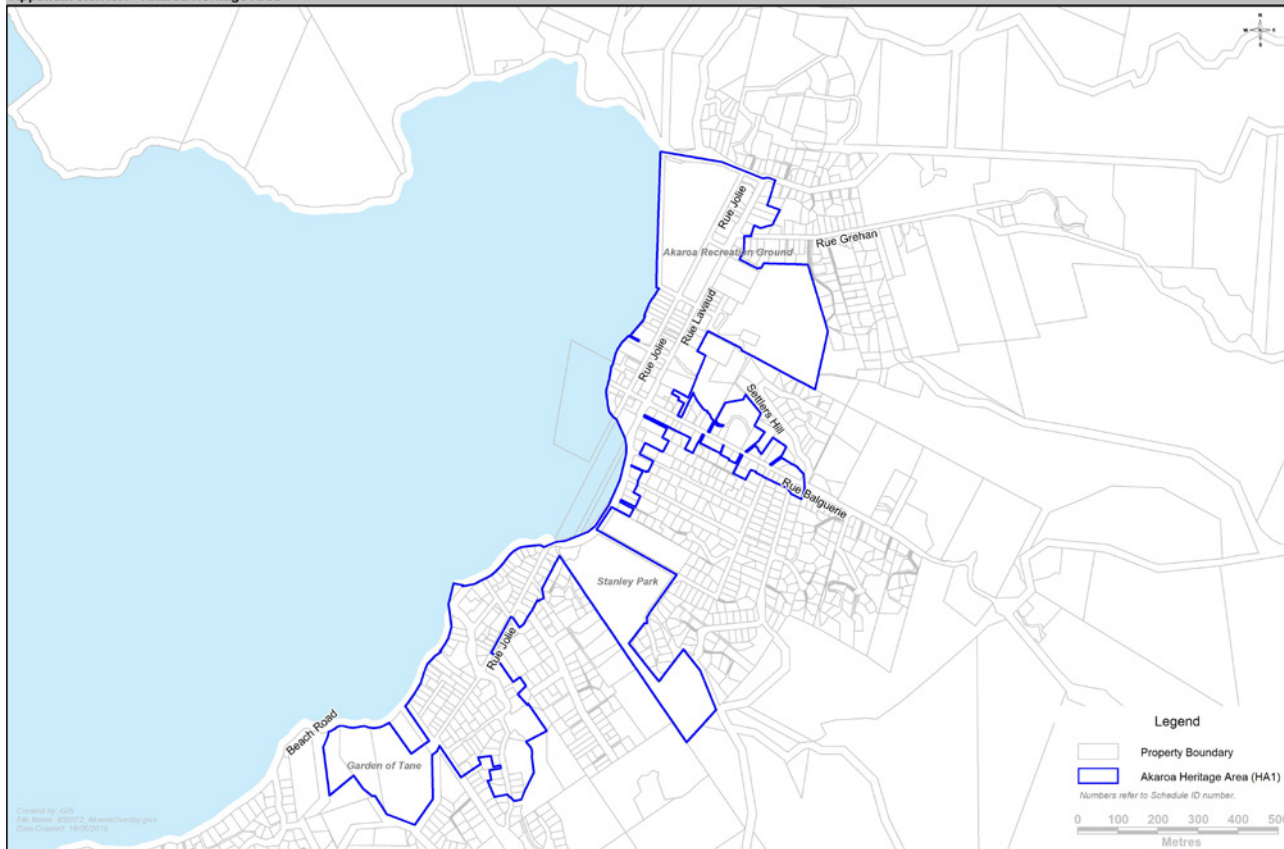


Figure 6.1 Map showing Akaroa Heritage Area (Source: Christchurch District Plan)

Section 6 does not prohibit development within important landscapes. Instead these matters are protected against 'inappropriate' subdivision, use and development. A landscape may be able to absorb some change without losing the values that make it outstanding or otherwise valued. For this reason, it is important that a landscape's values are identified and described. Whether development is appropriate or inappropriate depends on the facts and circumstances of each case (eg the landscape values that make the area outstanding or otherwise valued and whether the impacts of the proposed development detract from those values).

While section 6 matters are of national importance, the management of landscapes is not limited to these. The RMA also recognises the need to manage other landscapes. Section 7 of the RMA requires that, in achieving the Act's purpose, all persons exercising functions and powers under it, are required to "have particular regard to" the maintenance and enhancement of "amenity values" (s7(c)) and the "quality of the environment" (s7(f)).¹⁰ The term "amenity values" is defined in section 2 of the RMA as "those natural or physical qualities and characteristics of an area that contribute to people's appreciation of its pleasantness, aesthetic coherence, and cultural and recreational attributes".¹¹ The quality of the environment is not defined in the RMA. The upshot is that amenity landscapes can be significant, often more so at a local level, and the level of protection or management required for them is typically less than for other areas that fall under section 6. It is worth noting that

the wording of section 7 does not preclude 'protection' as an approach.

In order to provide for the protection of both ONLs and amenity values, as well as other resource management matters, a cascade of planning documents is created under the RMA, all of which must give effect to the provisions of the RMA (see Figure 6.2). These give substance to the RMA's purpose by identifying objectives, policies, rules and methods relating to land, marine and resource use activities. The hierarchy is intended to move from the more general to the more specific both in content and locality, with each level required to 'give effect to' (meaning implement) the level above it. These documents will be addressed in turn below.¹²



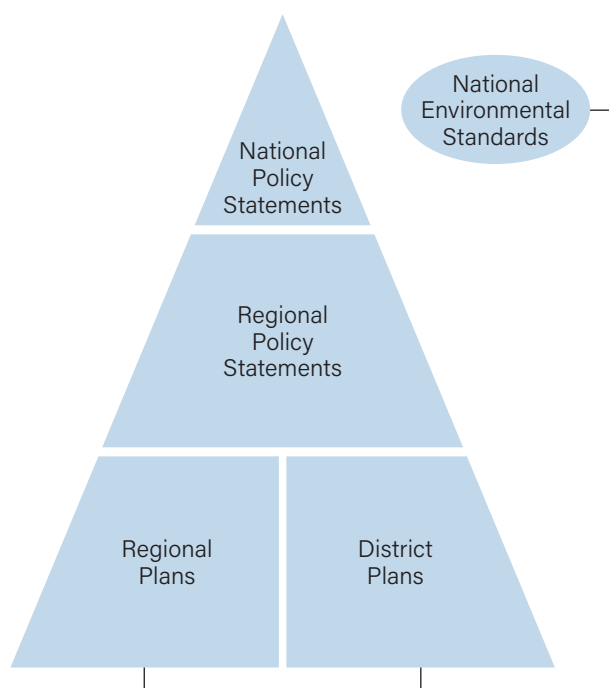


Figure 6.2 *Hierarchy of RMA plans*

Although the protection of ONLs has been identified as a matter of national significance in the RMA there remains no national policy statement on landscape. However, the NZCPS which came into effect on 3 December 2010, applies to areas that are within the coastal environment. These are mapped in the Christchurch District Plan as being within the Coastal Environment Overlay.

The NZCPS requires that adverse effects of activities on ONLs or areas with outstanding natural character must be avoided. In all other areas within the coastal environment, significant adverse effects on natural character and

landscapes must be avoided, and all other adverse effects avoided, remedied or mitigated.¹³

This provides a graduated scheme of protection and preservation of areas of natural character and landscape within the coastal environment, with a clear bottom line for areas identified as being outstanding. A similar bottom line is provided to protect threatened indigenous biodiversity and threatened or naturally rare ecosystems and vegetation types within the coastal environment.¹⁴

A spotlight on differences between the 1994 and 2010 NZCPS

The NZCPS 2010 took a significant change in approach to the previous 1994 document. Conflicting policies in the NZCPS 1994, that sought to protect the environment while at the same time avoiding, remedying or mitigating adverse effects of activities where practicable, were replaced by strong bottom-line directives. The new policies require avoidance of adverse effects on matters of national importance such as outstanding natural character, areas of significant vegetation and habitat, and ONLs and ONFs; and to avoid significant adverse effects on second tier matters such as amenity landscapes, indigenous vegetation that isn't significant, and natural character in the coastal environment.¹⁵ The NZCPS 2010 also reversed the approach of identifying areas that are 'appropriate' for subdivision, use and development to identifying areas that are 'inappropriate' for development.¹⁶ It introduced the concept of identifying outstanding natural character,¹⁷ which has also changed the way assessments have been undertaken at the district level.



Lyttelton Harbour

6.2 Landscape assessment of Banks Peninsula

Over the last three decades, there have been a number of studies focused on the identification of significant landscapes on Banks Peninsula. Somewhat surprisingly, these have come to very different conclusions.

In 1993, the entire Banks Peninsula was identified as a regionally outstanding landscape in a study carried out by Boffa Miskell Limited and Lucas Associates (see Figure 6.3). However, it was neither formally adopted by Environment Canterbury nor incorporated into the Canterbury RPS. Subsequent planning documents and processes have eroded this designation, resulting in the minimal ONL overlay that is currently found in the district plan. This process is described further below.

The Banks Peninsula District Plan was notified in early 1997 and identified around half (some 50,000 of approximately 96,000ha) of the rural land in Banks Peninsula as either landscape protection areas (with high or moderate sensitivity) or coastal protection areas. These areas were subject to stricter controls on buildings and production forestry. No landscape assessment was carried out to directly inform the plan. Instead, the findings of a visual assessment undertaken in 1991,¹⁸ after some reanalysis, were used as a basis for identifying important landscapes on the district plan maps. The proposed plan did not address cultural landscapes. However, it did provide for

the protection of sites and features of significance to mana whenua through the identification of 'silent file' areas.

The proposed plan provisions for coastal and landscape protection areas were heavily criticised, particularly by the farming community mobilised under the umbrella of the North Canterbury Branch of Federated Farmers. Federated Farmers threatened High Court action if the Council continued processing these sections of the proposed plan. In response, the Council decided not to hear submissions on the rural chapter of the plan, but instead it established a Rural Task Force to investigate ways of resolving the conflict. The Task Force recommended adopting interim provisions for landscape protection while a further and comprehensive landscape study was undertaken.

In response to this advice, the Council notified Variation 2 in 2002, which significantly reduced the areas of protection.¹⁹ Under Variation 2, only 30,000ha of landscape and coastal protection areas were mapped. This resulted in a number of appeals; however, these were put on hold pending the finalisation of the landscape study.²⁰

The 2007 Boffa Miskell landscape study has been described by the Environment Court as the "most comprehensive approach to a district landscape undertaken in New Zealand to date".²¹ It identified landscape character areas, and mapped and described them using existing physical, biological, cultural and land use data and field work observation. The study included three phases: character descriptions, landscape values and management mechanisms. It sought to maximise the

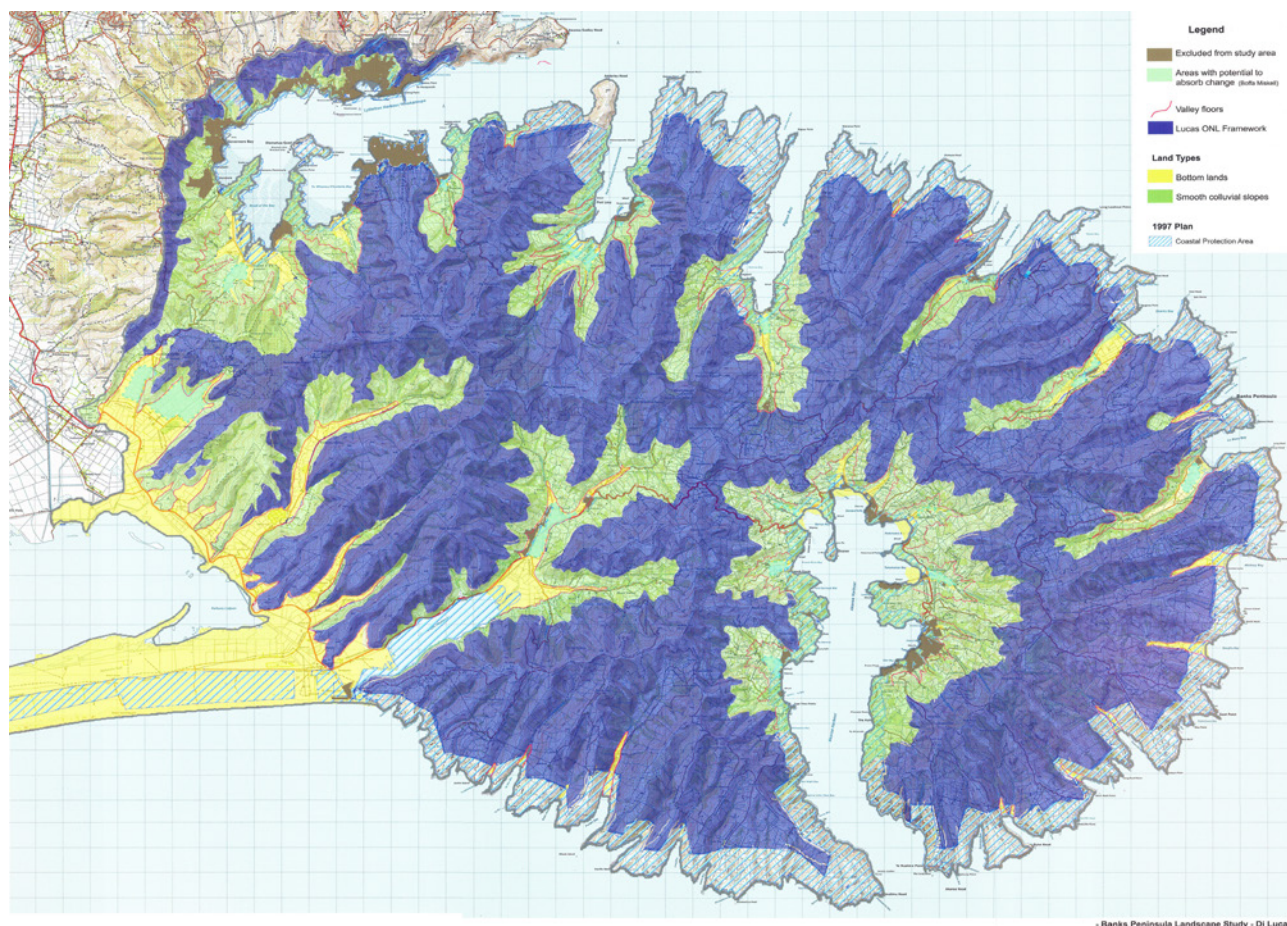


Figure 6.3 Extent of ONL mapped by the landscape architects for the Regional Council in response to Variation 2 (Source: Di Lucas)

objective information which could be incorporated into the layers, before assessing values, and integrating the layers into the various landscape protection categories.²²

Although comprehensive, the study made a fundamental error. In mapping the various landscape values (the *Pigeon Bay* criteria described above), it failed to adopt a landscape-scale approach and instead focused on individual landscape units. The landscape values within each unit were mapped and these were amalgamated to create a composite overlay. In adopting this methodology, the study failed to recognise the significance of the larger landscape as an integral entity.

The Banks Peninsula landscape has no equal in New Zealand. It is a highly legible landscape. It has a distinctive landform pattern of overlapping volcanoes with their radial spurs splaying out from the crater rims to the sea. By distilling the area down to landscape units, the significance of the radial spur-headland pattern, which is the distinctive essence of the Peninsula, was lost. Similarly, by identifying specific natural science and aesthetic values, and layering them to create the overlay, the study failed to recognise the inter-related nature of these elements. The study mapped parts of a landscape but did not fully integrate them or recognise the landscape as a whole.

The Environment Court has since rejected this landscape unit approach. "At a district level smaller landscapes may nest within a larger landscape. But there comes a point where that no longer applies. Care needs to be taken by local authorities not to divide a landscape into its units and then to treat units as landscapes."²³

The Boffa Miskell study identified and mapped four different categories of landscape: ONLs, coastal natural character landscapes, visual amenity landscapes and heritage landscapes. The protected landscape and coastal areas of the Boffa Miskell study encompassed an area of some 23,000ha, limited mostly to prominent upper ridgelines and the coastal edge.²⁴ The balance of the Peninsula was covered in a 'visual amenity' zone with a reliance on assessment criteria rather than the use of non-complying activity status.

Following an Environment Court process, described in the spotlight below, the Court adopted the findings in the Boffa Miskell study, and following the lengthy district plan review process, the Banks Peninsula District Plan became operative in 2007.



A spotlight on the Environment Court decision on Variation 2

The central issue for the Court's determination on Variation 2 was which delineation would better achieve the objectives and policies of the Proposed Plan, Canterbury RPS and NZCPS.²⁵ The degree of ONL protection sought by the various parties differed significantly. The Christchurch City Council, in support of the Boffa Miskell study, sought protection of 23,000ha. This was a significant reduction from the Council's Variation 2 as notified which protected 30,000ha and the District Council's originally notified plan that sought protection of 50,000ha. The Regional Council sought the protection of a larger area comprising 75,000ha.²⁶ It argued that the rural amenity zone would not adequately protect the landscape, and that these areas should more appropriately be categorised as ONL.

There were seven landscape witnesses in the case, with "as many opinions as to what represents outstanding natural landscape and coastal natural character landscapes as there are experts".²⁷ The Court accepted that the Banks Peninsula landscape was an ONL at a regional scale, given the geomorphology of the area, but considered the question it had to address was whether it constituted an ONL at a district level.²⁸ In the end, the Court accepted the ONL areas delineated by Boffa Miskell. Important to that finding (as opposed to accepting a more comprehensive coverage of ONL and coastal protection areas) was the presence of human modification on the Banks Peninsula, specifically prominent houses and forestry. As the Court stated:²⁹

One of the unfortunate consequences of being a member of the Court is that repeated analysis of landscapes leads to a critical evaluation of landscapes that many take for granted. For the general population there appears to be a conceit of the mind which occurs when it views pleasant landscapes. This appears to subtract from the view the incongruent elements such as large square forestry blocks up to and including ridges, prominent buildings situated on high points, roads, telephone poles and trees and other clear indications of extensive modification with the result that the individual reads the environment as natural.

In this manner, the scale at which the Court 'read' the landscape was important to its (and Boffa Miskell's) conclusions. Other landscape experts 'reading' the landscape within a wider (regional) context did not ascribe the same significance to the cultural modifications of the land (in terms of their detracting from its naturalness). The differences in approach dictated where the landscape and coastal protection lines on the maps went and, ultimately, what

landscapes were afforded protection under section 6(b) and what landscapes were not. Landscapes that are afforded protection can evolve in a significantly different manner to those that are not.

Subsequent to the Court's decision on Variation 2, there have been a number of decisions that discuss the spatial scale at which to assess ONLs. The Court of Appeal in *Man O'War* has confirmed that ONLs should be those that are outstanding in terms of the region's natural environment.³⁰ *Man O'War* was determined in the context of a regional policy statement. However, there is similar sentiment with respect to identifying ONLs and ONFs for the purpose of district plans. As the Court recently stated; while a local authority may be confined to its knowledge of its district, the Environment Court need not be, as it has built up its knowledge of New Zealand's landscapes.³¹ In light of these cases, it is likely that a more extensive ONL overlay would have been afforded to Banks Peninsula if the Court had adopted a broader scale of assessment.

Two subsequent studies were conducted as part of the Christchurch City Plan review, both resulting in minor variations to the overlay. As the 2007 Boffa Miskell study did not include the Port Hills, a supplementary Ōtautahi Christchurch Landscape Study was prepared by Boffa Miskell to bridge the gap. The report found that, at a district level, the upper slopes and unbuilt ridgelines were generally outstanding and the lower slopes significant.³²

In 2015, the City Council also prepared an addendum to the 2007 Banks Peninsula Landscape Study to address areas that were initially out of scope, to address changes to the landscape since 2007, and to provide further information in response to the NZCPS 2010. The addendum resulted in the reclassification of areas of high natural character to high, very high or outstanding. Many parts of the coastline, islands, crater rims, upper slopes and ridgelines were found to be ONLs or ONFs and much of the coastline was found to have high natural character. The balance of rural zoned land on Banks Peninsula was found to be a rural amenity landscape.

6.3 Protection of natural landscapes in regional and district planning documents

Canterbury RPS and Coastal Environment Plan

In 2010, the Canterbury Regional Council embarked on the Canterbury Regional Landscape Study Review³³ in order to update the 1993 Study. The Study identifies and describes ONL values at a regional scale and these have been included in an appendix to the Canterbury RPS. The Canterbury RPS identifies the Banks Peninsula and Port Hills as an ONL with the following description:

Banks Peninsula consists of the eroded remnants of two overlapping and long extinct volcanoes – the

Lyttelton Volcano and the Akaroa Volcano. The Peninsula is a landscape which is highly expressive of its geological formation. This is a landscape of harbours, rocky cliffs and headlands. The Port Hills provide a significant backdrop to the City of Christchurch when viewed from areas of the Canterbury Plains. The rocky skyline is important as a contrast to the city and its lowland surrounds.³⁴

The Canterbury RPS also acknowledges that the Banks Peninsula landscape occurs at a very large geographic scale, and the maintenance of working landscapes in it, is appropriate to maintain social and economic wellbeing.³⁵

Given the timing of the 2010 NZCPS release, the Canterbury RPS was not drafted to give effect to its provisions. Environment Canterbury undertook to prepare a collaborative region-wide strategy to implement the NZCPS within five years but this has not yet been undertaken. As a result, the Canterbury RPS still does not give effect to the NZCPS and this remains a significant gap.

The Canterbury Regional Coastal Environment Plan 2005 (Coastal Plan) also does not give effect to the NZCPS. It was notified in 1994 and made operative in November 2005. Review of the Coastal Plan is now underway, with notification of a new Plan expected in 2023/24. Further work will likely be required to ensure that the landscape assessment that has been undertaken on the landward side of the coastal marine area boundary, is well integrated with provisions for the seascapes of Banks Peninsula, including the marine area. This includes the two important harbours at Akaroa and Lyttelton, as well as the multitude of bays surrounding the Peninsula.

Christchurch District Plan

As already indicated, in 2006, the Banks Peninsula District Council was disestablished and the Peninsula came under the control of Christchurch City Council. Following amalgamation and the earthquakes of 2010–2011, Christchurch City Council came under pressure to review its district plan with a view to quickly enabling rebuild activities to take place. This review was fast-tracked, with a board of inquiry type process, and as such was undertaken over a short timeframe. The review involved the amalgamation of the Banks Peninsula District Plan and the Christchurch City Plan. However, as a result of the speed of the process, with only three years to produce a new plan, little reworking of the provisions of the Banks Peninsula District Plan was undertaken. The provisions were largely carried over unchanged, including the extent of the ONL delineation.

The proposed Plan was notified in stages, with hearings held during 2015 to 2017. The full plan became operative (except for some coastal environment hazard provisions which were excluded from the review) in December 2017.

The Plan contains a suite of objectives, policies and rules that manage the landscapes of Banks Peninsula. The Plan identifies and seeks to protect ONLs and outstanding natural features; maintain significant features and Rural Amenity Landscapes; and preserve the natural character

of the coastal environment, wetlands, lakes, rivers and their margins.³⁶ It describes the qualities of the ONLs and ONFs for Banks Peninsula as:

- A. Distinctive volcanic landforms with highly legible crater rims, prominent ridgelines and peaks extending to a series of gullies, spurs and lower slopes, and continuity of rock outcrops;
- B. The presence of important areas of indigenous vegetation;
- C. Important landscapes and features to Ngāi Tahu, especially those prominent peaks and passes, streams and vegetation that relate to key tribal origin, migration and settlement traditions, including mahinga kai;
- D. A predominance of natural features, processes and patterns and farming with existing development generally well-integrated into the landscape; and
- E. Within the coastal environment, the presence of generally high legible landforms, visually impressive cliffs, islands and caves.

The Coastal Environment chapter of the District Plan includes an objective to maintain and protect the values of the coastal environment, including natural features and landscapes.³⁷ The plan includes policies that seek to ensure that subdivision, use and development is of a scale, and located, to maintain and protect the values of the coastal environment, including natural features

and landscapes. However, the District Plan policy about managing adverse effects within the coastal environment does not reflect the clear NZCPS directive to avoid adverse effects on section 6 matters (including ONLs) and/or avoid significant adverse effects on matters of local importance (Rural Amenity Landscapes and significant features).³⁸

As shown in Figure 6.4 below, Banks Peninsula is largely classified as a Rural Amenity Landscape, with ONLs mainly confined to the ridgetops and coastal edges.

In line with section 6 of the RMA, the district plan includes objectives to protect identified ONFs and ONLs. The associated policy is based around recognising the qualities of the areas and protecting those qualities from inappropriate subdivision, use and development (see Figure 6.5).

High risk activities in these areas such as (most) residential units, quarrying and plantation forestry are listed as non-complying activities, signalling that they are inappropriate and unlikely to be approved within an ONL or ONF. Farm buildings with a footprint $\leq 100\text{m}^2$ are permitted but, unless they comply with the relevant standards, most other buildings are discretionary activities. Built form standards apply to permitted buildings.³⁹ The standards say that buildings shall be located at an elevation of at least 20 vertical metres below that of any adjoining identified important ridgeline and have a reflectivity no greater than 40 per cent.⁴⁰ To manage scarring on the landscape, new access tracks

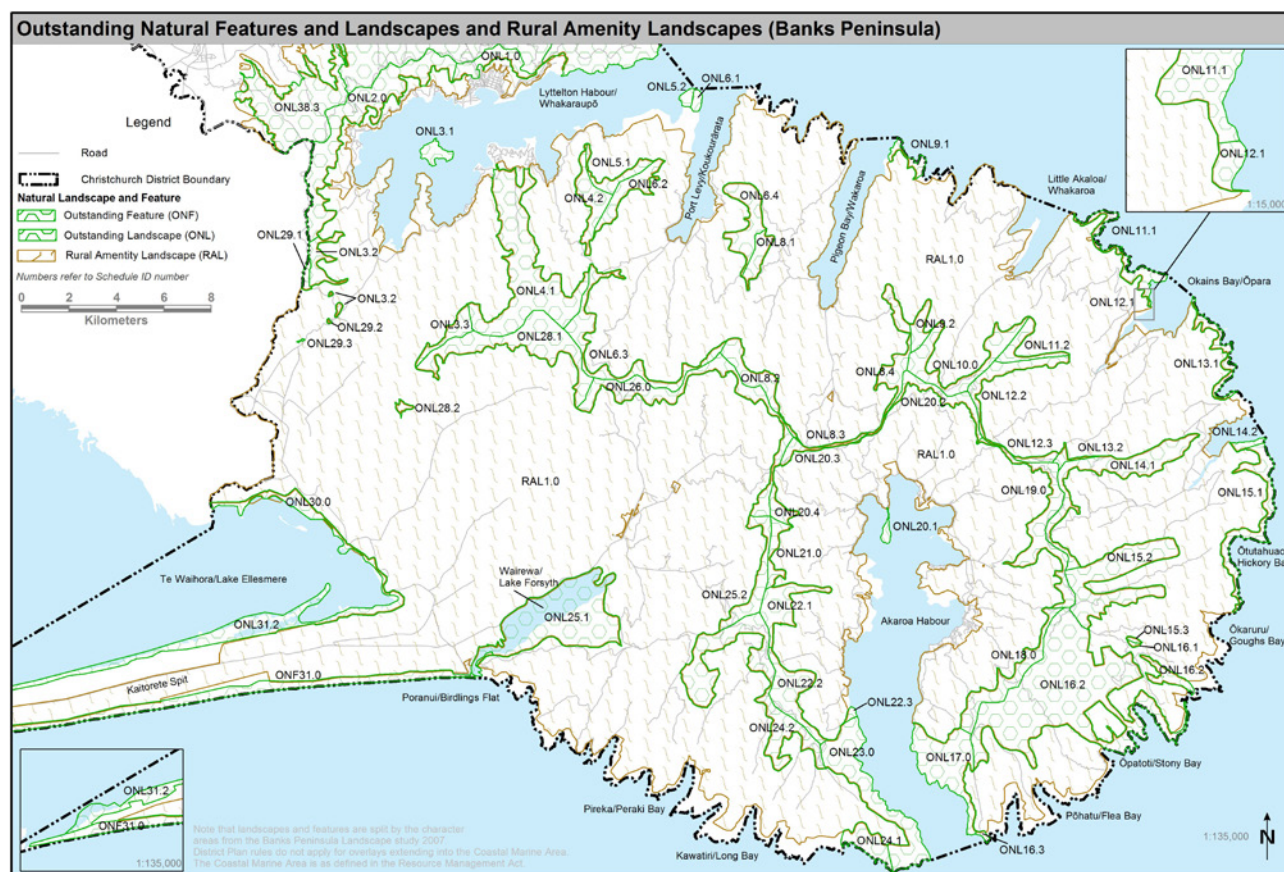


Figure 6.4 Banks Peninsula's identified ONLs, ONFs and Rural Amenity Landscapes (Source: Christchurch District Plan, Appendix 9.2.9.2.6)

with a width greater than 5m are restricted discretionary activities and new roads are discretionary activities.⁴¹

In terms of earthworks, more broadly, the chapter 8 earthworks rules trigger matters of discretion for managing effects on landscape. These include the effects that result from earthworks in terms of visual amenity, landscape context and character, views, outlook, overlooking and privacy. In addition, as relevant, the specific ONL and ONF matters of discretion apply to earthworks.

There are no specific rules in the coastal environment chapter but, relevantly, the matters of discretion refer to the location, scale and intensity of the activity and/or buildings and the extent to which the proposal will adversely affect the values of the coastal environment, including natural character, natural landscapes and features, visual qualities and amenity values.⁴²

Most of Banks Peninsula is identified as a Rural Amenity Landscape outside the settlements and those areas identified as ONLs or ONFs. The district plan includes objectives to maintain identified Rural Amenity Landscapes and significant natural features. The qualities of Rural Amenity Landscapes for Banks Peninsula are described as follows:

- A. Predominantly farming, rural working landscape;
- B. Distinctive volcanic landforms with highly legible crater rims, prominent ridgelines and peaks

extending to a series of gullies, spurs and lower slopes, and continuity of rock outcrops;

- C. The presence of important areas of indigenous vegetation;
- D. Important landscapes and features to Ngāi Tahu, especially those prominent peaks and passes, streams and vegetation that relate to key tribal origin, migration and settlement traditions, including mahinga kai;
- F. A predominance of natural features, processes and patterns and farming with existing development generally well-integrated into the landscape; and
- G. Within the coastal environment, the presence of generally high legible landforms, visually impressive cliffs, islands and caves.
- H. Port Hills:
 - a. An important visual backdrop to Christchurch District.
 - b. A major recreation resource for the city's residents and visitors.⁴³

For Rural Amenity Landscapes with an underlying Rural Banks Peninsula zoning, the rural amenity landscape rules (in the rural chapter) apply. The rural chapter of the District Plan includes a specific objective for Banks



ONL policies	Rural Amenity Landscapes policies	Significant Natural Features policies
Avoiding use and development that breaks the skyline, including the crater rim, ridgelines on Banks Peninsula and the radial spurs of the Port Hills	Avoiding use and development that breaks the skyline, including the crater rim, ridgelines on Banks Peninsula and radial spurs of the Port Hills Avoiding visually prominent development Requiring development to be separated from identified important ridgelines on Banks Peninsula, taking into account visual separation and horizontal and vertical separation	Restricting visually prominent uses and development
Avoiding subdivision, use and development in those parts of outstanding natural landscapes with little or no capacity to absorb change, and allowing limited subdivision, use and development in those areas with higher potential to absorb change	Ensuring subdivision, use and development does not result in over domestication of the landscape	Limiting urban encroachment, particularly on waterway corridors
Recognising and providing protection for Ngāi Tahu values in locations of special significance to tāngata whenua		Recognising Ngāi Tahu values, particularly in relation to mahinga kai, and avoiding disturbance to known pā sites and urupā
Recognising the existence of working farms and their contribution to the openness of outstanding natural landscapes	Enabling farming, conservation activities and recreation activities which contribute to rural landscape character of Banks Peninsula	Promoting restoration and ecological enhancement initiatives, including appropriately designed ecological protection measures
		Ensuring activities are carried out in a way that maintains or enhances water quality

Figure 6.5 Policies for natural landscapes and features in the Christchurch District Plan

Peninsula, providing that activities on rural land maintain and enhance the distinctive amenity values of Banks Peninsula and the Port Hills, including indigenous biodiversity, Ngāi Tahu cultural values, open space, natural features and landscapes, and coastal environment values.⁴⁴ It also includes a policy about Banks Peninsula which encourages the protection, maintenance and enhancement of natural features and landscapes. This seeks to:

- Ensure that subdivision, use and development in the Rural Banks Peninsula Zone recognises, maintains and, where practicable, enhances the quality of the rural working environment by:
- restricting the scale, location and reflectivity of buildings to maintain a low density of built form that is not visually dominant and does not detract from views of cultural landscapes identified in the District Plan, sites of Ngāi Tahu cultural significance, or natural landforms and features;

- encouraging the protection, maintenance and enhancement of indigenous biodiversity, natural features and landscapes, historic heritage, coastal environment values, and open space; and
- encouraging public walking and cycling access connections where appropriate.⁴⁵

In recognition of the threats associated with plantation forestry and wildings, the rural chapter also includes a policy to ensure new plantation forestry is located and managed to: maintain the natural landforms and features and high visual amenity of the Banks Peninsula as well as its open rural character; not obscure views from the Summit Road; maintain the views to cultural landscapes identified in the District Plan on the Banks Peninsula; avoid adverse effects on areas of significant indigenous vegetation and significant habitats of indigenous fauna and promote enhancement, restoration and retention of indigenous biodiversity; and avoid wilding tree spread on the Banks Peninsula and promote land

management that contains or eradicates wilding trees.⁴⁶ To support that policy, the rural chapter rules provide that plantation forestry is a restricted discretionary activity and the adverse effects on landscape values are included as a matter of discretion.⁴⁷ However as discussed above, these rules have been superseded by the NESPF and plantation forestry in the rural zone, outside of an ONL or Coastal Environment overlay, is a permitted activity. This means that the values described above will have little protection against the adverse impacts of plantation forestry.

Residential activities on a site less than 4ha are a non-complying activity, except in limited circumstances. Restricted discretionary activities trigger the matters of discretion specific to significant natural features and rural amenity landscapes.⁴⁸ We were told that the increased use of non-notified consents is concerning, as it relies on Council officers alone assessing effects on landscape and natural character. As there is no guidance in the Plan about how to assess such effects, there can be inconsistent outcomes.

Under the District Plan, rural land can be subdivided down to the minimum net site requirement of 40ha below the 160m contour or 100ha if the site is entirely above the 160m contour as a controlled activity.⁴⁹ Subdivision down to 4ha is a discretionary activity⁵⁰ and below that it is non-complying.⁵¹ An exception to these minimum size requirements is provided for if the balance of the property is protected. The exception provides for a 1-4ha subdivision, if when combined with the balance area, it meets the minimum net site requirements.⁵² Only one residential development may be created and the balance area must be subject to an open space covenant that prevents the erection of any further residential units in perpetuity.

Following a 2012 subdivision under the Banks Peninsula District Plan, that resulted in a cluster of small lots surrounded by one large covenanted area, Christchurch City has now strengthened the language of the policies and conditions attached. There is now a requirement that the balance allotment be contiguous with the residential allotment⁵³ and that the layout of the subdivision maintains the rural character and avoids a form of development that results in a character of residential large-lot development.⁵⁴ As such, it is unlikely a similar proposal will be successful in the future.



A spotlight on consents data from Christchurch City Council 2005-2015

As part of the development of the Christchurch District Plan, consents data was compiled and included in the Council's section 32 analysis.⁵⁵ This provides a snapshot of landscape-related consents on Banks Peninsula over the decade from 2005 to 2015. During this period, the economy was considerably affected by both the Global Financial Crisis in 2007-8 and the Canterbury Earthquake Sequence from 2010-2013.⁵⁶

The total number of landscape consents during the period 1 January 2005 to 31 August 2015 within the Banks Peninsula (former District area) was 1,174. Of these, a total of 45 consents (4 per cent of the total) fall within an ONL, ONF or high coastal natural character area (see Figure 6.7). These increase to 116 consents (10 per cent of the total) when areas that are now within the Rural Banks Peninsula Zone are included (see Figure 6.6).

The majority of applications impacting ONLs, ONFs or high coastal natural character areas in Figure 6.6 occur near settlements such as Akaroa (13 per cent), Diamond Harbour (16 per cent), Governors Bay (13 per cent) and Lyttelton (22 per cent). The majority of consents related to dwellings and accessory buildings (53 per cent), with the balance of consents primarily made up of access ways, telecommunications, subdivision applications and earthworks. During this period, only two consents were notified, and only one was declined.

The majority of applications in the area now zoned as Rural Banks Peninsula, occurred near settlements, with Akaroa (18 per cent) and Wainui (17 per cent) having the largest number. Only two consent applications were declined (both in 2007). One was in relation to a dwelling on an undersized lot in a highly visible location. This was declined due to adverse effects on landscape character. The other consent was for a visitor accommodation complex that was also proposed to be in an open, elevated area currently free from other buildings. Again it was deemed that the effects on landscape character were too great. Eleven consent applications were notified including limited notification (11 per cent of the total). In 2014, there were six suspended consents (6 per cent of total applications), some of these dating back as far as 2006. The consents that were notified were for a range of activities including dwellings (45 per cent), recreation facilities (27 per cent) and subdivisions (27 per cent).

Effectiveness of provisions is always difficult to monitor, particularly on the basis of consents notified and declined. There will be situations where, due to discussions with Council, an application is dropped entirely or significant changes made prior to being formally lodged.

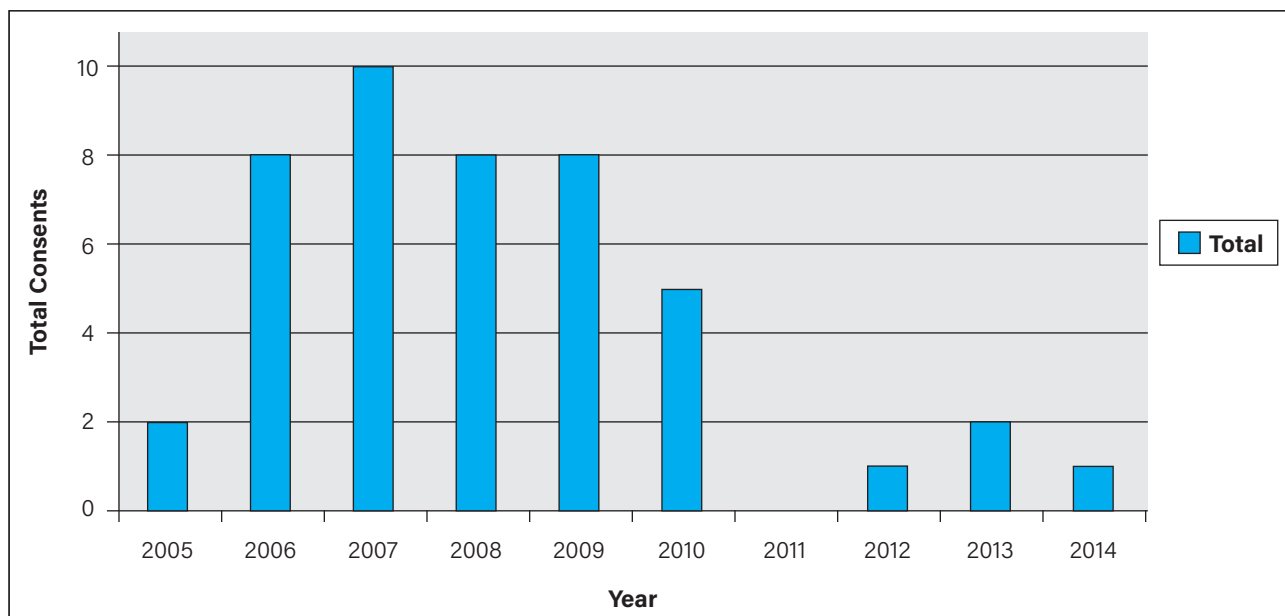


Figure 6.6 Consents impacting ONL, ONF or high coastal natural character 2005-2015 (Source: Christchurch City Council)

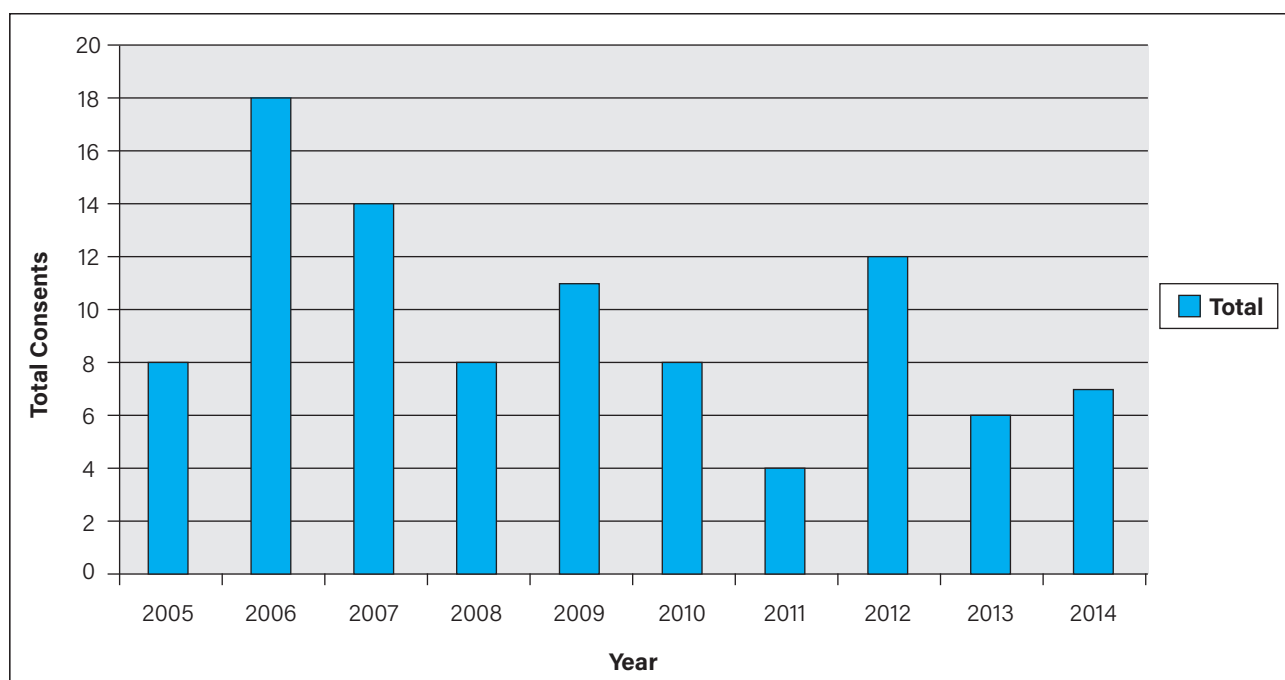


Figure 6.7 Consents impacting the now Rural Banks Peninsula Zone 2005-2015 (Source: Christchurch City Council)

6.4 Protection of cultural landscapes

As mentioned above, the Banks Peninsula District Plan did not undertake a cultural landscape assessment, and as such did not identify cultural landscapes to be protected under section 6(e) of the RMA. It did, however, use 'silent files' in order to protect sites that are culturally important to tangata whenua such as wāhi tapu. Rather than identifying the precise location of a site, a silent file gives a general indication of its location and triggers the need for engagement with mana whenua over a development proposal that might affect that locality.

As already described, following notification of the plan and the issues that arose as a result of the proposed landscape overlays, a Rural Task Force was established

primarily to investigate options to resolve the conflict as to the extent of the ONL overlay. Rūnanga did not participate in this process, and as such, the tangata whenua cultural landscape issues were not addressed in Variation 2.⁵⁷

In 2015, following amalgamation of Banks Peninsula District Council with Christchurch City Council, the Christchurch District Plan was notified. The notified plan did not update the mapping of cultural sites of significance and simply included the 'silent file' sites that had been included in the previous Banks Peninsula District Plan. The notified plan also included a policy on working with Ngāi Tahu and Papatipu Rūnanga to identify and assess sites for listing and protection and some related rules for the consideration of resource consents for activities in areas that included those sites. The current plan is largely

the result of what was proposed in submissions rather than what was notified.

Ngāi Tahu made a submission seeking a significantly more comprehensive approach to the identification and protection of sites of cultural significance. Its submission identified a greater, more representative range of cultural sites, and noted that the mapping 'recognises' sites of significance but does not 'provide for' their protection from inappropriate use and development as required under section 6 of the RMA. To address this, a new sub-chapter entitled 'Ngāi Tahu Values and the Natural Environment' was proposed which included extensive objectives, policies and rules. The submission was based on the directions in Part 2 of the RMA that relate to enabling Ngāi Tahu to provide for their social and cultural wellbeing. In particular, it was submitted that section 6(e) provided for a significantly strengthened safeguard of the relationship, including cultural relationship, of Māori to ancestral lands, water sites, wāhi tapu and other taonga.⁵⁸

At the commencement of the hearing in front of the independent hearings panel, Council and Ngāi Tahu were largely agreed on the framework for identifying cultural sites of significance. However, no agreement had been reached on how to provide for protection of the identified values. More time was sought to jointly undertake a work programme to refine the mapping of cultural sites and consider the provisions applicable to these. This process significantly narrowed the differences between the parties.⁵⁹ At the reconvened hearing, parties were largely agreed on the proposed objectives and policies. With regard to the schedules and related rules for sites of significance, a two-tiered approach was adopted by Council, which was supported (albeit with qualifiers) by Ngāi Tahu.

A spotlight on the importance of a cultural landscape approach

"A cultural landscape is a geographical area with particular (often related) traditional, historical, spiritual and ecological value to Ngāi Tahu ... Cultural landscapes are integral to Ngāi Tahu culture, identity and history, and are testament to the relationship of tāngata whenua with the land over time. They are intergenerational: providing future generations (our tamariki and mokopuna) the opportunity to experience and engage with the landscape as their tūpuna once did."⁶⁰

"A cultural landscape approach is used by Papatipu Rūnanga to identify and protect tāngata whenua values and interests from the effects of subdivision, land use change and development. While many specific sites (e.g. pā sites) are protected as recognised historic heritage, the wider contexts, settings or landscapes in which they occur are not. A cultural landscape approach enables a holistic identification and assessment of sites of significance, and other values of importance such as waterways, wetlands and waipuna."⁶¹

Through this process, in order to recognise tangata whenua values, a range of cultural layers have now been incorporated into the District Plan. At the broader level, they include Ngā Tūranga Tūpuna – a cultural landscape layer. The Plan provides a set of policies that apply to the area (see below) but there are no rules attached. However, if a resource consent is required by other rules in the



Plan, then this needs to address cultural values which in turn triggers consultation with rūnanga. The rūnanga can then provide practical advice on how their values can be protected. The Ngā Tūranga Tūpuna areas identified in the District Plan on Banks Peninsula include:

- *Ōkeina (Okains Bay) and Kawatea (Little Okains Bay)*: which are associated with several important pā settlements and events and are ancestrally and contemporarily significant as a mahinga kai for Ngāi Tahu.
- *Ōhinehou (Lyttelton)*: which has been occupied by Māori for many centuries. A pā, nohoanga and a former Māori hostel were located here. It is also important as a mahinga kai. Recent Port activities related with port recovery have revealed taonga and archaeological sites of Māori origin.
- *Pūrau, southern side of Whakaraupō (Lyttelton Harbour)*: with the catchment of Purau being home to Māori for many centuries. Pā sites, kāinga, urupā and a former Māori reserve are located here. It is also significant as a mahinga kai.
- *Whakaraupō (Lyttelton)*: which has areas of cultural landscapes with large concentrations of significant tribal history and archaeological sites, and prominent natural features that form landmarks.
- *Koukourārata -Te Ara Whānui o Makawhiua (Port Levy)*: which has areas of cultural landscapes with large concentrations of significant tribal history and archaeological sites, and prominent natural features that form landmarks.
- *Akaroa Harbour*: which has areas of cultural landscapes with large concentrations of significant tribal history and archaeological sites, and prominent natural features that form landmarks.



A spotlight on Christchurch District Plan Policy 9.5.2.2.2 Ngā Tūranga Tūpuna

- a. Recognise the historic and contemporary relationship of Ngāi Tahu with the areas and landscapes identified as Ngā Tūranga Tūpuna and:
 - i. facilitate opportunities to provide information about the historic occupation or use of these areas by Ngāi Tahu and associated values;
 - ii. incorporate representation of the association of Ngāi Tahu with these areas into the design of public buildings and/or community facilities, and in the subdivision and development or redevelopment of residential or commercial areas;
 - iii. manage earthworks involving disturbance of soils below a depth not previously disturbed by cultivation or building foundations;
 - iv. facilitate opportunities to enhance mahinga kai and other customary use of taonga species through planting and landscaping;
 - v. enhance the natural character and cultural values of water bodies, waipuna/springs, repo/wetlands and coastal waters, including reinstating original water courses where practicable;
 - vi. maintain or restore natural features with cultural values within these areas; and
 - vii. ensure that cultural values are recognised and provided for in the design, location and installation of utilities, while enabling their safe, secure and efficient installation.

Below these broader cultural landscape layers more discrete culturally-important sites are identified in the Plan as Wāhi Tapu/Wāhi Taonga or silent files. In addition, there are Nga Wai which cover estuaries, rivers and the coastal margin around the Peninsula. These reflect the statutory acknowledgement areas under the Ngai Tahu Claims Settlement Act 1998. "Nga wai are water bodies, including wetlands, tupuna (springs) and coastal waters which are significant areas of mahinga kai or other customary use for Ngāi Tahu. These areas include Te Tai o Mahaanui – the coastal area of Te Pātaka o Rākaihautū/ Banks Peninsula and Wairewa/Lake Forsyth..."⁶²

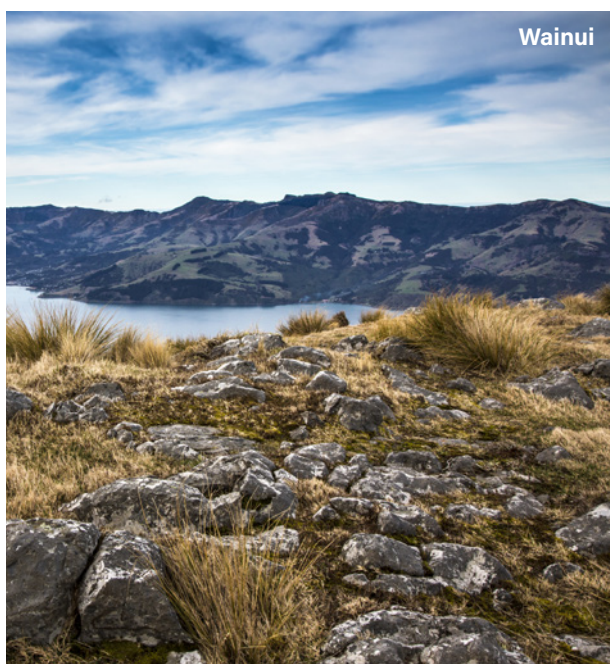
Wāhi Tapu and Wāhi Taonga areas must be protected from inappropriate development, disturbance, damage or destruction.⁶³ Similar to the case with Ngā Tūranga Tūpuna, consultation with rūnanga is required for activities proposed within or adjacent to these areas. In addition, it is a restricted discretionary activity to subdivide, construct a dwelling or undertake earthworks. The potential adverse effects, including on sensitive tangible and/or intangible Ngāi Tahu values as identified by engagement with the relevant Papatipu Rūnanga, is listed as a matter of discretion.

A spotlight on Ngā Tūtohu Whenua

As part of the refinement work undertaken between Council and Ngāi Tahu, a third category of sites was proposed – Ngā Tūtohu Whenua. These sites were significantly different from all the other sites identified as they were larger and covered multiple values and sites across whole catchments. An example of this is the Ngā Tūranga Tūpuna site of Te Whakaraupō (Lyttelton Harbour) which is a catchment-based site with many significant areas and wahi tāpu. However, this approach was rejected by the independent hearings panel on the basis of lack of scope.⁶⁴

The concept of Ngā Tūtohu Whenua (cultural landscapes) comes from the Mahaanui Iwi Management Plan which includes them as a concept to recognise areas and places of particular importance. Ngā Tūtohu Whenua can provide a culturally meaningful and effective framework for the identification, protection and management of sites and places of significance, the multiple values associated with those sites and places, and the relationship of tangata whenua with them. They differ from other sites identified as Ngā Tūranga Tūpuna, Wāhi Tapu/Wāhi Taonga or Ngā Wai which cover much smaller and more specific areas associated with a particular cultural landscape feature such as a pā or kāinga.⁶⁵

Cultural landscapes may be identified due to the concentration of values in a particular location, the particular importance of the area to Ngāi Tahu culture, history or identity, or the need to manage an area as a particular landscape unit. This is an interesting concept, and one that could also be applied to natural landscapes – recognising the feature or landscape to be protected, but also the buffer area that it.



6.5 Protection of sites of ecological significance

The process for identifying Significant Ecological Sites (SES) on Banks Peninsula has been particularly contentious. It started in 2008, as the result of an Environment Court consent order in 2007, which required the Christchurch City Council to “identify areas of significant indigenous vegetation and significant habitat of indigenous fauna” in the Banks Peninsula District Plan. In response, the Council established the Banks Peninsula Significance Ecological Steering Group to work alongside the Council planning team and landowners to identify sites. The work was designed to provide an update of historical data which included the botanical surveys undertaken by Hugh Wilson between 1983 and 1990 and the subsequent 1992 Protected Natural Areas Programme report for the Banks Ecological Region.⁶⁶

“When asked by the Panel whether stewardship on the part of the farmer is critical to the success of biodiversity goals, the Council’s ecologist for Banks Peninsula, Mr Hooson, answered ‘absolutely’... we heard a significant amount of evidence about the proactive roles that farmers at Banks Peninsula have played, over many years, in stewardship of their land.”⁶⁷

As part of the process, Council staff started talking to farmers on the Peninsula to obtain their approval to undertake ecological studies on their land. Many landowners agreed, with the expectation that they would work constructively with Council to identify the best ways to manage areas found to be of ecological importance.

“They said they wanted to do ecological studies and work with the farmers as to how to look after areas of good biodiversity. The majority of landowners agreed to having the studies done. They gave us a good understanding of what was there and made us proud of it.”

Then the earthquakes struck in 2010-2011 and the process was delayed. As described above, during the post-quake reconstruction period there was an imperative to fast track the preparation of a new Christchurch District Plan, which brought together the provisions of the Christchurch City Plan and the Banks Peninsula District Plan.

In 2014, the Government passed the Canterbury Earthquake (Christchurch Replacement District Plan) Order 2014 under the Canterbury Earthquake Recovery Act 2011. This set out a very tight timeframe for the preparation of a proposed plan. So instead of continuing the iterative process of working with landowners on the identification and management of the sites, the council listed 58 SES sites on the Banks Peninsula in the notified version of the Plan.⁶⁸ In many cases, these sites had not been ground-truthed and were based solely on the earlier ecological survey.

"[T]he Council has acknowledged, the evidence revealed that its processes fell well short of what could be regarded as proper engagement with landowners. It did not accord with the Council's own biodiversity strategy or the processes the Council committed to undertaking for the BDP [Banks Peninsula District Plan] plan change that was signalled by the 2007 Consent Order but never eventuated. In addition, the Council's evidence showed other serious deficiencies in its ecological assessment, such as a lack of adequate ground-truthing fieldwork."⁶⁹

The Plan provisions were challenged in front of the independent hearings panel by many of the landowners who had not been consulted on them. It was undisputed by all parties that there had been a failure by the Council to properly engage with landowners.⁷⁰ We were told that farmers considered this to be a breach of trust. They had allowed ecologists onto their land on the understanding that they would be fully involved in any subsequent processes. Farmers were told that they would be notified and consulted should their property have an SES identified on it. They were assured no lines or areas would appear on public maps without their prior knowledge and that full consultation around boundaries and areas

would happen prior to anything being included in the new district plan.⁷¹

"Trust between farmers and the Council has gone back 30 years."

"This did huge damage to the attitude of farmers who were proactive and wanted to work with the council. The whole room full of people [in the hearing] was being paid over a period of days with money that could have been used to fence areas off. It was counter-productive."

"Fencing is really expensive so when we first saw a SES on our property we were pretty pissed off."

Some of the most contentious issues were the requirement to take areas out of farm production and the high cost of fencing them. Because many of the farms on the Peninsula are quite small, with some verging on being uneconomic in size, removing significant areas of land from production as SESs can impact the viability of the farm. In one 'exceptional' case the notified Plan identified 95 per cent of a farmer's grazeable land as significant, giving the landowner only 40ha of usable farm.⁷²



A spotlight on SESs on Balcarres

Balcarres is a 710ha sheep and beef farm in Pigeon Bay. Around 40ha (just under 6 per cent) of the land is cultivated. 600ha of steep grassland is regularly top-dressed. In 2008, Christchurch City Council undertook an ecological survey of the property. This identified 15 SESs, from 0.3 to 80ha in size, totalling 181ha. Subsequently, the proposed Christchurch District Plan identified two of the larger SESs on the property totalling 168ha, a larger area than the 156ha originally mapped for them, as the Plan included pasture land to provide an ecological buffer.⁷³

Currently, 27ha of the property is fenced off and protected in perpetuity under a BPCT covenant. Stock graze amongst the other regenerating indigenous vegetation in the SESs. The farm's owners Pam and Ian Richardson argue that it would take around 13km of fencing to enclose the boundaries of the SESs identified in the District Plan. This would not only take around 20 per cent of their land out of production, but they estimate that the fences would cost between \$400,000-\$500,000 to construct.

After hearing all the competing evidence, the independent hearings panel "found that the evidence overwhelmingly demonstrated that the notified version was inappropriate".⁷⁴ The panel decided to adopt an alternative option which replaced the single schedule of SESs in the notified plan with two different schedules: Schedules A and B. Schedule A includes sites on public land or where, following collaboration, private landowners have agreed to their inclusion. Schedule B includes all sites on private land that require further collaboration with landowners.

Rules were imposed only for the management of SESs listed in Schedule A. The significant indigenous vegetation and habitats in Schedule A SESs is recognised and protected by requiring the avoidance of adverse effects of vegetation clearance as far as practicable (and if effects occur they needed to be remedied, mitigated or offset),⁷⁵ In the coastal environment, all adverse effects must be avoided in certain circumstances. Indigenous vegetation clearance within Schedule A sites is largely a non-complying activity (subject to certain exceptions). Outside Schedule A SESs, indigenous vegetation clearance is largely a restricted discretionary activity.⁷⁶

Schedule B was included specifically 'for information only' and identifies sites where the council intends to continue



Banks Peninsula south coast

discussions with landowners about the ecological values that are present. It is intended that Schedule A be added to in future plan changes, and as such, it does not constitute a comprehensive list.⁷⁷ This is reflected in the policy direction in the district plan. The identification of sites listed in Schedule B to be included in Schedule A is prioritised.⁷⁸ The Council is also required to initiate a plan change within 6 years (by 2022) to update Schedule A.⁷⁹

The Plan also includes an Appendix of areas outside Schedule A sites that potentially contain significant vegetation and habitat (Appendix 9.1.6.6).⁸⁰ These areas are controlled through the use of vegetation clearance rules. Council is required to advise and work with landowners where they consider that a property may contain the indigenous vegetation and habitat types listed in the Appendix, file a written record of the property file, and use rules to manage any adverse effects of the clearance or disturbance of the identified indigenous vegetation. In the coastal environment, adverse effects on certain elements (such as indigenous biodiversity that is threatened or at risk, or indigenous ecosystems and vegetation types that are threatened or are naturally rare) must be avoided.⁸¹ This provides a system for the protection of significant vegetation in areas that are not subject to the rules for Schedule A SESs.

To incentivise and assist landowners to maintain and enhance indigenous biodiversity, the Plan also includes a policy that provides a support package to landowners with incentives, advice and guidance for those managing

Schedule A SESs or those that may contain significant indigenous vegetation and habitat listed in Appendix 9.1.6.6.⁸² This is provided from the Christchurch Biodiversity Fund, which was set up with the purpose of providing support to the custodians of biodiversity in the district, and is described further below.⁸³

As part of the plan change to update Schedule A, it is proposed that Appendix 9.1.6.6 be removed. This is concerning, as it is important that the Plan retains rules for the protection of potentially significant vegetation and habitat, even if a full assessment of significance has not been undertaken. This also recognises that biodiversity is not static, and that some areas that have not been listed in Schedule A may be significant in the future, and should be managed as such.

"The Council instigated the SES programme as a statutory duty but this struck me as being risky given the background of landowner resistance to biodiversity protection on the Peninsula. We nearly had street marches when the Banks Peninsula plan was launched so the council took a non-regulatory approach and the Banks Peninsula Conservation Trust came out of that. Now the council is stepping back to a regulatory framework and has met with resistance again. I'm not sure how you unpick that. It is always my view that we have to work with landowners and not impose a top down tool set."

Godley Head



Although non-regulatory methods have largely produced positive results on Banks Peninsula for the protection of biodiversity, this is not the case where intensification is possible. This was demonstrated in the case of the Kaitorete Spit which has large areas of flat land which can be irrigated (see spotlight).

The district plan allows for indigenous vegetation clearance (both within and outside SESs) for the purpose of maintaining improved pasture as a permitted activity. The definition of 'improved pasture' in the district plan includes a visual assessment limb: "where exotic pasture grass and herb species are the visually predominant vegetation cover" and has either been grazed or cultivated since 1996.⁸⁴ The problem with applying this rule to a drylands area, like the Kaitorete Spit, is that many of the important indigenous plant species, like moss and cushion fields, are cryptic and aren't identified in a visual assessment. The definition of improved pasture has also been an issue in the Mackenzie Basin, which is detailed in the EDS case study on that area.

Ambiguity over the district plan rules has resulted in clearance and irrigation on Kaitorete spit, threatening the area's native biodiversity. This is explained further in the spotlight below.

A spotlight on development on Kaitorete Spit

Kaitorete Spit was formed approximately 6,000 years ago by gravels drifting north from the Rakaia River. It is ecologically unique and home to rare and threatened species of grasslands and shrublands as well as invertebrates, lizards and birds. For over 100 years, ownership of much of the spit has been passed down through multi-generational farming families. It has been farmed lightly so as to preserve the ecological values including being a national stronghold for the nationally endangered shrub tororaro (*Muehlenbeckia astonii*), 95 per cent of which is found on Kaitorete spit.

However, in 2018, a 968ha property on the spit was sold to Wangan Hills Limited, a farming enterprise run by a Banks Peninsula farmer. Most of the surviving population of tororaro was growing on the property. Before DOC had the opportunity to consult with the new landowner, three of the eight paddocks were sprayed and oversown. This occurred notwithstanding the property being within the coastal environment and the direction in Policy 11 of the NZCPS to avoid adverse effects on threatened or at-risk species in the coastal environment.⁸⁵

While the landowner obtained legal and farm management advice on development of the property, it had failed to obtain ecological advice. The ambiguity in the definition of 'improved pasture' meant that the landowner had proceeded with the clearance on the basis that it was a permitted activity. However, we

understand that Christchurch City Council formed the view that it was not permitted – as indigenous, not exotic, vegetation was visually predominant and therefore the clearance was unlawful. This clearance resulted in the destruction of up to a third of the viable wild population of tororaro nationally.⁸⁶

Forest and Bird has subsequently been successful in obtaining interim enforcement orders from the Environment Court preventing further intensification on the property.⁸⁷ It also sought declarations regarding the improved pasture rule and definition, including that the rule was so uncertain that it could not be enforceable.⁸⁸ We understand that these declaration proceedings have since been withdrawn on the basis that Christchurch City Council undertakes a focused plan change on this issue.

In an effort to strengthen non-regulatory support to landowners, in 2017, Christchurch City Council set up a new biodiversity fund. The funding is only available for SESs with a management plan. It covers activities such as fencing, planting and pest control. The Council will fund up to \$40,000 per property over a one-year period from a total contestable fund of \$200,000 per annum. The fund only covers 50 per cent of costs but landowner contributions for the other half can be in kind.⁸⁹ This is a positive move to further support farmers to protect special areas.

Key messages

- Protection of landscapes, cultural sites of significance and SESs have been contentious issues in the development of planning instruments for the Banks Peninsula.
- Landscape assessments of Banks Peninsula have resulted in very different outcomes. The current ONL overlay contained in the District Plan is inadequate to protect the area's significant landscape values. This is particularly so when national instruments are at play.
- Marine landscapes – the seascapes – have as yet not been comprehensively assessed.
- Sites of cultural significance to tangata whenua are now identified and provided for in the District Plan. Two tiers of protection are provided, both requiring consultation with rūnanga about protection of cultural values. More discrete areas of cultural significance trigger a restricted discretionary activity resource consent.
- SESs remain largely unprotected. However, policies in the District Plan direct that a review needs to be undertaken by 2022 with the aim of providing regulatory protection for those sites still considered to be significant.

ENDNOTES

- 1 RMA, Section 6
- 2 Salmond, 2005, 5
- 3 See *Waiareka Valley Preservation Society Inc and others v Waitaki District Council & others* C058/2009 at [230]–[231]; *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211 at [185]; *T W Reed Estate v Far North District Council* [2014] NZHC 3328 at [65]; *Gavin H Wallace Ltd v Auckland Council* [2012] NZEnvC 120
- 4 *Waiareka Valley Preservation Society Inc and others v Waitaki District Council & others* C058/2009 at [232]
- 5 *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211 at [193]
- 6 *Gavin H Wallace Ltd v Auckland Council* [2012] NZEnvC 120 at [67]
- 7 Kenderdine, 2005, 39
- 8 Christchurch District Plan, Chapter 15 (Commercial), Appendix 15.15.6 and 15.15.7
- 9 <https://ccc.govt.nz/assets/Documents/Consents-and-Licences/resource-consents/Forms/Character-Areas/Akaroa-Residential-Character-Area-Design-Guide.pdf>
- 10 RMA, Section 7
- 11 RMA, Section 2
- 12 In *Environmental Defence Society v New Zealand King Salmon*, the Supreme Court determined that ‘give effect’ is a strongly worded directive, meaning to implement
- 13 NZCPS, Policies 13 and 15
- 14 NZCPS, Policy 11(1)
- 15 NZCPS, Policies 11, 13 and 1.
- 16 See NZCPS, Policies 3.2.1 and 7.1(b)
- 17 NZCPS, Policy 13.
- 18 Glasson, 1991
- 19 Christchurch City Council, 2015, *Stage 3, Section 32, chapter 9, natural and cultural heritage, Appendix 3.4, Banks Peninsula Landscape Study 2007, 4*
- 20 *Briggs v Christchurch City Council* C045/08 at [12]
- 21 *Briggs v Christchurch City Council* C045/08 at [122]
- 22 *Briggs v Christchurch City Council* C045/08 at [130]
- 23 *Upper Clutha Tracks Trust v Queenstown Lakes District Council* [2010] NZEnvC 432 at [52]
- 24 *Briggs v Christchurch City Council* C045/08 at [22] and Annexure 1:E
- 25 *Briggs v Christchurch City Council* C045/08 at [52]
- 26 *Briggs v Christchurch City Council* C045/08 at [22]
- 27 *Briggs v Christchurch City Council* C045/08 at [123]
- 28 *Briggs v Christchurch City Council* C045/08 at [125]
- 29 *Briggs v Christchurch City Council* C045/08 at [84]
- 30 *Man O’War v Auckland Council* 19 ELRNZ 662 at [93]
- 31 *Maniototo Environmental Society Inc v Central Otago District Council* C103/2009 at [323]
- 32 Christchurch City Council, 2015, *Stage 3, Section 32, Chapter 9, Natural and Cultural Heritage, Appendix 3.1, Technical Landscape Overview Report*
- 33 Boffa Miskell, 2010
- 34 Canterbury RPS, Appendix 4, 230
- 35 See the discussion of the Banks Peninsula in *Canterbury RPS 2013*, Policy 12.3.2, principal reasons and explanations.
- 36 See Christchurch District Plan, Chapter 9.2 (landscapes and natural character)
- 37 See Christchurch District Plan, Chapter 9.6 (coastal environment)
- 38 See Christchurch District Plan, Chapter 9.6 (coastal environment), part 9.6.2.2.1
- 39 Christchurch District Plan, chapter 9.2 (landscapes and natural character), part 9.2.4.1
- 40 Christchurch District Plan, chapter 9.2 (landscapes and natural character), part 9.2.7.2
- 41 Christchurch District Plan, chapter 9.2 (landscapes and natural character), part 9.2.4.1
- 42 Christchurch District Plan, Chapter 9.2 (landscapes and natural character), part 9.6.3
- 43 Christchurch District Plan, Chapter 9.2 (landscapes and natural character), Appendix 9.2.9.1.4
- 44 Christchurch District Plan, Chapter 17 (rural), part 17.2.1.1
- 45 Christchurch District Plan, Chapter 17 (rural), part 17.2.2.8
- 46 Christchurch District Plan, Chapter 17 (rural), part 17.2.2.2.9
- 47 Christchurch District Plan, Chapter 17 (rural), part 17.11.2.4
- 48 Christchurch District Plan, Chapter 17 (rural), part 17.4
- 49 Christchurch District Plan, Rule 8.5.1.2 C6
- 50 Christchurch District Plan, Rule 8.5.1.4 D1
- 51 Christchurch District Plan, Rule 8.5.1.5 NC4
- 52 Christchurch District Plan, Rule 8.5.1.2 C7 and 8.6.1 (Table 5)
- 53 Christchurch District Plan, Rule 8.5.1.2 C7
- 54 Christchurch District Plan, Rule 8.7.6(h)
- 55 <http://resources.ccc.govt.nz/files/policiesreportsstrategies/chapter9-naturalandculturalheritage-s32-appendix3.1-technicalandscapeoverviewreport.pdf>, 31
- 56 The earthquake sequence continues but the period of highest impact and uncertainty relates to the period between 2010 and 2013
- 57 Peart, 2004, 77
- 58 Independent Hearings Panel, 2016a, [111] and [24]
- 59 Independent Hearings Panel, 2016a, [7]
- 60 Ngāi Tūāhuriri Rūnanga et al, 2013, 165
- 61 Ngāi Tūāhuriri Rūnanga et al, 2013, 105–106
- 62 <http://www.chchplan.ihp.govt.nz/wp-content/uploads/2015/12/3722-Ngāi-Tahu-S.32AA-Report-corrected-tracked.pdf>
- 63 Christchurch District Plan, Policy 9.5.2.2.1(b)
- 64 Independent Hearings Panel, 2016a, [235]
- 65 Pauling, 2016, [4.1]–[4.6]
- 66 Appelton, 2015, 8.2
- 67 Independent Hearings Panel, 2016b, [23]
- 68 Hogan, 2015, [9.1]; Hooson, 2015
- 69 Independent Hearings Panel, 2016b, [196]
- 70 Independent Hearings Panel, 2016b, [25]
- 71 Independent Hearings Panel, 2016b, [105]
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7. CONCLUSIONS AND RECOMMENDATIONS

Banks Peninsula is a unique and iconic landscape, quite different to anywhere else in the country. Its very steep and deeply incised terrain has created a vast array of microclimates which support rare and endemic species. It has a fascinating history which is closely intertwined with the rise of Ngāi Tahu in the South Island, early encounters between Māori and Europeans, and the formation of New Zealand as a new nation. It has experienced the very worst excesses of an exploitative approach to nature, with over 99 per cent of its forest being destroyed by humans. But it is also the location of a range of innovative initiatives aimed at healing the land and regenerating the forest.

The constantly shifting economics of pastoral farming has driven different land management strategies over the decades. But they have all been strongly bound by the natural constraints of the land. Ploughing is not possible on the steep terrain and there are insufficient natural water sources to support irrigation. Farmers have largely worked within the bounds of nature, applying low inputs and modest stocking regimes.

The recent collapse of coarse wool prices, coupled with the rise in red meat values, has driven Banks Peninsula farmers to carry fewer and better-fed stock. This has made keeping in production steep marginal land, which provides poorer nutrition, less financially viable. The key question for the future of the Banks Peninsula landscapes, is what use this land will be put to. This in turn will be heavily influenced by the relative economics of competing land uses.

Fenced off to exclude stock, and left to itself, the land rapidly regenerates through exotic scrub into native forest. But this does not, on its own, directly provide a financial return to landowners. There is the potential to derive carbon credits for regenerating forest under the ETS, but the rules around achieving this appear clunky and bureaucratic. It is not clear that the full sequestration and other benefits of permanent indigenous forest are properly recognised under the system. It is a similar story with the government's One Billion Trees programme which is not well configured for native regeneration.

The same cannot be said for exotic plantation forestry. To date, plantation forestry has only occurred in a small way on the Peninsula due to its rugged terrain, marginal roading network and high cost of extracting logs and getting them to market. The district plan also has strong provisions to manage the impacts of plantation forestry on biodiversity, natural character and landscape values. However, all this has changed with recent government policy.

The ETS, coupled with a now rising price for carbon credits, has created a financial incentive to establish new forestry plantations on the Peninsula, which can be financially viable even without harvest. At the

same time, the NESPF has swept away most council controls on plantation forestry on the Peninsula. It is now permitted as of right in most areas, risking wilding spread, fire proneness and water scarcity as well as significantly degrading landscape values. In seeking to create a national rule system for such forestry, government has failed to recognise the special and often unique characteristics of different landscapes in different localities. In the case of the Banks Peninsula, this deficiency has been coupled with a failure by Christchurch City Council to adequately identify ONLs, resulting in a gaping loophole in the rule framework. This needs to be urgently fixed.

Another large loophole in the regulatory framework, which needs to be remedied, is the failure of the Canterbury RPS and regional coastal plan to give effect to the NZCPS. We understand work is belatedly underway to address this.

Tourism has the potential to contribute positively to the landscapes of Banks Peninsula. Initiatives such as the Banks Peninsula Track have demonstrated how landowners can make money out of conservation through tourism activities. However, bulk tourism has the potential to overwhelm the small community and struggling infrastructure in Akaroa. The recent impacts of cruise ships was telling. The current downturn in the tourism industry due to the Covid-19 pandemic creates an opportunity to rethink the tourism model for the town and broader Banks Peninsula area. 'Slow tourism' which more deeply engages with the local rūnanga, residents and landscapes, and is more sympathetic to the area's natural values, might generate greater benefits with less negative impacts. It can help incentivise landowners to better look after the natural values on their land.

This case study has highlighted the important role that rūnanga, as kaitiaki, play in restoring landscapes both on land and in the sea. In addition, non-governmental charitable entities can bring together mana whenua, landowners and statutory authorities to work collaboratively in order to achieve ambitious landscape-scale conservation goals. Landowners can be suspicious of statutory authorities, particularly when there has been a history of conflict over regulatory provisions, and independent entities can play an important role in brokering agreement across divides. Philanthropic funding to support such charitable entities can be important to resource such initiatives.

Overall, there is considerable potential on Banks Peninsula to continue to heal the land and create a thriving, biodiverse and sustainable working landscape that supports the community and nature. Government policy needs to be aligned to this end.

Key recommendations

- Urgently revise the NESPF so that councils have the ability to control forestry landscape impacts outside designated ONLs
- Simplify the requirements for indigenous regenerating forest under the ETS and ensure its benefits are fully recognised in the allocation of credits
- Review the One Billion Tree programme requirements to ensure they are fit for purpose to support indigenous regenerating forest on the Peninsula
- Review the Canterbury RPS and regional coastal plan to give effect to the NZCPS
- Review the Christchurch District Plan, following active engagement with the community, to ensure it fully recognises cultural and natural landscapes including more comprehensively mapping the ONL areas
- Continue community engagement on the assessment and status of SESs in Schedule B of the District Plan
- Develop a tourism destination management plan for the Akaroa area which prioritises slow tourism and deeper engagement of visitors in the cultural, historical and natural landscape. Support initiatives such as the work of the Rod Donald Banks Peninsula Trust and Te Pātaka o Rākaihautū/ Banks Peninsula Geopark Trust to provide walking opportunities and interpretation to enhance the appreciation by visitors of the landscape.
- Continue to support the work of the BPCT and others in covenanting and facilitating broader landscape-scale restoration projects



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Te Pātaka o Rākaihautu Banks Peninsula is a unique and iconic landscape, quite different to anywhere else in Aotearoa. Its very steep and deeply incised terrain has created a vast array of microclimates which support rare and endemic species. It has a fascinating history which is closely intertwined with the rise of Ngāi Tahu in the South Island, early encounters between Māori and Europeans, and the formation of New Zealand as a new nation. It has experienced the worst excesses of an exploitative approach to nature, with over 99 per cent of its indigenous forest being destroyed by humans. A wide array of forest-dependant species have suffered as a result. But Banks Peninsula is now the location of a range of innovative initiatives aimed at healing the land and regenerating the forest.

The case study explores the history of this landscape, current and future pressures on it, and how the management system has responded. It investigates the impacts of farming, forestry, urban development, tourism and marine farming. It charts how central and local government policy has impacted land use, often in unintended ways. It finds that there is considerable potential to continue to heal the land on Banks Peninsula and create a thriving, biodiverse and sustainable working landscape that supports the community and nature. But for this to happen, government policy needs to be aligned in support. The report contains a set of recommendations on how to achieve this.